

Expanded Medical Reference

Topic 1 – Review Version 1

Patient-centred care is an approach that respects and responds to individual patient preferences, needs, and values, ensuring that patient values guide all clinical decisions. Its core dimensions include respect for patient values and expressed needs, emotional support and relief of fear and anxiety, information and communication in accessible language, coordination and integration of care across settings, physical comfort and pain management, involvement of family and friends as appropriate, and access to care and continuity. Evidence consistently shows that patient-centred care improves patient satisfaction, treatment adherence, health outcomes, and reduces unnecessary investigations and readmissions.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 2 – Review Version 1

Prediabetes is characterized by fasting glucose 100-125 mg/dL, 2-hour OGTT glucose 140-199 mg/dL, or HbA1c 5.7-6.4 percent. Without intervention, 5 to 10 percent progress to type 2 diabetes each year. Intensive lifestyle intervention with 5 to 7 percent weight loss and 150 minutes weekly of moderate exercise reduces progression by up to 58 percent, making early identification and intervention critical.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 3 – Review Version 1

Hypertension, defined as blood pressure at or above 130/80 mmHg by ACC/AHA guidelines, affects over 1.3 billion adults worldwide and is the leading modifiable risk factor for cardiovascular disease, stroke, and renal failure. Primary hypertension has no identifiable single cause, with genetic predisposition, dietary sodium, obesity, physical inactivity, and chronic stress all contributing. Secondary causes account for 5 to 10 percent of cases.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 4 – Review Version 1

Thiazide diuretics such as chlorthalidone and hydrochlorothiazide are effective antihypertensives that reduce blood volume through natriuresis. They are particularly effective in Black patients and elderly patients with isolated systolic hypertension. Metabolic side effects include hypokalemia, hyperuricemia, and mild glucose and lipid disturbances. Chlorthalidone has longer duration of action and better evidence for cardiovascular event reduction compared with hydrochlorothiazide.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 5 – Review Version 1

SGLT-2 inhibitors including empagliflozin, dapagliflozin, and canagliflozin reduce blood glucose by inhibiting renal glucose reabsorption, causing glycosuria. Beyond glycemic control, they demonstrate heart failure hospitalisation reduction and nephroprotective effects. They are now

recommended early in patients with established cardiovascular disease or chronic kidney disease regardless of glycemic status.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 6 – Review Version 1

Oedema represents excess fluid accumulation in the interstitial compartment and is classified by distribution and cause. Bilateral leg oedema suggests systemic causes including heart failure (elevated venous pressure with sodium retention), nephrotic syndrome (protein loss reducing oncotic pressure), liver cirrhosis (reduced albumin synthesis and portal hypertension), or hypothyroidism causing myxoedema. Unilateral leg oedema suggests local causes including deep vein thrombosis, venous insufficiency, lymphoedema, or infection. Investigation follows the distribution: bilateral oedema warrants cardiac, renal, liver, and thyroid assessment; unilateral oedema warrants duplex ultrasound to exclude DVT.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 7 – Review Version 1

The DASH diet reduces systolic blood pressure by 8 to 14 mmHg in large randomized trials. It emphasises fruits, vegetables, whole grains, low-fat dairy, and reduced saturated fat and sodium. Limiting sodium below 2.3g daily provides additional reduction. Regular aerobic exercise reduces systolic pressure by a further 5 to 8 mmHg. These lifestyle measures can reduce or delay pharmacological treatment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 8 – Review Version 1

Urinary tract infections are classified as uncomplicated (affecting otherwise healthy, non-pregnant women with normal urinary tracts) or complicated (occurring in men, pregnant women, those with structural abnormalities, immunosuppression, or catheter-related). *Escherichia coli* causes approximately 80 percent of community-acquired UTIs. Uncomplicated lower UTI in women is treated with a 3-day course of trimethoprim or nitrofurantoin based on local resistance patterns. Recurrent UTIs (three or more per year) warrant investigation for predisposing factors and may be managed with low-dose prophylactic antibiotics, post-coital prophylaxis, or self-start therapy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 9 – Review Version 1

These agents stimulate insulin secretion in a glucose-dependent manner, suppress glucagon, slow gastric emptying, and promote weight loss. Cardiovascular outcome trials have shown significant reductions in major adverse cardiovascular events in high-risk patients. GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated significant cardiovascular benefits beyond glycemic control.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 10 – Review Version 1

Screening is performed between 24 and 28 weeks using an oral glucose tolerance test. Management prioritises dietary modification and blood glucose monitoring, with insulin or metformin added if glycemic targets are not achieved.. Gestational diabetes mellitus complicates 6 to 9 percent of pregnancies and is associated with adverse outcomes including macrosomia, birth trauma, and neonatal hypoglycemia.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 11 – Review Version 1

COPD is characterised by progressive largely irreversible airflow limitation, predominantly caused by cigarette smoking or biomass fuel exposure. Spirometric confirmation with FEV1/FVC below 0.70 post-bronchodilator is required for diagnosis. Emphysema involves alveolar wall destruction reducing gas exchange, while chronic bronchitis is productive cough for at least three months in two consecutive years. Smoking cessation is the only intervention that alters disease progression.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 12 – Review Version 1

WHO recommends adults perform 150 to 300 minutes of moderate-intensity aerobic activity or 75 to 150 minutes of vigorous-intensity activity weekly, plus muscle-strengthening activities twice weekly. Regular physical activity reduces cardiovascular disease, type 2 diabetes, depression, several cancers, and all-cause mortality. Even modest increases from a sedentary baseline provide significant health benefits.. Physical inactivity contributes to 3.2 million deaths annually globally.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 13 – Review Version 1

Medication reconciliation is the process of creating the most accurate possible list of all medications a patient is taking, including dose, frequency, and route, and comparing it against the physician's admission, transfer, or discharge orders to identify and resolve discrepancies. It reduces medication errors at care transitions, which are responsible for approximately 60 percent of serious medication errors in hospital settings. Discrepancies found during reconciliation include omitted medications, wrong doses, duplicated medications, and medications continued that should have been stopped. All newly hospitalised patients should have medication reconciliation performed by a pharmacist within 24 hours of admission.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 14 – Review Version 1

Each class has independently demonstrated mortality reduction in randomised trials. Achieving all four drug classes at target doses reduces all-cause mortality by more than 70 percent compared with no treatment.. Heart failure with reduced ejection fraction is managed with quadruple therapy: an ACE inhibitor, ARB, or ARNI (sacubitril/valsartan), a beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT-2 inhibitor.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 15 – Review Version 1

Hypertensive emergencies involve markedly elevated blood pressure above 180/120 mmHg with acute target organ damage including hypertensive encephalopathy, aortic dissection, acute coronary syndrome, or acute kidney injury. They require immediate ICU admission and parenteral antihypertensive therapy with careful organ-specific blood pressure targets. Hypertensive urgency involves severely elevated pressure without organ damage and can be managed with oral agents and close follow-up.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 16 – Review Version 1

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection. The Surviving Sepsis Campaign Hour-1 Bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, starting 30 mL/kg intravenous crystalloid for hypotension or elevated lactate, and vasopressors for refractory hypotension. Each hour of delay in antibiotic administration increases mortality by approximately 7 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 17 – Review Version 1

High-intensity statins including atorvastatin 40 to 80mg and rosuvastatin 20 to 40mg are recommended for patients with established cardiovascular disease, targeting LDL-C below 1.4 mmol/L in very high-risk patients. Beyond cholesterol lowering, statins have plaque-stabilising and anti-inflammatory effects.. Statins inhibit HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis, reducing LDL cholesterol by 30 to 55 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 18 – Review Version 1

It progresses from non-proliferative stages with microaneurysms and retinal hemorrhages to proliferative retinopathy with neovascularization risking vitreous hemorrhage. Annual dilated fundal examination enables early detection. Intravitreal anti-VEGF injections and laser photocoagulation are effective treatments for sight-threatening retinopathy.. Diabetic retinopathy is the leading cause of new blindness in working-age adults.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 19 – Review Version 1

Pulmonary rehabilitation is a supervised multidisciplinary programme for patients with chronic respiratory disease combining exercise training, education, and psychosocial support. It consistently improves exercise capacity, quality of life, and dyspnoea scores without altering FEV1. It reduces acute exacerbations and hospital readmissions. It is recommended for COPD patients

with MRC dyspnoea grade 3 or above and following hospitalisation for acute exacerbation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 20 – Review Version 1

Palliative care focuses on improving quality of life for patients and families facing life-limiting illness through prevention and relief of suffering by early identification and impeccable assessment and treatment of pain and other physical, psychosocial, and spiritual problems. It is applicable from diagnosis of a life-limiting condition alongside curative treatment, not only at end of life. The World Health Organization emphasises that palliative care affirms life and regards dying as a normal process, neither hastening nor postponing death. Advance care planning, including discussion of resuscitation preferences, preferred place of death, and lasting power of attorney for health and welfare, is an important component.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 21 – Review Version 1

Cognitive behavioural therapy is a structured evidence-based psychological treatment helping patients identify connections between thoughts, emotions, and behaviours. In depression, CBT targets negative automatic thoughts and behavioural withdrawal through behavioural activation. In anxiety disorders, it incorporates exposure and response prevention, worry postponement, and attentional training. Its benefits persist beyond treatment completion, making it superior to medication alone for long-term relapse prevention.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 22 – Review Version 1

Statins inhibit HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis, reducing LDL cholesterol by 30 to 55 percent. High-intensity statins including atorvastatin 40 to 80mg and rosuvastatin 20 to 40mg are recommended for patients with established cardiovascular disease, targeting LDL-C below 1.4 mmol/L in very high-risk patients. Beyond cholesterol lowering, statins have plaque-stabilising and anti-inflammatory effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 23 – Review Version 1

Dehydration occurs when fluid losses exceed intake, reducing total body water. Mild dehydration (1-2 percent body weight) causes thirst and reduced concentration; moderate dehydration (3-5 percent) causes headache, dizziness, and reduced urine output; severe dehydration (above 6 percent) causes tachycardia, hypotension, and confusion. In adults, urine colour is a practical guide: pale yellow indicates adequate hydration; dark amber indicates dehydration. Oral rehydration is appropriate for mild to moderate dehydration using water or oral rehydration salts. Intravenous fluid resuscitation is required for severe dehydration or when oral intake is impossible. Elderly patients are at particular risk due to reduced thirst sensation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications,

contraindications, and multidisciplinary care approaches.

Topic 24 – Review Version 1

Key principles include prescribing antibiotics only when bacterial infection is confirmed or highly likely, selecting the narrowest spectrum effective agent, using the shortest evidence-based duration, and reviewing at 48 to 72 hours once microbiological results are available. Antimicrobial resistance is a global crisis threatening the foundations of modern medicine.. Antibiotic stewardship promotes appropriate antimicrobial use to improve patient outcomes while limiting resistance selection pressure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 25 – Review Version 1

ACE inhibitors and ARBs are preferred antihypertensives in patients with diabetes, chronic kidney disease, or heart failure with reduced ejection fraction. ACE inhibitors block conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. ARBs block angiotensin II at its receptor. Both reduce proteinuria and slow nephropathy progression. They should not be combined due to increased adverse effects without additional blood pressure benefit.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 26 – Review Version 1

The principles of medical ethics provide a framework for clinical decision-making when values conflict. Autonomy refers to the patient's right to make informed decisions about their care, requiring valid informed consent. Beneficence is the duty to act in the patient's best interest. Non-maleficence requires avoiding unnecessary harm and balancing risks against benefits. Justice involves fair allocation of healthcare resources and equitable treatment of all patients. When these principles conflict, as in a patient refusing beneficial treatment (autonomy versus beneficence), the framework guides structured ethical analysis. Mental capacity to make specific decisions must be assessed before overriding patient choice, using the Mental Capacity Act 2005 framework in England and Wales.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 27 – Review Version 1

Peripheral diabetic neuropathy affects up to 50 percent of patients with long-standing diabetes, presenting as distal symmetric polyneuropathy with pain, tingling, numbness, and loss of protective sensation. Annual foot examinations and patient education on foot care are essential preventive measures. Pharmacological management of neuropathic pain includes duloxetine, pregabalin, and amitriptyline.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 28 – Review Version 1

Polypharmacy, the concurrent use of five or more medications, affects approximately 40 percent of adults over 65 and significantly increases risk of adverse drug reactions, drug-drug interactions,

medication non-adherence, falls, cognitive impairment, and preventable hospitalisation. Regular structured medication reviews using the Beers Criteria or STOPP/START criteria identify potentially inappropriate medications and opportunities for safe deprescribing.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 29 – Review Version 1

Deep vein thrombosis and pulmonary embolism constitute venous thromboembolism, a major cause of preventable hospital mortality. DVT typically involves the leg deep veins, with clot propagation risking embolisation to the pulmonary vasculature. Risk factors include immobility, surgery, malignancy, pregnancy, oral contraceptive use, thrombophilia, and obesity. Treatment with anticoagulation prevents clot extension and embolisation while the body's fibrinolytic system dissolves the thrombus.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 30 – Review Version 1

Calcium channel blockers such as amlodipine are effective antihypertensives with particular benefit in elderly patients, Black patients, and isolated systolic hypertension. They reduce vascular smooth muscle tone by blocking voltage-gated calcium channels. Non-dihydropyridine CCBs such as diltiazem and verapamil also reduce heart rate, useful in atrial fibrillation but contraindicated in heart failure with reduced ejection fraction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 31 – Review Version 1

Dyspepsia (upper abdominal discomfort, bloating, early satiety, or nausea) affects up to 40 percent of the population and has many causes. Functional dyspepsia (no organic cause on investigation) is most common, managed with lifestyle advice, PPI or H2 blocker trial, and consideration of *Helicobacter pylori* testing and eradication if positive. Alarm features requiring urgent endoscopy include unintentional weight loss, dysphagia, persistent vomiting, haematemesis or melaena, iron-deficiency anaemia, and epigastric mass. Any new onset dyspepsia over age 55 warrants early endoscopy. *H. pylori* eradication using triple therapy (PPI, amoxicillin, and clarithromycin for 7 days) achieves eradication in approximately 80 percent of cases.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 32 – Review Version 1

Diabetic retinopathy is the leading cause of new blindness in working-age adults. It progresses from non-proliferative stages with microaneurysms and retinal hemorrhages to proliferative retinopathy with neovascularization risking vitreous hemorrhage. Annual dilated fundal examination enables early detection. Intravitreal anti-VEGF injections and laser photocoagulation are effective treatments for sight-threatening retinopathy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 33 – Review Version 1

BMI provides simple screening: overweight 25 to 29.9, obesity class I 30 to 34.9, class II 35 to 39.9, class III 40 or above. Waist circumference better captures central adiposity and metabolic risk. Management involves comprehensive lifestyle intervention, pharmacotherapy in eligible patients including GLP-1 receptor agonists which can reduce weight by 10 to 20 percent, and bariatric surgery for severe obesity.. Obesity is characterised by excess adiposity impairing health.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 34 – Review Version 1

COX-1 inhibition impairs gastric mucosal protection and platelet aggregation, increasing gastrointestinal bleeding risk particularly in elderly patients, those on corticosteroids or anticoagulants, or those with previous peptic ulcer disease. All NSAIDs carry cardiovascular and renal risks, particularly in volume-depleted patients or those with pre-existing kidney disease.. NSAIDs including ibuprofen, naproxen, and diclofenac inhibit COX-1 and COX-2 enzymes, reducing prostaglandin synthesis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 35 – Review Version 1

It works primarily by inhibiting hepatic gluconeogenesis through activation of AMP-activated protein kinase (AMPK), reducing fasting plasma glucose without causing hypoglycemia. Common side effects include gastrointestinal disturbances, which can be minimized by titrating the dose gradually and taking it with food.. Metformin is the first-line pharmacological agent for type 2 diabetes in most international guidelines.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 36 – Review Version 1

The WHO five moments framework identifies critical junctures: before patient contact, before an aseptic procedure, after body fluid exposure, after patient contact, and after contact with patient surroundings. Alcohol-based hand rub is preferred for most clinical situations due to superior speed and convenience, while soap and water is required for *Clostridioides difficile* and norovirus.. Hand hygiene is the single most effective measure to prevent healthcare-associated infections.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 37 – Review Version 1

They reduce vascular smooth muscle tone by blocking voltage-gated calcium channels. Non-dihydropyridine CCBs such as diltiazem and verapamil also reduce heart rate, useful in atrial fibrillation but contraindicated in heart failure with reduced ejection fraction.. Calcium channel blockers such as amlodipine are effective antihypertensives with particular benefit in elderly patients, Black patients, and isolated systolic hypertension.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications,

contraindications, and multidisciplinary care approaches.

Topic 38 – Review Version 1

Physical inactivity contributes to 3.2 million deaths annually globally. WHO recommends adults perform 150 to 300 minutes of moderate-intensity aerobic activity or 75 to 150 minutes of vigorous-intensity activity weekly, plus muscle-strengthening activities twice weekly. Regular physical activity reduces cardiovascular disease, type 2 diabetes, depression, several cancers, and all-cause mortality. Even modest increases from a sedentary baseline provide significant health benefits.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 39 – Review Version 1

DVT typically involves the leg deep veins, with clot propagation risking embolisation to the pulmonary vasculature. Risk factors include immobility, surgery, malignancy, pregnancy, oral contraceptive use, thrombophilia, and obesity. Treatment with anticoagulation prevents clot extension and embolisation while the body's fibrinolytic system dissolves the thrombus.. Deep vein thrombosis and pulmonary embolism constitute venous thromboembolism, a major cause of preventable hospital mortality.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 40 – Review Version 1

Atrial fibrillation is the most common sustained cardiac arrhythmia, causing irregular ventricular response from chaotic atrial electrical activity. The primary risks are stroke from atrial thrombus and heart failure from tachycardia-mediated cardiomyopathy. Management involves rate control, rhythm control in appropriate patients, and anticoagulation with direct oral anticoagulants guided by CHA2DS2-VASc scoring.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 41 – Review Version 1

The primary risks are stroke from atrial thrombus and heart failure from tachycardia-mediated cardiomyopathy. Management involves rate control, rhythm control in appropriate patients, and anticoagulation with direct oral anticoagulants guided by CHA2DS2-VASc scoring.. Atrial fibrillation is the most common sustained cardiac arrhythmia, causing irregular ventricular response from chaotic atrial electrical activity.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 42 – Review Version 1

SSRIs are the first-line pharmacological treatment for depression and anxiety disorders due to their balance of efficacy, tolerability, and safety in overdose compared with older tricyclics. Sertraline has the strongest evidence base in depression, while escitalopram and citalopram have good tolerability profiles. Full therapeutic benefit requires 4 to 6 weeks and treatment should continue for at least 6 to 12 months after remission to prevent relapse.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 43 – Review Version 1

Bipolar disorder features recurrent mania or hypomania alternating with depressive episodes. Lithium remains the gold standard mood stabiliser for long-term prevention of both manic and depressive episodes, with robust evidence for suicide risk reduction. Valproate is effective for mania prevention; lamotrigine particularly targets depressive episodes. Close monitoring of lithium levels, renal function, and thyroid function is essential given its narrow therapeutic index.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 44 – Review Version 1

Type 2 diabetes mellitus is a chronic metabolic disorder characterized by insulin resistance and progressive beta-cell dysfunction. The pathophysiology involves impaired insulin signaling in peripheral tissues, particularly skeletal muscle, liver, and adipose tissue, leading to chronic hyperglycemia. Lifestyle modifications including dietary changes, physical activity, and weight management remain the cornerstone of initial management before pharmacological intervention is considered.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 45 – Review Version 1

Annual foot examinations and patient education on foot care are essential preventive measures. Pharmacological management of neuropathic pain includes duloxetine, pregabalin, and amitriptyline. Peripheral diabetic neuropathy affects up to 50 percent of patients with long-standing diabetes, presenting as distal symmetric polyneuropathy with pain, tingling, numbness, and loss of protective sensation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 46 – Review Version 1

The renin-angiotensin-aldosterone system (RAAS) is a key hormonal cascade regulating blood pressure and fluid balance. Renin, secreted by juxtaglomerular cells in response to reduced renal perfusion, converts angiotensinogen to angiotensin I. ACE converts angiotensin I to the active angiotensin II, a potent vasoconstrictor that also stimulates aldosterone release causing sodium and water retention. RAAS blockade with ACE inhibitors or ARBs is central to managing hypertension, heart failure, and diabetic nephropathy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 47 – Review Version 1

Septic arthritis is a medical emergency requiring same-day joint aspiration and intravenous antibiotics. It typically presents with acute monoarthritis with fever, a hot swollen joint, and severely restricted movement. The knee is most commonly affected. *Staphylococcus aureus* is the most

common causative organism in adults. Diagnosis is confirmed by synovial fluid analysis showing turbid fluid with white cell count above 50,000 cells per microlitre with predominantly neutrophils, and Gram stain and culture. Delay in treatment risks rapid cartilage destruction and permanent joint damage.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 48 – Review Version 1

The WHO has identified antibiotic resistance as one of the greatest threats to global health. ESKAPE organisms (Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter species) account for the majority of serious healthcare-associated infections with resistance threatening treatment options.. Antimicrobial resistance is driven by overuse and misuse of antibiotics in human medicine and agriculture, selecting for resistant bacterial strains.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 49 – Review Version 1

They are particularly effective in Black patients and elderly patients with isolated systolic hypertension. Metabolic side effects include hypokalemia, hyperuricemia, and mild glucose and lipid disturbances. Chlorthalidone has longer duration of action and better evidence for cardiovascular event reduction compared with hydrochlorothiazide.. Thiazide diuretics such as chlorthalidone and hydrochlorothiazide are effective antihypertensives that reduce blood volume through natriuresis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 50 – Review Version 1

Inhaled corticosteroids are the cornerstone of controller therapy, reducing airway inflammation and risk of exacerbations. Short-acting beta-2 agonists provide rapid bronchodilation for symptom relief. Management follows a stepwise approach guided by symptom control and exacerbation frequency.. Asthma is a chronic inflammatory airway disease characterised by variable reversible airflow obstruction, bronchial hyperresponsiveness, and airway remodelling.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 51 – Review Version 1

Chronic kidney disease is classified by cause, GFR category, and albuminuria category using the CGA staging system. GFR categories range from G1 (normal, above 90 mL/min) to G5 (kidney failure, below 15 mL/min). Albuminuria categories range from A1 (normal to mildly increased, ACR below 3 mg/mmol) to A3 (severely increased, ACR above 30 mg/mmol). High risk of progression is conferred by lower GFR combined with higher albuminuria. Management aims to slow progression through blood pressure control (target below 130/80 mmHg), RAAS blockade in proteinuric CKD, and SGLT-2 inhibitors now recommended across CKD stages for their proven renoprotective effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 52 – Review Version 1

Community-acquired pneumonia severity is assessed using the CURB-65 score, assigning one point each for confusion, urea above 7 mmol/L, respiratory rate 30 or more, blood pressure below 90/60 mmHg, and age 65 or over. Score 0-1 indicates low mortality risk suitable for home treatment. Score 2 warrants hospital admission. Score 3 or above indicates high mortality risk requiring hospital admission and possible ICU assessment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 53 – Review Version 1

It consistently improves exercise capacity, quality of life, and dyspnoea scores without altering FEV1. It reduces acute exacerbations and hospital readmissions. It is recommended for COPD patients with MRC dyspnoea grade 3 or above and following hospitalisation for acute exacerbation.. Pulmonary rehabilitation is a supervised multidisciplinary programme for patients with chronic respiratory disease combining exercise training, education, and psychosocial support.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 54 – Review Version 1

Diabetic ketoacidosis is a life-threatening acute complication caused by absolute or relative insulin deficiency. The biochemical triad includes hyperglycemia, metabolic acidosis, and ketonemia. Management requires fluid resuscitation, insulin infusion after potassium correction, electrolyte replacement particularly potassium, and identification of the precipitating cause such as infection or omitted insulin.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 55 – Review Version 1

Acute myocardial infarction results from rupture of an atherosclerotic plaque in a coronary artery leading to thrombus formation and vessel occlusion. Time to reperfusion is the critical determinant of outcome: primary percutaneous coronary intervention is the preferred strategy when achievable within 120 minutes of first medical contact. Immediate treatment includes aspirin, a P2Y12 inhibitor, anticoagulation, and infarct-related artery reperfusion.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 56 – Review Version 1

The HbA1c target for most adults with type 2 diabetes is below 7.0 percent, though individualized targets should account for age, comorbidities, hypoglycemia risk, and patient preference. In elderly or frail patients, a less stringent target of 7.5 to 8.0 percent may be appropriate to avoid hypoglycemia and its associated falls and cardiac risks.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 57 – Review Version 1

They require immediate ICU admission and parenteral antihypertensive therapy with careful organ-specific blood pressure targets. Hypertensive urgency involves severely elevated pressure without organ damage and can be managed with oral agents and close follow-up.. Hypertensive emergencies involve markedly elevated blood pressure above 180/120 mmHg with acute target organ damage including hypertensive encephalopathy, aortic dissection, acute coronary syndrome, or acute kidney injury.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 58 – Review Version 1

CYP3A4 enzyme interactions are particularly clinically significant: clarithromycin inhibits CYP3A4 and dramatically raises plasma levels of statins (risking rhabdomyolysis), ciclosporin, and benzodiazepines. Warfarin has hundreds of documented interactions and requires careful monitoring with any medication change.. Drug-drug interactions occur through pharmacokinetic mechanisms altering absorption, distribution, metabolism, or excretion, or pharmacodynamic mechanisms with additive or antagonistic effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 59 – Review Version 1

Vitamin D deficiency is highly prevalent, affecting approximately 1 billion people globally, and results from inadequate sun exposure, dietary insufficiency, or malabsorption. It causes rickets in children and osteomalacia in adults, presenting with bone pain, proximal muscle weakness, and pathological fractures. Severe vitamin D deficiency also impairs immune function and has been associated with increased risk of autoimmune disease, cardiovascular disease, and certain cancers. Supplementation guidelines recommend 400 IU daily for all adults in the UK and higher doses (1000-4000 IU) for those at high risk including elderly care home residents, those with dark skin or little sun exposure, and pregnant women.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 60 – Review Version 1

Spirometric confirmation with FEV1/FVC below 0.70 post-bronchodilator is required for diagnosis. Emphysema involves alveolar wall destruction reducing gas exchange, while chronic bronchitis is productive cough for at least three months in two consecutive years. Smoking cessation is the only intervention that alters disease progression.. COPD is characterised by progressive largely irreversible airflow limitation, predominantly caused by cigarette smoking or biomass fuel exposure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 61 – Review Version 1

Ambulatory blood pressure monitoring over 24 hours provides a more accurate assessment of true blood pressure burden. Masked hypertension, the converse, carries cardiovascular risk similar to sustained hypertension and should not be overlooked.. White coat hypertension affects 15 to 30 percent of referred patients and involves elevated readings in clinical settings but normal levels at home or on ambulatory monitoring.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 62 – Review Version 1

Chronic pain is defined as pain persisting beyond the normal expected healing time, typically more than 3 months. It affects approximately 28 million adults in the UK and has a significant biopsychosocial impact. It is now understood as a complex condition involving peripheral and central sensitisation, psychological factors including fear-avoidance beliefs and catastrophising, and social factors including work and relationship impacts. The modern pain management paradigm has shifted from a largely pharmacological approach to a multidisciplinary biopsychosocial model incorporating physiotherapy, psychological therapies particularly CBT and acceptance and commitment therapy, lifestyle optimisation, and judicious use of analgesics at the lowest effective dose.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 63 – Review Version 1

It emphasises fruits, vegetables, whole grains, low-fat dairy, and reduced saturated fat and sodium. Limiting sodium below 2.3g daily provides additional reduction. Regular aerobic exercise reduces systolic pressure by a further 5 to 8 mmHg. These lifestyle measures can reduce or delay pharmacological treatment.. The DASH diet reduces systolic blood pressure by 8 to 14 mmHg in large randomized trials.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 64 – Review Version 1

Informed consent is a legal and ethical requirement for medical procedures and treatment. For consent to be valid, the patient must have capacity to make the decision, the consent must be given voluntarily without coercion, and the patient must be provided with sufficient information about the proposed treatment including its benefits, risks, alternatives, and consequences of no treatment. Montgomery v Lanarkshire Health Board (2015) established in UK law that patients must be informed of all material risks that a reasonable patient would consider significant. Documentation of the consent process is essential. Consent for research requires additional safeguards including Research Ethics Committee approval.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 65 – Review Version 1

Inflammatory markers including C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR) are non-specific indicators of inflammation used to support diagnosis and monitor treatment response. CRP rises and falls rapidly within hours to days, making it more useful for monitoring acute conditions. ESR rises more slowly and reflects plasma viscosity. Both are elevated in

infection, autoimmune disease, malignancy, and tissue necrosis. Very high CRP above 100 mg/L suggests bacterial infection or severe inflammation. Procalcitonin is more specific for bacterial infection and is used in some settings to guide antibiotic prescribing. Normal inflammatory markers do not exclude serious pathology, particularly in immunocompromised patients.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 66 – Review Version 1

Continuous glucose monitoring systems provide real-time glucose readings every few minutes, enabling patients and clinicians to make more informed treatment decisions. Time-in-range, defined as glucose between 70 and 180 mg/dL, is emerging as a key glycemic metric alongside HbA1c. Higher time-in-range is associated with reduced rates of diabetic microvascular and macrovascular complications.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 67 – Review Version 1

GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated significant cardiovascular benefits beyond glycemic control. These agents stimulate insulin secretion in a glucose-dependent manner, suppress glucagon, slow gastric emptying, and promote weight loss. Cardiovascular outcome trials have shown significant reductions in major adverse cardiovascular events in high-risk patients.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 68 – Review Version 1

Antibiotic stewardship promotes appropriate antimicrobial use to improve patient outcomes while limiting resistance selection pressure. Key principles include prescribing antibiotics only when bacterial infection is confirmed or highly likely, selecting the narrowest spectrum effective agent, using the shortest evidence-based duration, and reviewing at 48 to 72 hours once microbiological results are available. Antimicrobial resistance is a global crisis threatening the foundations of modern medicine.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 69 – Review Version 1

Anaemia is defined as haemoglobin below the lower limit of normal for age and sex: below 130 g/L in adult men and below 120 g/L in adult women. The main causes are classified by mechanism: iron deficiency (most common worldwide, from blood loss, poor intake, or malabsorption), vitamin B12 or folate deficiency (macrocytic anaemia from impaired DNA synthesis), haemolysis (destruction of red blood cells from autoimmune, mechanical, or hereditary causes), and bone marrow failure or infiltration. History, full blood count, reticulocyte count, and iron studies guide initial investigation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 70 – Review Version 1

Primary hypertension has no identifiable single cause, with genetic predisposition, dietary sodium, obesity, physical inactivity, and chronic stress all contributing. Secondary causes account for 5 to 10 percent of cases.. Hypertension, defined as blood pressure at or above 130/80 mmHg by ACC/AHA guidelines, affects over 1.3 billion adults worldwide and is the leading modifiable risk factor for cardiovascular disease, stroke, and renal failure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 71 – Review Version 1

The PREDIMED trial demonstrated significant reductions in major cardiovascular events in high-risk patients following a Mediterranean diet compared with a low-fat diet. It also reduces risk of type 2 diabetes, colorectal and breast cancer, cognitive decline, and all-cause mortality.. The Mediterranean diet is characterised by high consumption of plant foods, olive oil as the primary fat, moderate fish and seafood, and low red and processed meat consumption.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 72 – Review Version 1

Occupational health medicine focuses on the relationship between work and health in both directions: how work can cause or worsen health conditions (occupational disease), and how health conditions affect the ability to work. Common occupational diseases include occupational asthma (from isocyanates, flour dust, or latex), vibration white finger from hand-arm vibration syndrome, noise-induced hearing loss, mesothelioma from asbestos exposure, contact dermatitis, and musculoskeletal disorders from manual handling or repetitive strain. Fitness for work assessments consider functional capacity, job demands, reasonable workplace adjustments, and prognosis. Statutory sick pay and fit notes (formerly sick notes) are important administrative tools in managing work absence.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 73 – Review Version 1

Hand hygiene is the single most effective measure to prevent healthcare-associated infections. The WHO five moments framework identifies critical junctures: before patient contact, before an aseptic procedure, after body fluid exposure, after patient contact, and after contact with patient surroundings. Alcohol-based hand rub is preferred for most clinical situations due to superior speed and convenience, while soap and water is required for *Clostridioides difficile* and norovirus.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 74 – Review Version 1

Diabetic nephropathy affects approximately 40 percent of diabetic patients and is the leading cause of end-stage renal disease globally. Early detection through annual screening for urinary albumin-to-creatinine ratio and eGFR is essential. ACE inhibitors and ARBs are first-line for hypertension in diabetic patients with albuminuria due to their independent nephroprotective properties beyond blood pressure lowering.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 75 – Review Version 1

Asthma is a chronic inflammatory airway disease characterised by variable reversible airflow obstruction, bronchial hyperresponsiveness, and airway remodelling. Inhaled corticosteroids are the cornerstone of controller therapy, reducing airway inflammation and risk of exacerbations. Short-acting beta-2 agonists provide rapid bronchodilation for symptom relief. Management follows a stepwise approach guided by symptom control and exacerbation frequency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 76 – Review Version 1

Time-in-range, defined as glucose between 70 and 180 mg/dL, is emerging as a key glycemic metric alongside HbA1c. Higher time-in-range is associated with reduced rates of diabetic microvascular and macrovascular complications.. Continuous glucose monitoring systems provide real-time glucose readings every few minutes, enabling patients and clinicians to make more informed treatment decisions.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 77 – Review Version 1

ACE inhibitors block conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. ARBs block angiotensin II at its receptor. Both reduce proteinuria and slow nephropathy progression. They should not be combined due to increased adverse effects without additional blood pressure benefit.. ACE inhibitors and ARBs are preferred antihypertensives in patients with diabetes, chronic kidney disease, or heart failure with reduced ejection fraction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 78 – Review Version 1

The Surviving Sepsis Campaign Hour-1 Bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, starting 30 mL/kg intravenous crystalloid for hypotension or elevated lactate, and vasopressors for refractory hypotension. Each hour of delay in antibiotic administration increases mortality by approximately 7 percent.. Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 79 – Review Version 1

Sertraline has the strongest evidence base in depression, while escitalopram and citalopram have good tolerability profiles. Full therapeutic benefit requires 4 to 6 weeks and treatment should continue for at least 6 to 12 months after remission to prevent relapse.. SSRIs are the first-line pharmacological treatment for depression and anxiety disorders due to their balance of efficacy, tolerability, and safety in overdose compared with older tricyclics.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 80 – Review Version 1

Wound healing progresses through four overlapping phases: haemostasis (immediate platelet plug and clot formation), inflammation (neutrophil and macrophage infiltration clearing debris and bacteria), proliferation (fibroblast migration producing collagen, angiogenesis, and re-epithelialisation), and remodelling (collagen reorganisation over months to years). Factors impairing wound healing include diabetes (neuropathy, ischaemia, and impaired immune response), malnutrition (particularly protein, zinc, and vitamin C deficiency), corticosteroids, smoking (causing vasoconstriction and reduced oxygen delivery), infection, and foreign bodies. Assessment should determine the wound bed, edges, surrounding skin, exudate, infection signs, and underlying cause.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 81 – Review Version 1

True resistance must be distinguished from pseudoresistance caused by poor adherence, white coat effect, or suboptimal measurement. Secondary causes should be investigated. Spironolactone (an aldosterone antagonist) as a fourth agent provides the most evidence for treating true resistant hypertension.. Resistant hypertension is blood pressure above target despite optimal doses of three antihypertensive agents including a diuretic.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 82 – Review Version 1

Antimicrobial resistance is driven by overuse and misuse of antibiotics in human medicine and agriculture, selecting for resistant bacterial strains. The WHO has identified antibiotic resistance as one of the greatest threats to global health. ESKAPE organisms (Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter species) account for the majority of serious healthcare-associated infections with resistance threatening treatment options.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 83 – Review Version 1

Pain assessment in clinical practice uses validated tools appropriate to the patient's communication ability. The numerical rating scale (0-10) is the most commonly used in cognitively intact adults. The visual analogue scale uses a 100mm line from no pain to worst imaginable pain. The Wong-Baker FACES scale is appropriate for children and patients with cognitive impairment. Multidimensional assessment tools such as the Brief Pain Inventory assess pain intensity and its impact on function and quality of life. Chronic pain assessment should include physical, psychological, and social dimensions using a biopsychosocial framework, with regular reassessment using the same validated tool for consistency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 84 – Review Version 1

Chronic liver disease progresses through stages from simple steatosis through steatohepatitis, fibrosis, cirrhosis, and finally hepatocellular carcinoma. Common causes include alcohol-related liver disease, non-alcoholic fatty liver disease (increasingly prevalent with obesity epidemic), chronic hepatitis B and C, autoimmune hepatitis, primary biliary cholangitis, and haemochromatosis. Cirrhosis complications include portal hypertension causing varices and ascites, hepatic encephalopathy, spontaneous bacterial peritonitis, hepatorenal syndrome, and coagulopathy. Six-monthly ultrasound and AFP for hepatocellular carcinoma surveillance is recommended in all patients with established cirrhosis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 85 – Review Version 1

The Mediterranean diet is characterised by high consumption of plant foods, olive oil as the primary fat, moderate fish and seafood, and low red and processed meat consumption. The PREDIMED trial demonstrated significant reductions in major cardiovascular events in high-risk patients following a Mediterranean diet compared with a low-fat diet. It also reduces risk of type 2 diabetes, colorectal and breast cancer, cognitive decline, and all-cause mortality.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 86 – Review Version 1

The biochemical triad includes hyperglycemia, metabolic acidosis, and ketonemia. Management requires fluid resuscitation, insulin infusion after potassium correction, electrolyte replacement particularly potassium, and identification of the precipitating cause such as infection or omitted insulin.. Diabetic ketoacidosis is a life-threatening acute complication caused by absolute or relative insulin deficiency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 87 – Review Version 1

Heart failure with reduced ejection fraction is managed with quadruple therapy: an ACE inhibitor, ARB, or ARNI (sacubitril/valsartan), a beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT-2 inhibitor. Each class has independently demonstrated mortality reduction in randomised trials. Achieving all four drug classes at target doses reduces all-cause mortality by more than 70 percent compared with no treatment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 88 – Review Version 1

Depression affects over 280 million people globally and is characterised by persistent low mood, anhedonia, fatigue, cognitive impairment, sleep disturbance, appetite changes, and suicidal ideation in severe cases. The biopsychosocial model recognises neurobiological contributions including dysregulation of serotonin, norepinephrine, and dopamine pathways, alongside psychological factors such as negative cognitive schemas and social factors including adverse life events and isolation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 89 – Review Version 1

Score 0-1 indicates low mortality risk suitable for home treatment. Score 2 warrants hospital admission. Score 3 or above indicates high mortality risk requiring hospital admission and possible ICU assessment.. Community-acquired pneumonia severity is assessed using the CURB-65 score, assigning one point each for confusion, urea above 7 mmol/L, respiratory rate 30 or more, blood pressure below 90/60 mmHg, and age 65 or over.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 90 – Review Version 1

Fibromyalgia is a centralised pain syndrome characterised by widespread musculoskeletal pain, fatigue, sleep disturbance, cognitive difficulties (fibro fog), and often comorbid depression and anxiety. It results from central sensitisation where the nervous system amplifies pain signals. Diagnosis is clinical using the 2010 ACR diagnostic criteria, requiring widespread pain index and symptom severity scores above thresholds. Management is multidisciplinary: aerobic exercise and physiotherapy are the most evidence-based interventions, CBT addresses maladaptive pain behaviours, and low-dose amitriptyline or duloxetine can improve pain and sleep. Opioids are not recommended and may worsen central sensitisation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 91 – Review Version 1

Thyroid hormones T3 (triiodothyronine) and T4 (thyroxine) regulate metabolic rate, growth, and development. Hypothyroidism causes fatigue, weight gain, cold intolerance, constipation, dry skin, and bradycardia, and is most commonly caused by Hashimoto's thyroiditis (autoimmune destruction). Hyperthyroidism causes weight loss despite increased appetite, heat intolerance, palpitations, tremor, and anxiety, most commonly from Graves' disease (TSH receptor stimulating antibodies). TSH is the most sensitive screening test for thyroid dysfunction, rising in hypothyroidism and falling in hyperthyroidism.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 92 – Review Version 1

Lithium remains the gold standard mood stabiliser for long-term prevention of both manic and depressive episodes, with robust evidence for suicide risk reduction. Valproate is effective for mania prevention; lamotrigine particularly targets depressive episodes. Close monitoring of lithium levels, renal function, and thyroid function is essential given its narrow therapeutic index.. Bipolar disorder features recurrent mania or hypomania alternating with depressive episodes.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 93 – Review Version 1

Delirium is an acute neuropsychiatric syndrome characterised by disturbance of attention and awareness with fluctuating course, developing acutely over hours to days. It affects up to 30 percent of hospitalised elderly patients and is associated with increased mortality, longer hospital stay, and accelerated cognitive decline. Precipitating factors include infection, metabolic disturbance, medications particularly anticholinergics and opioids, pain, urinary retention, constipation, dehydration, and unfamiliar environment. Management prioritises identifying and treating underlying causes, non-pharmacological reorientation strategies, and early mobilisation. Antipsychotics are used cautiously for severe agitation only.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 94 – Review Version 1

Obesity is characterised by excess adiposity impairing health. BMI provides simple screening: overweight 25 to 29.9, obesity class I 30 to 34.9, class II 35 to 39.9, class III 40 or above. Waist circumference better captures central adiposity and metabolic risk. Management involves comprehensive lifestyle intervention, pharmacotherapy in eligible patients including GLP-1 receptor agonists which can reduce weight by 10 to 20 percent, and bariatric surgery for severe obesity.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 95 – Review Version 1

Early detection through annual screening for urinary albumin-to-creatinine ratio and eGFR is essential. ACE inhibitors and ARBs are first-line for hypertension in diabetic patients with albuminuria due to their independent nephroprotective properties beyond blood pressure lowering. Diabetic nephropathy affects approximately 40 percent of diabetic patients and is the leading cause of end-stage renal disease globally.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 96 – Review Version 1

White coat hypertension affects 15 to 30 percent of referred patients and involves elevated readings in clinical settings but normal levels at home or on ambulatory monitoring. Ambulatory blood pressure monitoring over 24 hours provides a more accurate assessment of true blood pressure burden. Masked hypertension, the converse, carries cardiovascular risk similar to sustained hypertension and should not be overlooked.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 97 – Review Version 1

NSAIDs including ibuprofen, naproxen, and diclofenac inhibit COX-1 and COX-2 enzymes, reducing prostaglandin synthesis. COX-1 inhibition impairs gastric mucosal protection and platelet aggregation, increasing gastrointestinal bleeding risk particularly in elderly patients, those on corticosteroids or anticoagulants, or those with previous peptic ulcer disease. All NSAIDs carry cardiovascular and renal risks, particularly in volume-depleted patients or those with pre-existing kidney disease.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 98 – Review Version 1

In depression, CBT targets negative automatic thoughts and behavioural withdrawal through behavioural activation. In anxiety disorders, it incorporates exposure and response prevention, worry postponement, and attentional training. Its benefits persist beyond treatment completion, making it superior to medication alone for long-term relapse prevention.. Cognitive behavioural therapy is a structured evidence-based psychological treatment helping patients identify connections between thoughts, emotions, and behaviours.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 99 – Review Version 1

The pathophysiology involves impaired insulin signaling in peripheral tissues, particularly skeletal muscle, liver, and adipose tissue, leading to chronic hyperglycemia. Lifestyle modifications including dietary changes, physical activity, and weight management remain the cornerstone of initial management before pharmacological intervention is considered.. Type 2 diabetes mellitus is a chronic metabolic disorder characterized by insulin resistance and progressive beta-cell dysfunction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 100 – Review Version 1

Metformin is the first-line pharmacological agent for type 2 diabetes in most international guidelines. It works primarily by inhibiting hepatic gluconeogenesis through activation of AMP-activated protein kinase (AMPK), reducing fasting plasma glucose without causing hypoglycemia. Common side effects include gastrointestinal disturbances, which can be minimized by titrating the dose gradually and taking it with food.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 101 – Review Version 1

Beyond glycemic control, they demonstrate heart failure hospitalisation reduction and nephroprotective effects. They are now recommended early in patients with established cardiovascular disease or chronic kidney disease regardless of glycemic status.. SGLT-2 inhibitors including empagliflozin, dapagliflozin, and canagliflozin reduce blood glucose by inhibiting renal glucose reabsorption, causing glycosuria.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 102 – Review Version 1

Without intervention, 5 to 10 percent progress to type 2 diabetes each year. Intensive lifestyle intervention with 5 to 7 percent weight loss and 150 minutes weekly of moderate exercise reduces progression by up to 58 percent, making early identification and intervention critical.. Prediabetes is

characterized by fasting glucose 100-125 mg/dL, 2-hour OGTT glucose 140-199 mg/dL, or HbA1c 5.7-6.4 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 103 – Review Version 1

Gestational diabetes mellitus complicates 6 to 9 percent of pregnancies and is associated with adverse outcomes including macrosomia, birth trauma, and neonatal hypoglycemia. Screening is performed between 24 and 28 weeks using an oral glucose tolerance test. Management prioritises dietary modification and blood glucose monitoring, with insulin or metformin added if glycemic targets are not achieved.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 104 – Review Version 1

Time to reperfusion is the critical determinant of outcome: primary percutaneous coronary intervention is the preferred strategy when achievable within 120 minutes of first medical contact. Immediate treatment includes aspirin, a P2Y₁₂ inhibitor, anticoagulation, and infarct-related artery reperfusion.. Acute myocardial infarction results from rupture of an atherosclerotic plaque in a coronary artery leading to thrombus formation and vessel occlusion.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 105 – Review Version 1

Drug-drug interactions occur through pharmacokinetic mechanisms altering absorption, distribution, metabolism, or excretion, or pharmacodynamic mechanisms with additive or antagonistic effects. CYP3A4 enzyme interactions are particularly clinically significant: clarithromycin inhibits CYP3A4 and dramatically raises plasma levels of statins (risking rhabdomyolysis), ciclosporin, and benzodiazepines. Warfarin has hundreds of documented interactions and requires careful monitoring with any medication change.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 106 – Review Version 1

The Glasgow Coma Scale assesses conscious level through three components: eye opening (1-4), verbal response (1-5), and motor response (1-6), giving a total of 3 to 15. GCS 15 indicates full consciousness; GCS 8 or below indicates coma and the need to protect the airway. It is used in acute neurological and trauma assessment, with trends over time being more informative than a single measurement. The AVPU scale (Alert, Voice, Pain, Unresponsive) provides a simpler bedside assessment for initial triage. Causes of reduced conscious level are memorised using the AEIOU TIPS acronym: Alcohol/Acidosis, Epilepsy/Electrolytes, Infection, Opioids/Overdose, Uraemia, Trauma/Temperature, Insulin, Psychiatric, Stroke/Shock.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 107 – Review Version 1

Resistant hypertension is blood pressure above target despite optimal doses of three antihypertensive agents including a diuretic. True resistance must be distinguished from pseudoresistance caused by poor adherence, white coat effect, or suboptimal measurement. Secondary causes should be investigated. Spironolactone (an aldosterone antagonist) as a fourth agent provides the most evidence for treating true resistant hypertension.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 108 – Review Version 1

The coagulation cascade involves two pathways converging on a common final pathway. The extrinsic pathway, triggered by tissue factor exposure after vessel injury, activates factor X through factor VII. The intrinsic pathway, activated by contact with damaged surfaces, activates factor X through factors XII, XI, IX, and VIII. Factor Xa with factor Va forms prothrombinase, converting prothrombin to thrombin, which converts fibrinogen to fibrin and amplifies coagulation through positive feedback. Anticoagulants including warfarin, heparins, and direct oral anticoagulants target different steps in this cascade.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 109 – Review Version 1

The liver is the primary site of drug metabolism, predominantly through cytochrome P450 enzyme-mediated oxidation, reduction, and hydrolysis reactions (Phase I) followed by conjugation reactions making metabolites more water-soluble for excretion (Phase II). First-pass metabolism reduces bioavailability of oral medications before they reach systemic circulation. Liver disease impairs drug metabolism, requiring dose reductions for hepatically metabolised drugs. Enzyme inducers such as rifampicin and carbamazepine increase drug metabolism, reducing efficacy of co-medications. Enzyme inhibitors such as ciprofloxacin and fluoxetine reduce metabolism, risking toxicity of co-medications.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 110 – Review Version 1

Skin integrity assessment is fundamental to pressure ulcer prevention in at-risk patients. The Braden Scale assesses six risk factors: sensory perception, moisture, activity level, mobility, nutrition, and friction and shear. Scores range from 6 to 23, with lower scores indicating higher risk. Preventive interventions for high-risk patients include regular repositioning every 2 hours, pressure-redistributing mattresses and cushions, skin hygiene and moisturisation, optimisation of nutrition and hydration, and protection of heels with foam boots. Pressure ulcers are classified by depth from Stage 1 (non-blanchable erythema of intact skin) to Stage 4 (full thickness tissue loss exposing bone, tendon, or muscle).

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 1 – Review Version 2

Patient-centred care is an approach that respects and responds to individual patient preferences, needs, and values, ensuring that patient values guide all clinical decisions. Its core dimensions include respect for patient values and expressed needs, emotional support and relief of fear and anxiety, information and communication in accessible language, coordination and integration of care across settings, physical comfort and pain management, involvement of family and friends as appropriate, and access to care and continuity. Evidence consistently shows that patient-centred care improves patient satisfaction, treatment adherence, health outcomes, and reduces unnecessary investigations and readmissions.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 2 – Review Version 2

Prediabetes is characterized by fasting glucose 100-125 mg/dL, 2-hour OGTT glucose 140-199 mg/dL, or HbA1c 5.7-6.4 percent. Without intervention, 5 to 10 percent progress to type 2 diabetes each year. Intensive lifestyle intervention with 5 to 7 percent weight loss and 150 minutes weekly of moderate exercise reduces progression by up to 58 percent, making early identification and intervention critical.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 3 – Review Version 2

Hypertension, defined as blood pressure at or above 130/80 mmHg by ACC/AHA guidelines, affects over 1.3 billion adults worldwide and is the leading modifiable risk factor for cardiovascular disease, stroke, and renal failure. Primary hypertension has no identifiable single cause, with genetic predisposition, dietary sodium, obesity, physical inactivity, and chronic stress all contributing. Secondary causes account for 5 to 10 percent of cases.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 4 – Review Version 2

Thiazide diuretics such as chlorthalidone and hydrochlorothiazide are effective antihypertensives that reduce blood volume through natriuresis. They are particularly effective in Black patients and elderly patients with isolated systolic hypertension. Metabolic side effects include hypokalemia, hyperuricemia, and mild glucose and lipid disturbances. Chlorthalidone has longer duration of action and better evidence for cardiovascular event reduction compared with hydrochlorothiazide.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 5 – Review Version 2

SGLT-2 inhibitors including empagliflozin, dapagliflozin, and canagliflozin reduce blood glucose by inhibiting renal glucose reabsorption, causing glycosuria. Beyond glycemic control, they demonstrate heart failure hospitalisation reduction and nephroprotective effects. They are now recommended early in patients with established cardiovascular disease or chronic kidney disease regardless of glycemic status.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications,

contraindications, and multidisciplinary care approaches.

Topic 6 – Review Version 2

Oedema represents excess fluid accumulation in the interstitial compartment and is classified by distribution and cause. Bilateral leg oedema suggests systemic causes including heart failure (elevated venous pressure with sodium retention), nephrotic syndrome (protein loss reducing oncotic pressure), liver cirrhosis (reduced albumin synthesis and portal hypertension), or hypothyroidism causing myxoedema. Unilateral leg oedema suggests local causes including deep vein thrombosis, venous insufficiency, lymphoedema, or infection. Investigation follows the distribution: bilateral oedema warrants cardiac, renal, liver, and thyroid assessment; unilateral oedema warrants duplex ultrasound to exclude DVT.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 7 – Review Version 2

The DASH diet reduces systolic blood pressure by 8 to 14 mmHg in large randomized trials. It emphasises fruits, vegetables, whole grains, low-fat dairy, and reduced saturated fat and sodium. Limiting sodium below 2.3g daily provides additional reduction. Regular aerobic exercise reduces systolic pressure by a further 5 to 8 mmHg. These lifestyle measures can reduce or delay pharmacological treatment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 8 – Review Version 2

Urinary tract infections are classified as uncomplicated (affecting otherwise healthy, non-pregnant women with normal urinary tracts) or complicated (occurring in men, pregnant women, those with structural abnormalities, immunosuppression, or catheter-related). *Escherichia coli* causes approximately 80 percent of community-acquired UTIs. Uncomplicated lower UTI in women is treated with a 3-day course of trimethoprim or nitrofurantoin based on local resistance patterns. Recurrent UTIs (three or more per year) warrant investigation for predisposing factors and may be managed with low-dose prophylactic antibiotics, post-coital prophylaxis, or self-start therapy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 9 – Review Version 2

These agents stimulate insulin secretion in a glucose-dependent manner, suppress glucagon, slow gastric emptying, and promote weight loss. Cardiovascular outcome trials have shown significant reductions in major adverse cardiovascular events in high-risk patients. GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated significant cardiovascular benefits beyond glycemic control.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 10 – Review Version 2

Screening is performed between 24 and 28 weeks using an oral glucose tolerance test. Management prioritises dietary modification and blood glucose monitoring, with insulin or metformin

added if glycemic targets are not achieved.. Gestational diabetes mellitus complicates 6 to 9 percent of pregnancies and is associated with adverse outcomes including macrosomia, birth trauma, and neonatal hypoglycemia.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 11 – Review Version 2

COPD is characterised by progressive largely irreversible airflow limitation, predominantly caused by cigarette smoking or biomass fuel exposure. Spirometric confirmation with FEV1/FVC below 0.70 post-bronchodilator is required for diagnosis. Emphysema involves alveolar wall destruction reducing gas exchange, while chronic bronchitis is productive cough for at least three months in two consecutive years. Smoking cessation is the only intervention that alters disease progression.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 12 – Review Version 2

WHO recommends adults perform 150 to 300 minutes of moderate-intensity aerobic activity or 75 to 150 minutes of vigorous-intensity activity weekly, plus muscle-strengthening activities twice weekly. Regular physical activity reduces cardiovascular disease, type 2 diabetes, depression, several cancers, and all-cause mortality. Even modest increases from a sedentary baseline provide significant health benefits.. Physical inactivity contributes to 3.2 million deaths annually globally.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 13 – Review Version 2

Medication reconciliation is the process of creating the most accurate possible list of all medications a patient is taking, including dose, frequency, and route, and comparing it against the physician's admission, transfer, or discharge orders to identify and resolve discrepancies. It reduces medication errors at care transitions, which are responsible for approximately 60 percent of serious medication errors in hospital settings. Discrepancies found during reconciliation include omitted medications, wrong doses, duplicated medications, and medications continued that should have been stopped. All newly hospitalised patients should have medication reconciliation performed by a pharmacist within 24 hours of admission.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 14 – Review Version 2

Each class has independently demonstrated mortality reduction in randomised trials. Achieving all four drug classes at target doses reduces all-cause mortality by more than 70 percent compared with no treatment.. Heart failure with reduced ejection fraction is managed with quadruple therapy: an ACE inhibitor, ARB, or ARNI (sacubitril/valsartan), a beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT-2 inhibitor.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 15 – Review Version 2

Hypertensive emergencies involve markedly elevated blood pressure above 180/120 mmHg with acute target organ damage including hypertensive encephalopathy, aortic dissection, acute coronary syndrome, or acute kidney injury. They require immediate ICU admission and parenteral antihypertensive therapy with careful organ-specific blood pressure targets. Hypertensive urgency involves severely elevated pressure without organ damage and can be managed with oral agents and close follow-up.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 16 – Review Version 2

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection. The Surviving Sepsis Campaign Hour-1 Bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, starting 30 mL/kg intravenous crystalloid for hypotension or elevated lactate, and vasopressors for refractory hypotension. Each hour of delay in antibiotic administration increases mortality by approximately 7 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 17 – Review Version 2

High-intensity statins including atorvastatin 40 to 80mg and rosuvastatin 20 to 40mg are recommended for patients with established cardiovascular disease, targeting LDL-C below 1.4 mmol/L in very high-risk patients. Beyond cholesterol lowering, statins have plaque-stabilising and anti-inflammatory effects.. Statins inhibit HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis, reducing LDL cholesterol by 30 to 55 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 18 – Review Version 2

It progresses from non-proliferative stages with microaneurysms and retinal hemorrhages to proliferative retinopathy with neovascularization risking vitreous hemorrhage. Annual dilated fundal examination enables early detection. Intravitreal anti-VEGF injections and laser photocoagulation are effective treatments for sight-threatening retinopathy.. Diabetic retinopathy is the leading cause of new blindness in working-age adults.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 19 – Review Version 2

Pulmonary rehabilitation is a supervised multidisciplinary programme for patients with chronic respiratory disease combining exercise training, education, and psychosocial support. It consistently improves exercise capacity, quality of life, and dyspnoea scores without altering FEV1. It reduces acute exacerbations and hospital readmissions. It is recommended for COPD patients with MRC dyspnoea grade 3 or above and following hospitalisation for acute exacerbation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications,

contraindications, and multidisciplinary care approaches.

Topic 20 – Review Version 2

Palliative care focuses on improving quality of life for patients and families facing life-limiting illness through prevention and relief of suffering by early identification and impeccable assessment and treatment of pain and other physical, psychosocial, and spiritual problems. It is applicable from diagnosis of a life-limiting condition alongside curative treatment, not only at end of life. The World Health Organization emphasises that palliative care affirms life and regards dying as a normal process, neither hastening nor postponing death. Advance care planning, including discussion of resuscitation preferences, preferred place of death, and lasting power of attorney for health and welfare, is an important component.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 21 – Review Version 2

Cognitive behavioural therapy is a structured evidence-based psychological treatment helping patients identify connections between thoughts, emotions, and behaviours. In depression, CBT targets negative automatic thoughts and behavioural withdrawal through behavioural activation. In anxiety disorders, it incorporates exposure and response prevention, worry postponement, and attentional training. Its benefits persist beyond treatment completion, making it superior to medication alone for long-term relapse prevention.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 22 – Review Version 2

Statins inhibit HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis, reducing LDL cholesterol by 30 to 55 percent. High-intensity statins including atorvastatin 40 to 80mg and rosuvastatin 20 to 40mg are recommended for patients with established cardiovascular disease, targeting LDL-C below 1.4 mmol/L in very high-risk patients. Beyond cholesterol lowering, statins have plaque-stabilising and anti-inflammatory effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 23 – Review Version 2

Dehydration occurs when fluid losses exceed intake, reducing total body water. Mild dehydration (1-2 percent body weight) causes thirst and reduced concentration; moderate dehydration (3-5 percent) causes headache, dizziness, and reduced urine output; severe dehydration (above 6 percent) causes tachycardia, hypotension, and confusion. In adults, urine colour is a practical guide: pale yellow indicates adequate hydration; dark amber indicates dehydration. Oral rehydration is appropriate for mild to moderate dehydration using water or oral rehydration salts. Intravenous fluid resuscitation is required for severe dehydration or when oral intake is impossible. Elderly patients are at particular risk due to reduced thirst sensation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 24 – Review Version 2

Key principles include prescribing antibiotics only when bacterial infection is confirmed or highly likely, selecting the narrowest spectrum effective agent, using the shortest evidence-based duration, and reviewing at 48 to 72 hours once microbiological results are available. Antimicrobial resistance is a global crisis threatening the foundations of modern medicine.. Antibiotic stewardship promotes appropriate antimicrobial use to improve patient outcomes while limiting resistance selection pressure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 25 – Review Version 2

ACE inhibitors and ARBs are preferred antihypertensives in patients with diabetes, chronic kidney disease, or heart failure with reduced ejection fraction. ACE inhibitors block conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. ARBs block angiotensin II at its receptor. Both reduce proteinuria and slow nephropathy progression. They should not be combined due to increased adverse effects without additional blood pressure benefit.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 26 – Review Version 2

The principles of medical ethics provide a framework for clinical decision-making when values conflict. Autonomy refers to the patient's right to make informed decisions about their care, requiring valid informed consent. Beneficence is the duty to act in the patient's best interest. Non-maleficence requires avoiding unnecessary harm and balancing risks against benefits. Justice involves fair allocation of healthcare resources and equitable treatment of all patients. When these principles conflict, as in a patient refusing beneficial treatment (autonomy versus beneficence), the framework guides structured ethical analysis. Mental capacity to make specific decisions must be assessed before overriding patient choice, using the Mental Capacity Act 2005 framework in England and Wales.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 27 – Review Version 2

Peripheral diabetic neuropathy affects up to 50 percent of patients with long-standing diabetes, presenting as distal symmetric polyneuropathy with pain, tingling, numbness, and loss of protective sensation. Annual foot examinations and patient education on foot care are essential preventive measures. Pharmacological management of neuropathic pain includes duloxetine, pregabalin, and amitriptyline.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 28 – Review Version 2

Polypharmacy, the concurrent use of five or more medications, affects approximately 40 percent of adults over 65 and significantly increases risk of adverse drug reactions, drug-drug interactions, medication non-adherence, falls, cognitive impairment, and preventable hospitalisation. Regular structured medication reviews using the Beers Criteria or STOPP/START criteria identify potentially inappropriate medications and opportunities for safe deprescribing.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 29 – Review Version 2

Deep vein thrombosis and pulmonary embolism constitute venous thromboembolism, a major cause of preventable hospital mortality. DVT typically involves the leg deep veins, with clot propagation risking embolisation to the pulmonary vasculature. Risk factors include immobility, surgery, malignancy, pregnancy, oral contraceptive use, thrombophilia, and obesity. Treatment with anticoagulation prevents clot extension and embolisation while the body's fibrinolytic system dissolves the thrombus.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 30 – Review Version 2

Calcium channel blockers such as amlodipine are effective antihypertensives with particular benefit in elderly patients, Black patients, and isolated systolic hypertension. They reduce vascular smooth muscle tone by blocking voltage-gated calcium channels. Non-dihydropyridine CCBs such as diltiazem and verapamil also reduce heart rate, useful in atrial fibrillation but contraindicated in heart failure with reduced ejection fraction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 31 – Review Version 2

Dyspepsia (upper abdominal discomfort, bloating, early satiety, or nausea) affects up to 40 percent of the population and has many causes. Functional dyspepsia (no organic cause on investigation) is most common, managed with lifestyle advice, PPI or H2 blocker trial, and consideration of *Helicobacter pylori* testing and eradication if positive. Alarm features requiring urgent endoscopy include unintentional weight loss, dysphagia, persistent vomiting, haematemesis or melaena, iron-deficiency anaemia, and epigastric mass. Any new onset dyspepsia over age 55 warrants early endoscopy. *H. pylori* eradication using triple therapy (PPI, amoxicillin, and clarithromycin for 7 days) achieves eradication in approximately 80 percent of cases.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 32 – Review Version 2

Diabetic retinopathy is the leading cause of new blindness in working-age adults. It progresses from non-proliferative stages with microaneurysms and retinal hemorrhages to proliferative retinopathy with neovascularization risking vitreous hemorrhage. Annual dilated fundal examination enables early detection. Intravitreal anti-VEGF injections and laser photocoagulation are effective treatments for sight-threatening retinopathy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 33 – Review Version 2

BMI provides simple screening: overweight 25 to 29.9, obesity class I 30 to 34.9, class II 35 to 39.9, class III 40 or above. Waist circumference better captures central adiposity and metabolic risk. Management involves comprehensive lifestyle intervention, pharmacotherapy in eligible patients including GLP-1 receptor agonists which can reduce weight by 10 to 20 percent, and bariatric surgery for severe obesity.. Obesity is characterised by excess adiposity impairing health.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 34 – Review Version 2

COX-1 inhibition impairs gastric mucosal protection and platelet aggregation, increasing gastrointestinal bleeding risk particularly in elderly patients, those on corticosteroids or anticoagulants, or those with previous peptic ulcer disease. All NSAIDs carry cardiovascular and renal risks, particularly in volume-depleted patients or those with pre-existing kidney disease.. NSAIDs including ibuprofen, naproxen, and diclofenac inhibit COX-1 and COX-2 enzymes, reducing prostaglandin synthesis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 35 – Review Version 2

It works primarily by inhibiting hepatic gluconeogenesis through activation of AMP-activated protein kinase (AMPK), reducing fasting plasma glucose without causing hypoglycemia. Common side effects include gastrointestinal disturbances, which can be minimized by titrating the dose gradually and taking it with food.. Metformin is the first-line pharmacological agent for type 2 diabetes in most international guidelines.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 36 – Review Version 2

The WHO five moments framework identifies critical junctures: before patient contact, before an aseptic procedure, after body fluid exposure, after patient contact, and after contact with patient surroundings. Alcohol-based hand rub is preferred for most clinical situations due to superior speed and convenience, while soap and water is required for *Clostridioides difficile* and norovirus.. Hand hygiene is the single most effective measure to prevent healthcare-associated infections.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 37 – Review Version 2

They reduce vascular smooth muscle tone by blocking voltage-gated calcium channels. Non-dihydropyridine CCBs such as diltiazem and verapamil also reduce heart rate, useful in atrial fibrillation but contraindicated in heart failure with reduced ejection fraction.. Calcium channel blockers such as amlodipine are effective antihypertensives with particular benefit in elderly patients, Black patients, and isolated systolic hypertension.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 38 – Review Version 2

Physical inactivity contributes to 3.2 million deaths annually globally. WHO recommends adults perform 150 to 300 minutes of moderate-intensity aerobic activity or 75 to 150 minutes of vigorous-intensity activity weekly, plus muscle-strengthening activities twice weekly. Regular physical activity reduces cardiovascular disease, type 2 diabetes, depression, several cancers, and all-cause mortality. Even modest increases from a sedentary baseline provide significant health benefits.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 39 – Review Version 2

DVT typically involves the leg deep veins, with clot propagation risking embolisation to the pulmonary vasculature. Risk factors include immobility, surgery, malignancy, pregnancy, oral contraceptive use, thrombophilia, and obesity. Treatment with anticoagulation prevents clot extension and embolisation while the body's fibrinolytic system dissolves the thrombus.. Deep vein thrombosis and pulmonary embolism constitute venous thromboembolism, a major cause of preventable hospital mortality.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 40 – Review Version 2

Atrial fibrillation is the most common sustained cardiac arrhythmia, causing irregular ventricular response from chaotic atrial electrical activity. The primary risks are stroke from atrial thrombus and heart failure from tachycardia-mediated cardiomyopathy. Management involves rate control, rhythm control in appropriate patients, and anticoagulation with direct oral anticoagulants guided by CHA2DS2-VASc scoring.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 41 – Review Version 2

The primary risks are stroke from atrial thrombus and heart failure from tachycardia-mediated cardiomyopathy. Management involves rate control, rhythm control in appropriate patients, and anticoagulation with direct oral anticoagulants guided by CHA2DS2-VASc scoring.. Atrial fibrillation is the most common sustained cardiac arrhythmia, causing irregular ventricular response from chaotic atrial electrical activity.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 42 – Review Version 2

SSRIs are the first-line pharmacological treatment for depression and anxiety disorders due to their balance of efficacy, tolerability, and safety in overdose compared with older tricyclics. Sertraline has the strongest evidence base in depression, while escitalopram and citalopram have good tolerability profiles. Full therapeutic benefit requires 4 to 6 weeks and treatment should continue for at least 6 to 12 months after remission to prevent relapse.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 43 – Review Version 2

Bipolar disorder features recurrent mania or hypomania alternating with depressive episodes. Lithium remains the gold standard mood stabiliser for long-term prevention of both manic and depressive episodes, with robust evidence for suicide risk reduction. Valproate is effective for mania prevention; lamotrigine particularly targets depressive episodes. Close monitoring of lithium levels, renal function, and thyroid function is essential given its narrow therapeutic index.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 44 – Review Version 2

Type 2 diabetes mellitus is a chronic metabolic disorder characterized by insulin resistance and progressive beta-cell dysfunction. The pathophysiology involves impaired insulin signaling in peripheral tissues, particularly skeletal muscle, liver, and adipose tissue, leading to chronic hyperglycemia. Lifestyle modifications including dietary changes, physical activity, and weight management remain the cornerstone of initial management before pharmacological intervention is considered.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 45 – Review Version 2

Annual foot examinations and patient education on foot care are essential preventive measures. Pharmacological management of neuropathic pain includes duloxetine, pregabalin, and amitriptyline. Peripheral diabetic neuropathy affects up to 50 percent of patients with long-standing diabetes, presenting as distal symmetric polyneuropathy with pain, tingling, numbness, and loss of protective sensation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 46 – Review Version 2

The renin-angiotensin-aldosterone system (RAAS) is a key hormonal cascade regulating blood pressure and fluid balance. Renin, secreted by juxtaglomerular cells in response to reduced renal perfusion, converts angiotensinogen to angiotensin I. ACE converts angiotensin I to the active angiotensin II, a potent vasoconstrictor that also stimulates aldosterone release causing sodium and water retention. RAAS blockade with ACE inhibitors or ARBs is central to managing hypertension, heart failure, and diabetic nephropathy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 47 – Review Version 2

Septic arthritis is a medical emergency requiring same-day joint aspiration and intravenous antibiotics. It typically presents with acute monoarthritis with fever, a hot swollen joint, and severely restricted movement. The knee is most commonly affected. *Staphylococcus aureus* is the most

common causative organism in adults. Diagnosis is confirmed by synovial fluid analysis showing turbid fluid with white cell count above 50,000 cells per microlitre with predominantly neutrophils, and Gram stain and culture. Delay in treatment risks rapid cartilage destruction and permanent joint damage.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 48 – Review Version 2

The WHO has identified antibiotic resistance as one of the greatest threats to global health. ESKAPE organisms (Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter species) account for the majority of serious healthcare-associated infections with resistance threatening treatment options.. Antimicrobial resistance is driven by overuse and misuse of antibiotics in human medicine and agriculture, selecting for resistant bacterial strains.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 49 – Review Version 2

They are particularly effective in Black patients and elderly patients with isolated systolic hypertension. Metabolic side effects include hypokalemia, hyperuricemia, and mild glucose and lipid disturbances. Chlorthalidone has longer duration of action and better evidence for cardiovascular event reduction compared with hydrochlorothiazide.. Thiazide diuretics such as chlorthalidone and hydrochlorothiazide are effective antihypertensives that reduce blood volume through natriuresis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 50 – Review Version 2

Inhaled corticosteroids are the cornerstone of controller therapy, reducing airway inflammation and risk of exacerbations. Short-acting beta-2 agonists provide rapid bronchodilation for symptom relief. Management follows a stepwise approach guided by symptom control and exacerbation frequency.. Asthma is a chronic inflammatory airway disease characterised by variable reversible airflow obstruction, bronchial hyperresponsiveness, and airway remodelling.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 51 – Review Version 2

Chronic kidney disease is classified by cause, GFR category, and albuminuria category using the CGA staging system. GFR categories range from G1 (normal, above 90 mL/min) to G5 (kidney failure, below 15 mL/min). Albuminuria categories range from A1 (normal to mildly increased, ACR below 3 mg/mmol) to A3 (severely increased, ACR above 30 mg/mmol). High risk of progression is conferred by lower GFR combined with higher albuminuria. Management aims to slow progression through blood pressure control (target below 130/80 mmHg), RAAS blockade in proteinuric CKD, and SGLT-2 inhibitors now recommended across CKD stages for their proven renoprotective effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 52 – Review Version 2

Community-acquired pneumonia severity is assessed using the CURB-65 score, assigning one point each for confusion, urea above 7 mmol/L, respiratory rate 30 or more, blood pressure below 90/60 mmHg, and age 65 or over. Score 0-1 indicates low mortality risk suitable for home treatment. Score 2 warrants hospital admission. Score 3 or above indicates high mortality risk requiring hospital admission and possible ICU assessment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 53 – Review Version 2

It consistently improves exercise capacity, quality of life, and dyspnoea scores without altering FEV1. It reduces acute exacerbations and hospital readmissions. It is recommended for COPD patients with MRC dyspnoea grade 3 or above and following hospitalisation for acute exacerbation.. Pulmonary rehabilitation is a supervised multidisciplinary programme for patients with chronic respiratory disease combining exercise training, education, and psychosocial support.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 54 – Review Version 2

Diabetic ketoacidosis is a life-threatening acute complication caused by absolute or relative insulin deficiency. The biochemical triad includes hyperglycemia, metabolic acidosis, and ketonemia. Management requires fluid resuscitation, insulin infusion after potassium correction, electrolyte replacement particularly potassium, and identification of the precipitating cause such as infection or omitted insulin.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 55 – Review Version 2

Acute myocardial infarction results from rupture of an atherosclerotic plaque in a coronary artery leading to thrombus formation and vessel occlusion. Time to reperfusion is the critical determinant of outcome: primary percutaneous coronary intervention is the preferred strategy when achievable within 120 minutes of first medical contact. Immediate treatment includes aspirin, a P2Y12 inhibitor, anticoagulation, and infarct-related artery reperfusion.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 56 – Review Version 2

The HbA1c target for most adults with type 2 diabetes is below 7.0 percent, though individualized targets should account for age, comorbidities, hypoglycemia risk, and patient preference. In elderly or frail patients, a less stringent target of 7.5 to 8.0 percent may be appropriate to avoid hypoglycemia and its associated falls and cardiac risks.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 57 – Review Version 2

They require immediate ICU admission and parenteral antihypertensive therapy with careful organ-specific blood pressure targets. Hypertensive urgency involves severely elevated pressure without organ damage and can be managed with oral agents and close follow-up.. Hypertensive emergencies involve markedly elevated blood pressure above 180/120 mmHg with acute target organ damage including hypertensive encephalopathy, aortic dissection, acute coronary syndrome, or acute kidney injury.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 58 – Review Version 2

CYP3A4 enzyme interactions are particularly clinically significant: clarithromycin inhibits CYP3A4 and dramatically raises plasma levels of statins (risking rhabdomyolysis), ciclosporin, and benzodiazepines. Warfarin has hundreds of documented interactions and requires careful monitoring with any medication change.. Drug-drug interactions occur through pharmacokinetic mechanisms altering absorption, distribution, metabolism, or excretion, or pharmacodynamic mechanisms with additive or antagonistic effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 59 – Review Version 2

Vitamin D deficiency is highly prevalent, affecting approximately 1 billion people globally, and results from inadequate sun exposure, dietary insufficiency, or malabsorption. It causes rickets in children and osteomalacia in adults, presenting with bone pain, proximal muscle weakness, and pathological fractures. Severe vitamin D deficiency also impairs immune function and has been associated with increased risk of autoimmune disease, cardiovascular disease, and certain cancers. Supplementation guidelines recommend 400 IU daily for all adults in the UK and higher doses (1000-4000 IU) for those at high risk including elderly care home residents, those with dark skin or little sun exposure, and pregnant women.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 60 – Review Version 2

Spirometric confirmation with FEV1/FVC below 0.70 post-bronchodilator is required for diagnosis. Emphysema involves alveolar wall destruction reducing gas exchange, while chronic bronchitis is productive cough for at least three months in two consecutive years. Smoking cessation is the only intervention that alters disease progression.. COPD is characterised by progressive largely irreversible airflow limitation, predominantly caused by cigarette smoking or biomass fuel exposure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 61 – Review Version 2

Ambulatory blood pressure monitoring over 24 hours provides a more accurate assessment of true blood pressure burden. Masked hypertension, the converse, carries cardiovascular risk similar to sustained hypertension and should not be overlooked.. White coat hypertension affects 15 to 30 percent of referred patients and involves elevated readings in clinical settings but normal levels at home or on ambulatory monitoring.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 62 – Review Version 2

Chronic pain is defined as pain persisting beyond the normal expected healing time, typically more than 3 months. It affects approximately 28 million adults in the UK and has a significant biopsychosocial impact. It is now understood as a complex condition involving peripheral and central sensitisation, psychological factors including fear-avoidance beliefs and catastrophising, and social factors including work and relationship impacts. The modern pain management paradigm has shifted from a largely pharmacological approach to a multidisciplinary biopsychosocial model incorporating physiotherapy, psychological therapies particularly CBT and acceptance and commitment therapy, lifestyle optimisation, and judicious use of analgesics at the lowest effective dose.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 63 – Review Version 2

It emphasises fruits, vegetables, whole grains, low-fat dairy, and reduced saturated fat and sodium. Limiting sodium below 2.3g daily provides additional reduction. Regular aerobic exercise reduces systolic pressure by a further 5 to 8 mmHg. These lifestyle measures can reduce or delay pharmacological treatment.. The DASH diet reduces systolic blood pressure by 8 to 14 mmHg in large randomized trials.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 64 – Review Version 2

Informed consent is a legal and ethical requirement for medical procedures and treatment. For consent to be valid, the patient must have capacity to make the decision, the consent must be given voluntarily without coercion, and the patient must be provided with sufficient information about the proposed treatment including its benefits, risks, alternatives, and consequences of no treatment. Montgomery v Lanarkshire Health Board (2015) established in UK law that patients must be informed of all material risks that a reasonable patient would consider significant. Documentation of the consent process is essential. Consent for research requires additional safeguards including Research Ethics Committee approval.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 65 – Review Version 2

Inflammatory markers including C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR) are non-specific indicators of inflammation used to support diagnosis and monitor treatment response. CRP rises and falls rapidly within hours to days, making it more useful for monitoring acute conditions. ESR rises more slowly and reflects plasma viscosity. Both are elevated in

infection, autoimmune disease, malignancy, and tissue necrosis. Very high CRP above 100 mg/L suggests bacterial infection or severe inflammation. Procalcitonin is more specific for bacterial infection and is used in some settings to guide antibiotic prescribing. Normal inflammatory markers do not exclude serious pathology, particularly in immunocompromised patients.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 66 – Review Version 2

Continuous glucose monitoring systems provide real-time glucose readings every few minutes, enabling patients and clinicians to make more informed treatment decisions. Time-in-range, defined as glucose between 70 and 180 mg/dL, is emerging as a key glycemic metric alongside HbA1c. Higher time-in-range is associated with reduced rates of diabetic microvascular and macrovascular complications.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 67 – Review Version 2

GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated significant cardiovascular benefits beyond glycemic control. These agents stimulate insulin secretion in a glucose-dependent manner, suppress glucagon, slow gastric emptying, and promote weight loss. Cardiovascular outcome trials have shown significant reductions in major adverse cardiovascular events in high-risk patients.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 68 – Review Version 2

Antibiotic stewardship promotes appropriate antimicrobial use to improve patient outcomes while limiting resistance selection pressure. Key principles include prescribing antibiotics only when bacterial infection is confirmed or highly likely, selecting the narrowest spectrum effective agent, using the shortest evidence-based duration, and reviewing at 48 to 72 hours once microbiological results are available. Antimicrobial resistance is a global crisis threatening the foundations of modern medicine.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 69 – Review Version 2

Anaemia is defined as haemoglobin below the lower limit of normal for age and sex: below 130 g/L in adult men and below 120 g/L in adult women. The main causes are classified by mechanism: iron deficiency (most common worldwide, from blood loss, poor intake, or malabsorption), vitamin B12 or folate deficiency (macrocytic anaemia from impaired DNA synthesis), haemolysis (destruction of red blood cells from autoimmune, mechanical, or hereditary causes), and bone marrow failure or infiltration. History, full blood count, reticulocyte count, and iron studies guide initial investigation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 70 – Review Version 2

Primary hypertension has no identifiable single cause, with genetic predisposition, dietary sodium, obesity, physical inactivity, and chronic stress all contributing. Secondary causes account for 5 to 10 percent of cases.. Hypertension, defined as blood pressure at or above 130/80 mmHg by ACC/AHA guidelines, affects over 1.3 billion adults worldwide and is the leading modifiable risk factor for cardiovascular disease, stroke, and renal failure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 71 – Review Version 2

The PREDIMED trial demonstrated significant reductions in major cardiovascular events in high-risk patients following a Mediterranean diet compared with a low-fat diet. It also reduces risk of type 2 diabetes, colorectal and breast cancer, cognitive decline, and all-cause mortality.. The Mediterranean diet is characterised by high consumption of plant foods, olive oil as the primary fat, moderate fish and seafood, and low red and processed meat consumption.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 72 – Review Version 2

Occupational health medicine focuses on the relationship between work and health in both directions: how work can cause or worsen health conditions (occupational disease), and how health conditions affect the ability to work. Common occupational diseases include occupational asthma (from isocyanates, flour dust, or latex), vibration white finger from hand-arm vibration syndrome, noise-induced hearing loss, mesothelioma from asbestos exposure, contact dermatitis, and musculoskeletal disorders from manual handling or repetitive strain. Fitness for work assessments consider functional capacity, job demands, reasonable workplace adjustments, and prognosis. Statutory sick pay and fit notes (formerly sick notes) are important administrative tools in managing work absence.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 73 – Review Version 2

Hand hygiene is the single most effective measure to prevent healthcare-associated infections. The WHO five moments framework identifies critical junctures: before patient contact, before an aseptic procedure, after body fluid exposure, after patient contact, and after contact with patient surroundings. Alcohol-based hand rub is preferred for most clinical situations due to superior speed and convenience, while soap and water is required for *Clostridioides difficile* and norovirus.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 74 – Review Version 2

Diabetic nephropathy affects approximately 40 percent of diabetic patients and is the leading cause of end-stage renal disease globally. Early detection through annual screening for urinary albumin-to-creatinine ratio and eGFR is essential. ACE inhibitors and ARBs are first-line for hypertension in diabetic patients with albuminuria due to their independent nephroprotective properties beyond blood pressure lowering.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 75 – Review Version 2

Asthma is a chronic inflammatory airway disease characterised by variable reversible airflow obstruction, bronchial hyperresponsiveness, and airway remodelling. Inhaled corticosteroids are the cornerstone of controller therapy, reducing airway inflammation and risk of exacerbations. Short-acting beta-2 agonists provide rapid bronchodilation for symptom relief. Management follows a stepwise approach guided by symptom control and exacerbation frequency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 76 – Review Version 2

Time-in-range, defined as glucose between 70 and 180 mg/dL, is emerging as a key glycemic metric alongside HbA1c. Higher time-in-range is associated with reduced rates of diabetic microvascular and macrovascular complications.. Continuous glucose monitoring systems provide real-time glucose readings every few minutes, enabling patients and clinicians to make more informed treatment decisions.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 77 – Review Version 2

ACE inhibitors block conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. ARBs block angiotensin II at its receptor. Both reduce proteinuria and slow nephropathy progression. They should not be combined due to increased adverse effects without additional blood pressure benefit.. ACE inhibitors and ARBs are preferred antihypertensives in patients with diabetes, chronic kidney disease, or heart failure with reduced ejection fraction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 78 – Review Version 2

The Surviving Sepsis Campaign Hour-1 Bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, starting 30 mL/kg intravenous crystalloid for hypotension or elevated lactate, and vasopressors for refractory hypotension. Each hour of delay in antibiotic administration increases mortality by approximately 7 percent.. Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 79 – Review Version 2

Sertraline has the strongest evidence base in depression, while escitalopram and citalopram have good tolerability profiles. Full therapeutic benefit requires 4 to 6 weeks and treatment should continue for at least 6 to 12 months after remission to prevent relapse.. SSRIs are the first-line pharmacological treatment for depression and anxiety disorders due to their balance of efficacy, tolerability, and safety in overdose compared with older tricyclics.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 80 – Review Version 2

Wound healing progresses through four overlapping phases: haemostasis (immediate platelet plug and clot formation), inflammation (neutrophil and macrophage infiltration clearing debris and bacteria), proliferation (fibroblast migration producing collagen, angiogenesis, and re-epithelialisation), and remodelling (collagen reorganisation over months to years). Factors impairing wound healing include diabetes (neuropathy, ischaemia, and impaired immune response), malnutrition (particularly protein, zinc, and vitamin C deficiency), corticosteroids, smoking (causing vasoconstriction and reduced oxygen delivery), infection, and foreign bodies. Assessment should determine the wound bed, edges, surrounding skin, exudate, infection signs, and underlying cause.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 81 – Review Version 2

True resistance must be distinguished from pseudoresistance caused by poor adherence, white coat effect, or suboptimal measurement. Secondary causes should be investigated. Spironolactone (an aldosterone antagonist) as a fourth agent provides the most evidence for treating true resistant hypertension.. Resistant hypertension is blood pressure above target despite optimal doses of three antihypertensive agents including a diuretic.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 82 – Review Version 2

Antimicrobial resistance is driven by overuse and misuse of antibiotics in human medicine and agriculture, selecting for resistant bacterial strains. The WHO has identified antibiotic resistance as one of the greatest threats to global health. ESKAPE organisms (Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter species) account for the majority of serious healthcare-associated infections with resistance threatening treatment options.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 83 – Review Version 2

Pain assessment in clinical practice uses validated tools appropriate to the patient's communication ability. The numerical rating scale (0-10) is the most commonly used in cognitively intact adults. The visual analogue scale uses a 100mm line from no pain to worst imaginable pain. The Wong-Baker FACES scale is appropriate for children and patients with cognitive impairment. Multidimensional assessment tools such as the Brief Pain Inventory assess pain intensity and its impact on function and quality of life. Chronic pain assessment should include physical, psychological, and social dimensions using a biopsychosocial framework, with regular reassessment using the same validated tool for consistency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 84 – Review Version 2

Chronic liver disease progresses through stages from simple steatosis through steatohepatitis, fibrosis, cirrhosis, and finally hepatocellular carcinoma. Common causes include alcohol-related liver disease, non-alcoholic fatty liver disease (increasingly prevalent with obesity epidemic), chronic hepatitis B and C, autoimmune hepatitis, primary biliary cholangitis, and haemochromatosis. Cirrhosis complications include portal hypertension causing varices and ascites, hepatic encephalopathy, spontaneous bacterial peritonitis, hepatorenal syndrome, and coagulopathy. Six-monthly ultrasound and AFP for hepatocellular carcinoma surveillance is recommended in all patients with established cirrhosis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 85 – Review Version 2

The Mediterranean diet is characterised by high consumption of plant foods, olive oil as the primary fat, moderate fish and seafood, and low red and processed meat consumption. The PREDIMED trial demonstrated significant reductions in major cardiovascular events in high-risk patients following a Mediterranean diet compared with a low-fat diet. It also reduces risk of type 2 diabetes, colorectal and breast cancer, cognitive decline, and all-cause mortality.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 86 – Review Version 2

The biochemical triad includes hyperglycemia, metabolic acidosis, and ketonemia. Management requires fluid resuscitation, insulin infusion after potassium correction, electrolyte replacement particularly potassium, and identification of the precipitating cause such as infection or omitted insulin.. Diabetic ketoacidosis is a life-threatening acute complication caused by absolute or relative insulin deficiency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 87 – Review Version 2

Heart failure with reduced ejection fraction is managed with quadruple therapy: an ACE inhibitor, ARB, or ARNI (sacubitril/valsartan), a beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT-2 inhibitor. Each class has independently demonstrated mortality reduction in randomised trials. Achieving all four drug classes at target doses reduces all-cause mortality by more than 70 percent compared with no treatment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 88 – Review Version 2

Depression affects over 280 million people globally and is characterised by persistent low mood, anhedonia, fatigue, cognitive impairment, sleep disturbance, appetite changes, and suicidal ideation in severe cases. The biopsychosocial model recognises neurobiological contributions including dysregulation of serotonin, norepinephrine, and dopamine pathways, alongside psychological factors such as negative cognitive schemas and social factors including adverse life events and isolation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 89 – Review Version 2

Score 0-1 indicates low mortality risk suitable for home treatment. Score 2 warrants hospital admission. Score 3 or above indicates high mortality risk requiring hospital admission and possible ICU assessment.. Community-acquired pneumonia severity is assessed using the CURB-65 score, assigning one point each for confusion, urea above 7 mmol/L, respiratory rate 30 or more, blood pressure below 90/60 mmHg, and age 65 or over.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 90 – Review Version 2

Fibromyalgia is a centralised pain syndrome characterised by widespread musculoskeletal pain, fatigue, sleep disturbance, cognitive difficulties (fibro fog), and often comorbid depression and anxiety. It results from central sensitisation where the nervous system amplifies pain signals. Diagnosis is clinical using the 2010 ACR diagnostic criteria, requiring widespread pain index and symptom severity scores above thresholds. Management is multidisciplinary: aerobic exercise and physiotherapy are the most evidence-based interventions, CBT addresses maladaptive pain behaviours, and low-dose amitriptyline or duloxetine can improve pain and sleep. Opioids are not recommended and may worsen central sensitisation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 91 – Review Version 2

Thyroid hormones T3 (triiodothyronine) and T4 (thyroxine) regulate metabolic rate, growth, and development. Hypothyroidism causes fatigue, weight gain, cold intolerance, constipation, dry skin, and bradycardia, and is most commonly caused by Hashimoto's thyroiditis (autoimmune destruction). Hyperthyroidism causes weight loss despite increased appetite, heat intolerance, palpitations, tremor, and anxiety, most commonly from Graves' disease (TSH receptor stimulating antibodies). TSH is the most sensitive screening test for thyroid dysfunction, rising in hypothyroidism and falling in hyperthyroidism.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 92 – Review Version 2

Lithium remains the gold standard mood stabiliser for long-term prevention of both manic and depressive episodes, with robust evidence for suicide risk reduction. Valproate is effective for mania prevention; lamotrigine particularly targets depressive episodes. Close monitoring of lithium levels, renal function, and thyroid function is essential given its narrow therapeutic index.. Bipolar disorder features recurrent mania or hypomania alternating with depressive episodes.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 93 – Review Version 2

Delirium is an acute neuropsychiatric syndrome characterised by disturbance of attention and awareness with fluctuating course, developing acutely over hours to days. It affects up to 30 percent of hospitalised elderly patients and is associated with increased mortality, longer hospital stay, and accelerated cognitive decline. Precipitating factors include infection, metabolic disturbance, medications particularly anticholinergics and opioids, pain, urinary retention, constipation, dehydration, and unfamiliar environment. Management prioritises identifying and treating underlying causes, non-pharmacological reorientation strategies, and early mobilisation. Antipsychotics are used cautiously for severe agitation only.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 94 – Review Version 2

Obesity is characterised by excess adiposity impairing health. BMI provides simple screening: overweight 25 to 29.9, obesity class I 30 to 34.9, class II 35 to 39.9, class III 40 or above. Waist circumference better captures central adiposity and metabolic risk. Management involves comprehensive lifestyle intervention, pharmacotherapy in eligible patients including GLP-1 receptor agonists which can reduce weight by 10 to 20 percent, and bariatric surgery for severe obesity.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 95 – Review Version 2

Early detection through annual screening for urinary albumin-to-creatinine ratio and eGFR is essential. ACE inhibitors and ARBs are first-line for hypertension in diabetic patients with albuminuria due to their independent nephroprotective properties beyond blood pressure lowering. Diabetic nephropathy affects approximately 40 percent of diabetic patients and is the leading cause of end-stage renal disease globally.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 96 – Review Version 2

White coat hypertension affects 15 to 30 percent of referred patients and involves elevated readings in clinical settings but normal levels at home or on ambulatory monitoring. Ambulatory blood pressure monitoring over 24 hours provides a more accurate assessment of true blood pressure burden. Masked hypertension, the converse, carries cardiovascular risk similar to sustained hypertension and should not be overlooked.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 97 – Review Version 2

NSAIDs including ibuprofen, naproxen, and diclofenac inhibit COX-1 and COX-2 enzymes, reducing prostaglandin synthesis. COX-1 inhibition impairs gastric mucosal protection and platelet aggregation, increasing gastrointestinal bleeding risk particularly in elderly patients, those on corticosteroids or anticoagulants, or those with previous peptic ulcer disease. All NSAIDs carry cardiovascular and renal risks, particularly in volume-depleted patients or those with pre-existing kidney disease.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 98 – Review Version 2

In depression, CBT targets negative automatic thoughts and behavioural withdrawal through behavioural activation. In anxiety disorders, it incorporates exposure and response prevention, worry postponement, and attentional training. Its benefits persist beyond treatment completion, making it superior to medication alone for long-term relapse prevention.. Cognitive behavioural therapy is a structured evidence-based psychological treatment helping patients identify connections between thoughts, emotions, and behaviours.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 99 – Review Version 2

The pathophysiology involves impaired insulin signaling in peripheral tissues, particularly skeletal muscle, liver, and adipose tissue, leading to chronic hyperglycemia. Lifestyle modifications including dietary changes, physical activity, and weight management remain the cornerstone of initial management before pharmacological intervention is considered.. Type 2 diabetes mellitus is a chronic metabolic disorder characterized by insulin resistance and progressive beta-cell dysfunction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 100 – Review Version 2

Metformin is the first-line pharmacological agent for type 2 diabetes in most international guidelines. It works primarily by inhibiting hepatic gluconeogenesis through activation of AMP-activated protein kinase (AMPK), reducing fasting plasma glucose without causing hypoglycemia. Common side effects include gastrointestinal disturbances, which can be minimized by titrating the dose gradually and taking it with food.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 101 – Review Version 2

Beyond glycemic control, they demonstrate heart failure hospitalisation reduction and nephroprotective effects. They are now recommended early in patients with established cardiovascular disease or chronic kidney disease regardless of glycemic status.. SGLT-2 inhibitors including empagliflozin, dapagliflozin, and canagliflozin reduce blood glucose by inhibiting renal glucose reabsorption, causing glycosuria.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 102 – Review Version 2

Without intervention, 5 to 10 percent progress to type 2 diabetes each year. Intensive lifestyle intervention with 5 to 7 percent weight loss and 150 minutes weekly of moderate exercise reduces progression by up to 58 percent, making early identification and intervention critical.. Prediabetes is

characterized by fasting glucose 100-125 mg/dL, 2-hour OGTT glucose 140-199 mg/dL, or HbA1c 5.7-6.4 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 103 – Review Version 2

Gestational diabetes mellitus complicates 6 to 9 percent of pregnancies and is associated with adverse outcomes including macrosomia, birth trauma, and neonatal hypoglycemia. Screening is performed between 24 and 28 weeks using an oral glucose tolerance test. Management prioritises dietary modification and blood glucose monitoring, with insulin or metformin added if glycemic targets are not achieved.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 104 – Review Version 2

Time to reperfusion is the critical determinant of outcome: primary percutaneous coronary intervention is the preferred strategy when achievable within 120 minutes of first medical contact. Immediate treatment includes aspirin, a P2Y₁₂ inhibitor, anticoagulation, and infarct-related artery reperfusion.. Acute myocardial infarction results from rupture of an atherosclerotic plaque in a coronary artery leading to thrombus formation and vessel occlusion.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 105 – Review Version 2

Drug-drug interactions occur through pharmacokinetic mechanisms altering absorption, distribution, metabolism, or excretion, or pharmacodynamic mechanisms with additive or antagonistic effects. CYP3A4 enzyme interactions are particularly clinically significant: clarithromycin inhibits CYP3A4 and dramatically raises plasma levels of statins (risking rhabdomyolysis), ciclosporin, and benzodiazepines. Warfarin has hundreds of documented interactions and requires careful monitoring with any medication change.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 106 – Review Version 2

The Glasgow Coma Scale assesses conscious level through three components: eye opening (1-4), verbal response (1-5), and motor response (1-6), giving a total of 3 to 15. GCS 15 indicates full consciousness; GCS 8 or below indicates coma and the need to protect the airway. It is used in acute neurological and trauma assessment, with trends over time being more informative than a single measurement. The AVPU scale (Alert, Voice, Pain, Unresponsive) provides a simpler bedside assessment for initial triage. Causes of reduced conscious level are memorised using the AEIOU TIPS acronym: Alcohol/Acidosis, Epilepsy/Electrolytes, Infection, Opioids/Overdose, Uraemia, Trauma/Temperature, Insulin, Psychiatric, Stroke/Shock.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 107 – Review Version 2

Resistant hypertension is blood pressure above target despite optimal doses of three antihypertensive agents including a diuretic. True resistance must be distinguished from pseudoresistance caused by poor adherence, white coat effect, or suboptimal measurement. Secondary causes should be investigated. Spironolactone (an aldosterone antagonist) as a fourth agent provides the most evidence for treating true resistant hypertension.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 108 – Review Version 2

The coagulation cascade involves two pathways converging on a common final pathway. The extrinsic pathway, triggered by tissue factor exposure after vessel injury, activates factor X through factor VII. The intrinsic pathway, activated by contact with damaged surfaces, activates factor X through factors XII, XI, IX, and VIII. Factor Xa with factor Va forms prothrombinase, converting prothrombin to thrombin, which converts fibrinogen to fibrin and amplifies coagulation through positive feedback. Anticoagulants including warfarin, heparins, and direct oral anticoagulants target different steps in this cascade.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 109 – Review Version 2

The liver is the primary site of drug metabolism, predominantly through cytochrome P450 enzyme-mediated oxidation, reduction, and hydrolysis reactions (Phase I) followed by conjugation reactions making metabolites more water-soluble for excretion (Phase II). First-pass metabolism reduces bioavailability of oral medications before they reach systemic circulation. Liver disease impairs drug metabolism, requiring dose reductions for hepatically metabolised drugs. Enzyme inducers such as rifampicin and carbamazepine increase drug metabolism, reducing efficacy of co-medications. Enzyme inhibitors such as ciprofloxacin and fluoxetine reduce metabolism, risking toxicity of co-medications.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 110 – Review Version 2

Skin integrity assessment is fundamental to pressure ulcer prevention in at-risk patients. The Braden Scale assesses six risk factors: sensory perception, moisture, activity level, mobility, nutrition, and friction and shear. Scores range from 6 to 23, with lower scores indicating higher risk. Preventive interventions for high-risk patients include regular repositioning every 2 hours, pressure-redistributing mattresses and cushions, skin hygiene and moisturisation, optimisation of nutrition and hydration, and protection of heels with foam boots. Pressure ulcers are classified by depth from Stage 1 (non-blanchable erythema of intact skin) to Stage 4 (full thickness tissue loss exposing bone, tendon, or muscle).

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 1 – Review Version 3

Patient-centred care is an approach that respects and responds to individual patient preferences, needs, and values, ensuring that patient values guide all clinical decisions. Its core dimensions include respect for patient values and expressed needs, emotional support and relief of fear and anxiety, information and communication in accessible language, coordination and integration of care across settings, physical comfort and pain management, involvement of family and friends as appropriate, and access to care and continuity. Evidence consistently shows that patient-centred care improves patient satisfaction, treatment adherence, health outcomes, and reduces unnecessary investigations and readmissions.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 2 – Review Version 3

Prediabetes is characterized by fasting glucose 100-125 mg/dL, 2-hour OGTT glucose 140-199 mg/dL, or HbA1c 5.7-6.4 percent. Without intervention, 5 to 10 percent progress to type 2 diabetes each year. Intensive lifestyle intervention with 5 to 7 percent weight loss and 150 minutes weekly of moderate exercise reduces progression by up to 58 percent, making early identification and intervention critical.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 3 – Review Version 3

Hypertension, defined as blood pressure at or above 130/80 mmHg by ACC/AHA guidelines, affects over 1.3 billion adults worldwide and is the leading modifiable risk factor for cardiovascular disease, stroke, and renal failure. Primary hypertension has no identifiable single cause, with genetic predisposition, dietary sodium, obesity, physical inactivity, and chronic stress all contributing. Secondary causes account for 5 to 10 percent of cases.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 4 – Review Version 3

Thiazide diuretics such as chlorthalidone and hydrochlorothiazide are effective antihypertensives that reduce blood volume through natriuresis. They are particularly effective in Black patients and elderly patients with isolated systolic hypertension. Metabolic side effects include hypokalemia, hyperuricemia, and mild glucose and lipid disturbances. Chlorthalidone has longer duration of action and better evidence for cardiovascular event reduction compared with hydrochlorothiazide.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 5 – Review Version 3

SGLT-2 inhibitors including empagliflozin, dapagliflozin, and canagliflozin reduce blood glucose by inhibiting renal glucose reabsorption, causing glycosuria. Beyond glycemic control, they demonstrate heart failure hospitalisation reduction and nephroprotective effects. They are now recommended early in patients with established cardiovascular disease or chronic kidney disease regardless of glycemic status.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications,

contraindications, and multidisciplinary care approaches.

Topic 6 – Review Version 3

Oedema represents excess fluid accumulation in the interstitial compartment and is classified by distribution and cause. Bilateral leg oedema suggests systemic causes including heart failure (elevated venous pressure with sodium retention), nephrotic syndrome (protein loss reducing oncotic pressure), liver cirrhosis (reduced albumin synthesis and portal hypertension), or hypothyroidism causing myxoedema. Unilateral leg oedema suggests local causes including deep vein thrombosis, venous insufficiency, lymphoedema, or infection. Investigation follows the distribution: bilateral oedema warrants cardiac, renal, liver, and thyroid assessment; unilateral oedema warrants duplex ultrasound to exclude DVT.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 7 – Review Version 3

The DASH diet reduces systolic blood pressure by 8 to 14 mmHg in large randomized trials. It emphasises fruits, vegetables, whole grains, low-fat dairy, and reduced saturated fat and sodium. Limiting sodium below 2.3g daily provides additional reduction. Regular aerobic exercise reduces systolic pressure by a further 5 to 8 mmHg. These lifestyle measures can reduce or delay pharmacological treatment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 8 – Review Version 3

Urinary tract infections are classified as uncomplicated (affecting otherwise healthy, non-pregnant women with normal urinary tracts) or complicated (occurring in men, pregnant women, those with structural abnormalities, immunosuppression, or catheter-related). *Escherichia coli* causes approximately 80 percent of community-acquired UTIs. Uncomplicated lower UTI in women is treated with a 3-day course of trimethoprim or nitrofurantoin based on local resistance patterns. Recurrent UTIs (three or more per year) warrant investigation for predisposing factors and may be managed with low-dose prophylactic antibiotics, post-coital prophylaxis, or self-start therapy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 9 – Review Version 3

These agents stimulate insulin secretion in a glucose-dependent manner, suppress glucagon, slow gastric emptying, and promote weight loss. Cardiovascular outcome trials have shown significant reductions in major adverse cardiovascular events in high-risk patients. GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated significant cardiovascular benefits beyond glycemic control.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 10 – Review Version 3

Screening is performed between 24 and 28 weeks using an oral glucose tolerance test. Management prioritises dietary modification and blood glucose monitoring, with insulin or metformin

added if glycemic targets are not achieved.. Gestational diabetes mellitus complicates 6 to 9 percent of pregnancies and is associated with adverse outcomes including macrosomia, birth trauma, and neonatal hypoglycemia.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 11 – Review Version 3

COPD is characterised by progressive largely irreversible airflow limitation, predominantly caused by cigarette smoking or biomass fuel exposure. Spirometric confirmation with FEV1/FVC below 0.70 post-bronchodilator is required for diagnosis. Emphysema involves alveolar wall destruction reducing gas exchange, while chronic bronchitis is productive cough for at least three months in two consecutive years. Smoking cessation is the only intervention that alters disease progression.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 12 – Review Version 3

WHO recommends adults perform 150 to 300 minutes of moderate-intensity aerobic activity or 75 to 150 minutes of vigorous-intensity activity weekly, plus muscle-strengthening activities twice weekly. Regular physical activity reduces cardiovascular disease, type 2 diabetes, depression, several cancers, and all-cause mortality. Even modest increases from a sedentary baseline provide significant health benefits.. Physical inactivity contributes to 3.2 million deaths annually globally.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 13 – Review Version 3

Medication reconciliation is the process of creating the most accurate possible list of all medications a patient is taking, including dose, frequency, and route, and comparing it against the physician's admission, transfer, or discharge orders to identify and resolve discrepancies. It reduces medication errors at care transitions, which are responsible for approximately 60 percent of serious medication errors in hospital settings. Discrepancies found during reconciliation include omitted medications, wrong doses, duplicated medications, and medications continued that should have been stopped. All newly hospitalised patients should have medication reconciliation performed by a pharmacist within 24 hours of admission.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 14 – Review Version 3

Each class has independently demonstrated mortality reduction in randomised trials. Achieving all four drug classes at target doses reduces all-cause mortality by more than 70 percent compared with no treatment.. Heart failure with reduced ejection fraction is managed with quadruple therapy: an ACE inhibitor, ARB, or ARNI (sacubitril/valsartan), a beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT-2 inhibitor.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 15 – Review Version 3

Hypertensive emergencies involve markedly elevated blood pressure above 180/120 mmHg with acute target organ damage including hypertensive encephalopathy, aortic dissection, acute coronary syndrome, or acute kidney injury. They require immediate ICU admission and parenteral antihypertensive therapy with careful organ-specific blood pressure targets. Hypertensive urgency involves severely elevated pressure without organ damage and can be managed with oral agents and close follow-up.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 16 – Review Version 3

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection. The Surviving Sepsis Campaign Hour-1 Bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, starting 30 mL/kg intravenous crystalloid for hypotension or elevated lactate, and vasopressors for refractory hypotension. Each hour of delay in antibiotic administration increases mortality by approximately 7 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 17 – Review Version 3

High-intensity statins including atorvastatin 40 to 80mg and rosuvastatin 20 to 40mg are recommended for patients with established cardiovascular disease, targeting LDL-C below 1.4 mmol/L in very high-risk patients. Beyond cholesterol lowering, statins have plaque-stabilising and anti-inflammatory effects.. Statins inhibit HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis, reducing LDL cholesterol by 30 to 55 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 18 – Review Version 3

It progresses from non-proliferative stages with microaneurysms and retinal hemorrhages to proliferative retinopathy with neovascularization risking vitreous hemorrhage. Annual dilated fundal examination enables early detection. Intravitreal anti-VEGF injections and laser photocoagulation are effective treatments for sight-threatening retinopathy.. Diabetic retinopathy is the leading cause of new blindness in working-age adults.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 19 – Review Version 3

Pulmonary rehabilitation is a supervised multidisciplinary programme for patients with chronic respiratory disease combining exercise training, education, and psychosocial support. It consistently improves exercise capacity, quality of life, and dyspnoea scores without altering FEV1. It reduces acute exacerbations and hospital readmissions. It is recommended for COPD patients with MRC dyspnoea grade 3 or above and following hospitalisation for acute exacerbation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications,

contraindications, and multidisciplinary care approaches.

Topic 20 – Review Version 3

Palliative care focuses on improving quality of life for patients and families facing life-limiting illness through prevention and relief of suffering by early identification and impeccable assessment and treatment of pain and other physical, psychosocial, and spiritual problems. It is applicable from diagnosis of a life-limiting condition alongside curative treatment, not only at end of life. The World Health Organization emphasises that palliative care affirms life and regards dying as a normal process, neither hastening nor postponing death. Advance care planning, including discussion of resuscitation preferences, preferred place of death, and lasting power of attorney for health and welfare, is an important component.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 21 – Review Version 3

Cognitive behavioural therapy is a structured evidence-based psychological treatment helping patients identify connections between thoughts, emotions, and behaviours. In depression, CBT targets negative automatic thoughts and behavioural withdrawal through behavioural activation. In anxiety disorders, it incorporates exposure and response prevention, worry postponement, and attentional training. Its benefits persist beyond treatment completion, making it superior to medication alone for long-term relapse prevention.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 22 – Review Version 3

Statins inhibit HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis, reducing LDL cholesterol by 30 to 55 percent. High-intensity statins including atorvastatin 40 to 80mg and rosuvastatin 20 to 40mg are recommended for patients with established cardiovascular disease, targeting LDL-C below 1.4 mmol/L in very high-risk patients. Beyond cholesterol lowering, statins have plaque-stabilising and anti-inflammatory effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 23 – Review Version 3

Dehydration occurs when fluid losses exceed intake, reducing total body water. Mild dehydration (1-2 percent body weight) causes thirst and reduced concentration; moderate dehydration (3-5 percent) causes headache, dizziness, and reduced urine output; severe dehydration (above 6 percent) causes tachycardia, hypotension, and confusion. In adults, urine colour is a practical guide: pale yellow indicates adequate hydration; dark amber indicates dehydration. Oral rehydration is appropriate for mild to moderate dehydration using water or oral rehydration salts. Intravenous fluid resuscitation is required for severe dehydration or when oral intake is impossible. Elderly patients are at particular risk due to reduced thirst sensation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 24 – Review Version 3

Key principles include prescribing antibiotics only when bacterial infection is confirmed or highly likely, selecting the narrowest spectrum effective agent, using the shortest evidence-based duration, and reviewing at 48 to 72 hours once microbiological results are available. Antimicrobial resistance is a global crisis threatening the foundations of modern medicine.. Antibiotic stewardship promotes appropriate antimicrobial use to improve patient outcomes while limiting resistance selection pressure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 25 – Review Version 3

ACE inhibitors and ARBs are preferred antihypertensives in patients with diabetes, chronic kidney disease, or heart failure with reduced ejection fraction. ACE inhibitors block conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. ARBs block angiotensin II at its receptor. Both reduce proteinuria and slow nephropathy progression. They should not be combined due to increased adverse effects without additional blood pressure benefit.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 26 – Review Version 3

The principles of medical ethics provide a framework for clinical decision-making when values conflict. Autonomy refers to the patient's right to make informed decisions about their care, requiring valid informed consent. Beneficence is the duty to act in the patient's best interest. Non-maleficence requires avoiding unnecessary harm and balancing risks against benefits. Justice involves fair allocation of healthcare resources and equitable treatment of all patients. When these principles conflict, as in a patient refusing beneficial treatment (autonomy versus beneficence), the framework guides structured ethical analysis. Mental capacity to make specific decisions must be assessed before overriding patient choice, using the Mental Capacity Act 2005 framework in England and Wales.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 27 – Review Version 3

Peripheral diabetic neuropathy affects up to 50 percent of patients with long-standing diabetes, presenting as distal symmetric polyneuropathy with pain, tingling, numbness, and loss of protective sensation. Annual foot examinations and patient education on foot care are essential preventive measures. Pharmacological management of neuropathic pain includes duloxetine, pregabalin, and amitriptyline.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 28 – Review Version 3

Polypharmacy, the concurrent use of five or more medications, affects approximately 40 percent of adults over 65 and significantly increases risk of adverse drug reactions, drug-drug interactions, medication non-adherence, falls, cognitive impairment, and preventable hospitalisation. Regular structured medication reviews using the Beers Criteria or STOPP/START criteria identify potentially inappropriate medications and opportunities for safe deprescribing.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 29 – Review Version 3

Deep vein thrombosis and pulmonary embolism constitute venous thromboembolism, a major cause of preventable hospital mortality. DVT typically involves the leg deep veins, with clot propagation risking embolisation to the pulmonary vasculature. Risk factors include immobility, surgery, malignancy, pregnancy, oral contraceptive use, thrombophilia, and obesity. Treatment with anticoagulation prevents clot extension and embolisation while the body's fibrinolytic system dissolves the thrombus.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 30 – Review Version 3

Calcium channel blockers such as amlodipine are effective antihypertensives with particular benefit in elderly patients, Black patients, and isolated systolic hypertension. They reduce vascular smooth muscle tone by blocking voltage-gated calcium channels. Non-dihydropyridine CCBs such as diltiazem and verapamil also reduce heart rate, useful in atrial fibrillation but contraindicated in heart failure with reduced ejection fraction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 31 – Review Version 3

Dyspepsia (upper abdominal discomfort, bloating, early satiety, or nausea) affects up to 40 percent of the population and has many causes. Functional dyspepsia (no organic cause on investigation) is most common, managed with lifestyle advice, PPI or H2 blocker trial, and consideration of *Helicobacter pylori* testing and eradication if positive. Alarm features requiring urgent endoscopy include unintentional weight loss, dysphagia, persistent vomiting, haematemesis or melaena, iron-deficiency anaemia, and epigastric mass. Any new onset dyspepsia over age 55 warrants early endoscopy. *H. pylori* eradication using triple therapy (PPI, amoxicillin, and clarithromycin for 7 days) achieves eradication in approximately 80 percent of cases.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 32 – Review Version 3

Diabetic retinopathy is the leading cause of new blindness in working-age adults. It progresses from non-proliferative stages with microaneurysms and retinal hemorrhages to proliferative retinopathy with neovascularization risking vitreous hemorrhage. Annual dilated fundal examination enables early detection. Intravitreal anti-VEGF injections and laser photocoagulation are effective treatments for sight-threatening retinopathy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 33 – Review Version 3

BMI provides simple screening: overweight 25 to 29.9, obesity class I 30 to 34.9, class II 35 to 39.9, class III 40 or above. Waist circumference better captures central adiposity and metabolic risk. Management involves comprehensive lifestyle intervention, pharmacotherapy in eligible patients including GLP-1 receptor agonists which can reduce weight by 10 to 20 percent, and bariatric surgery for severe obesity.. Obesity is characterised by excess adiposity impairing health.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 34 – Review Version 3

COX-1 inhibition impairs gastric mucosal protection and platelet aggregation, increasing gastrointestinal bleeding risk particularly in elderly patients, those on corticosteroids or anticoagulants, or those with previous peptic ulcer disease. All NSAIDs carry cardiovascular and renal risks, particularly in volume-depleted patients or those with pre-existing kidney disease.. NSAIDs including ibuprofen, naproxen, and diclofenac inhibit COX-1 and COX-2 enzymes, reducing prostaglandin synthesis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 35 – Review Version 3

It works primarily by inhibiting hepatic gluconeogenesis through activation of AMP-activated protein kinase (AMPK), reducing fasting plasma glucose without causing hypoglycemia. Common side effects include gastrointestinal disturbances, which can be minimized by titrating the dose gradually and taking it with food.. Metformin is the first-line pharmacological agent for type 2 diabetes in most international guidelines.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 36 – Review Version 3

The WHO five moments framework identifies critical junctures: before patient contact, before an aseptic procedure, after body fluid exposure, after patient contact, and after contact with patient surroundings. Alcohol-based hand rub is preferred for most clinical situations due to superior speed and convenience, while soap and water is required for *Clostridioides difficile* and norovirus.. Hand hygiene is the single most effective measure to prevent healthcare-associated infections.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 37 – Review Version 3

They reduce vascular smooth muscle tone by blocking voltage-gated calcium channels. Non-dihydropyridine CCBs such as diltiazem and verapamil also reduce heart rate, useful in atrial fibrillation but contraindicated in heart failure with reduced ejection fraction.. Calcium channel blockers such as amlodipine are effective antihypertensives with particular benefit in elderly patients, Black patients, and isolated systolic hypertension.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 38 – Review Version 3

Physical inactivity contributes to 3.2 million deaths annually globally. WHO recommends adults perform 150 to 300 minutes of moderate-intensity aerobic activity or 75 to 150 minutes of vigorous-intensity activity weekly, plus muscle-strengthening activities twice weekly. Regular physical activity reduces cardiovascular disease, type 2 diabetes, depression, several cancers, and all-cause mortality. Even modest increases from a sedentary baseline provide significant health benefits.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 39 – Review Version 3

DVT typically involves the leg deep veins, with clot propagation risking embolisation to the pulmonary vasculature. Risk factors include immobility, surgery, malignancy, pregnancy, oral contraceptive use, thrombophilia, and obesity. Treatment with anticoagulation prevents clot extension and embolisation while the body's fibrinolytic system dissolves the thrombus.. Deep vein thrombosis and pulmonary embolism constitute venous thromboembolism, a major cause of preventable hospital mortality.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 40 – Review Version 3

Atrial fibrillation is the most common sustained cardiac arrhythmia, causing irregular ventricular response from chaotic atrial electrical activity. The primary risks are stroke from atrial thrombus and heart failure from tachycardia-mediated cardiomyopathy. Management involves rate control, rhythm control in appropriate patients, and anticoagulation with direct oral anticoagulants guided by CHA2DS2-VASc scoring.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 41 – Review Version 3

The primary risks are stroke from atrial thrombus and heart failure from tachycardia-mediated cardiomyopathy. Management involves rate control, rhythm control in appropriate patients, and anticoagulation with direct oral anticoagulants guided by CHA2DS2-VASc scoring.. Atrial fibrillation is the most common sustained cardiac arrhythmia, causing irregular ventricular response from chaotic atrial electrical activity.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 42 – Review Version 3

SSRIs are the first-line pharmacological treatment for depression and anxiety disorders due to their balance of efficacy, tolerability, and safety in overdose compared with older tricyclics. Sertraline has the strongest evidence base in depression, while escitalopram and citalopram have good tolerability profiles. Full therapeutic benefit requires 4 to 6 weeks and treatment should continue for at least 6 to 12 months after remission to prevent relapse.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 43 – Review Version 3

Bipolar disorder features recurrent mania or hypomania alternating with depressive episodes. Lithium remains the gold standard mood stabiliser for long-term prevention of both manic and depressive episodes, with robust evidence for suicide risk reduction. Valproate is effective for mania prevention; lamotrigine particularly targets depressive episodes. Close monitoring of lithium levels, renal function, and thyroid function is essential given its narrow therapeutic index.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 44 – Review Version 3

Type 2 diabetes mellitus is a chronic metabolic disorder characterized by insulin resistance and progressive beta-cell dysfunction. The pathophysiology involves impaired insulin signaling in peripheral tissues, particularly skeletal muscle, liver, and adipose tissue, leading to chronic hyperglycemia. Lifestyle modifications including dietary changes, physical activity, and weight management remain the cornerstone of initial management before pharmacological intervention is considered.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 45 – Review Version 3

Annual foot examinations and patient education on foot care are essential preventive measures. Pharmacological management of neuropathic pain includes duloxetine, pregabalin, and amitriptyline. Peripheral diabetic neuropathy affects up to 50 percent of patients with long-standing diabetes, presenting as distal symmetric polyneuropathy with pain, tingling, numbness, and loss of protective sensation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 46 – Review Version 3

The renin-angiotensin-aldosterone system (RAAS) is a key hormonal cascade regulating blood pressure and fluid balance. Renin, secreted by juxtaglomerular cells in response to reduced renal perfusion, converts angiotensinogen to angiotensin I. ACE converts angiotensin I to the active angiotensin II, a potent vasoconstrictor that also stimulates aldosterone release causing sodium and water retention. RAAS blockade with ACE inhibitors or ARBs is central to managing hypertension, heart failure, and diabetic nephropathy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 47 – Review Version 3

Septic arthritis is a medical emergency requiring same-day joint aspiration and intravenous antibiotics. It typically presents with acute monoarthritis with fever, a hot swollen joint, and severely restricted movement. The knee is most commonly affected. *Staphylococcus aureus* is the most

common causative organism in adults. Diagnosis is confirmed by synovial fluid analysis showing turbid fluid with white cell count above 50,000 cells per microlitre with predominantly neutrophils, and Gram stain and culture. Delay in treatment risks rapid cartilage destruction and permanent joint damage.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 48 – Review Version 3

The WHO has identified antibiotic resistance as one of the greatest threats to global health. ESKAPE organisms (Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter species) account for the majority of serious healthcare-associated infections with resistance threatening treatment options.. Antimicrobial resistance is driven by overuse and misuse of antibiotics in human medicine and agriculture, selecting for resistant bacterial strains.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 49 – Review Version 3

They are particularly effective in Black patients and elderly patients with isolated systolic hypertension. Metabolic side effects include hypokalemia, hyperuricemia, and mild glucose and lipid disturbances. Chlorthalidone has longer duration of action and better evidence for cardiovascular event reduction compared with hydrochlorothiazide.. Thiazide diuretics such as chlorthalidone and hydrochlorothiazide are effective antihypertensives that reduce blood volume through natriuresis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 50 – Review Version 3

Inhaled corticosteroids are the cornerstone of controller therapy, reducing airway inflammation and risk of exacerbations. Short-acting beta-2 agonists provide rapid bronchodilation for symptom relief. Management follows a stepwise approach guided by symptom control and exacerbation frequency.. Asthma is a chronic inflammatory airway disease characterised by variable reversible airflow obstruction, bronchial hyperresponsiveness, and airway remodelling.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 51 – Review Version 3

Chronic kidney disease is classified by cause, GFR category, and albuminuria category using the CGA staging system. GFR categories range from G1 (normal, above 90 mL/min) to G5 (kidney failure, below 15 mL/min). Albuminuria categories range from A1 (normal to mildly increased, ACR below 3 mg/mmol) to A3 (severely increased, ACR above 30 mg/mmol). High risk of progression is conferred by lower GFR combined with higher albuminuria. Management aims to slow progression through blood pressure control (target below 130/80 mmHg), RAAS blockade in proteinuric CKD, and SGLT-2 inhibitors now recommended across CKD stages for their proven renoprotective effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 52 – Review Version 3

Community-acquired pneumonia severity is assessed using the CURB-65 score, assigning one point each for confusion, urea above 7 mmol/L, respiratory rate 30 or more, blood pressure below 90/60 mmHg, and age 65 or over. Score 0-1 indicates low mortality risk suitable for home treatment. Score 2 warrants hospital admission. Score 3 or above indicates high mortality risk requiring hospital admission and possible ICU assessment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 53 – Review Version 3

It consistently improves exercise capacity, quality of life, and dyspnoea scores without altering FEV1. It reduces acute exacerbations and hospital readmissions. It is recommended for COPD patients with MRC dyspnoea grade 3 or above and following hospitalisation for acute exacerbation.. Pulmonary rehabilitation is a supervised multidisciplinary programme for patients with chronic respiratory disease combining exercise training, education, and psychosocial support.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 54 – Review Version 3

Diabetic ketoacidosis is a life-threatening acute complication caused by absolute or relative insulin deficiency. The biochemical triad includes hyperglycemia, metabolic acidosis, and ketonemia. Management requires fluid resuscitation, insulin infusion after potassium correction, electrolyte replacement particularly potassium, and identification of the precipitating cause such as infection or omitted insulin.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 55 – Review Version 3

Acute myocardial infarction results from rupture of an atherosclerotic plaque in a coronary artery leading to thrombus formation and vessel occlusion. Time to reperfusion is the critical determinant of outcome: primary percutaneous coronary intervention is the preferred strategy when achievable within 120 minutes of first medical contact. Immediate treatment includes aspirin, a P2Y12 inhibitor, anticoagulation, and infarct-related artery reperfusion.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 56 – Review Version 3

The HbA1c target for most adults with type 2 diabetes is below 7.0 percent, though individualized targets should account for age, comorbidities, hypoglycemia risk, and patient preference. In elderly or frail patients, a less stringent target of 7.5 to 8.0 percent may be appropriate to avoid hypoglycemia and its associated falls and cardiac risks.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 57 – Review Version 3

They require immediate ICU admission and parenteral antihypertensive therapy with careful organ-specific blood pressure targets. Hypertensive urgency involves severely elevated pressure without organ damage and can be managed with oral agents and close follow-up.. Hypertensive emergencies involve markedly elevated blood pressure above 180/120 mmHg with acute target organ damage including hypertensive encephalopathy, aortic dissection, acute coronary syndrome, or acute kidney injury.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 58 – Review Version 3

CYP3A4 enzyme interactions are particularly clinically significant: clarithromycin inhibits CYP3A4 and dramatically raises plasma levels of statins (risking rhabdomyolysis), ciclosporin, and benzodiazepines. Warfarin has hundreds of documented interactions and requires careful monitoring with any medication change.. Drug-drug interactions occur through pharmacokinetic mechanisms altering absorption, distribution, metabolism, or excretion, or pharmacodynamic mechanisms with additive or antagonistic effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 59 – Review Version 3

Vitamin D deficiency is highly prevalent, affecting approximately 1 billion people globally, and results from inadequate sun exposure, dietary insufficiency, or malabsorption. It causes rickets in children and osteomalacia in adults, presenting with bone pain, proximal muscle weakness, and pathological fractures. Severe vitamin D deficiency also impairs immune function and has been associated with increased risk of autoimmune disease, cardiovascular disease, and certain cancers. Supplementation guidelines recommend 400 IU daily for all adults in the UK and higher doses (1000-4000 IU) for those at high risk including elderly care home residents, those with dark skin or little sun exposure, and pregnant women.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 60 – Review Version 3

Spirometric confirmation with FEV1/FVC below 0.70 post-bronchodilator is required for diagnosis. Emphysema involves alveolar wall destruction reducing gas exchange, while chronic bronchitis is productive cough for at least three months in two consecutive years. Smoking cessation is the only intervention that alters disease progression.. COPD is characterised by progressive largely irreversible airflow limitation, predominantly caused by cigarette smoking or biomass fuel exposure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 61 – Review Version 3

Ambulatory blood pressure monitoring over 24 hours provides a more accurate assessment of true blood pressure burden. Masked hypertension, the converse, carries cardiovascular risk similar to sustained hypertension and should not be overlooked.. White coat hypertension affects 15 to 30 percent of referred patients and involves elevated readings in clinical settings but normal levels at home or on ambulatory monitoring.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 62 – Review Version 3

Chronic pain is defined as pain persisting beyond the normal expected healing time, typically more than 3 months. It affects approximately 28 million adults in the UK and has a significant biopsychosocial impact. It is now understood as a complex condition involving peripheral and central sensitisation, psychological factors including fear-avoidance beliefs and catastrophising, and social factors including work and relationship impacts. The modern pain management paradigm has shifted from a largely pharmacological approach to a multidisciplinary biopsychosocial model incorporating physiotherapy, psychological therapies particularly CBT and acceptance and commitment therapy, lifestyle optimisation, and judicious use of analgesics at the lowest effective dose.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 63 – Review Version 3

It emphasises fruits, vegetables, whole grains, low-fat dairy, and reduced saturated fat and sodium. Limiting sodium below 2.3g daily provides additional reduction. Regular aerobic exercise reduces systolic pressure by a further 5 to 8 mmHg. These lifestyle measures can reduce or delay pharmacological treatment.. The DASH diet reduces systolic blood pressure by 8 to 14 mmHg in large randomized trials.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 64 – Review Version 3

Informed consent is a legal and ethical requirement for medical procedures and treatment. For consent to be valid, the patient must have capacity to make the decision, the consent must be given voluntarily without coercion, and the patient must be provided with sufficient information about the proposed treatment including its benefits, risks, alternatives, and consequences of no treatment. Montgomery v Lanarkshire Health Board (2015) established in UK law that patients must be informed of all material risks that a reasonable patient would consider significant. Documentation of the consent process is essential. Consent for research requires additional safeguards including Research Ethics Committee approval.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 65 – Review Version 3

Inflammatory markers including C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR) are non-specific indicators of inflammation used to support diagnosis and monitor treatment response. CRP rises and falls rapidly within hours to days, making it more useful for monitoring acute conditions. ESR rises more slowly and reflects plasma viscosity. Both are elevated in

infection, autoimmune disease, malignancy, and tissue necrosis. Very high CRP above 100 mg/L suggests bacterial infection or severe inflammation. Procalcitonin is more specific for bacterial infection and is used in some settings to guide antibiotic prescribing. Normal inflammatory markers do not exclude serious pathology, particularly in immunocompromised patients.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 66 – Review Version 3

Continuous glucose monitoring systems provide real-time glucose readings every few minutes, enabling patients and clinicians to make more informed treatment decisions. Time-in-range, defined as glucose between 70 and 180 mg/dL, is emerging as a key glycemic metric alongside HbA1c. Higher time-in-range is associated with reduced rates of diabetic microvascular and macrovascular complications.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 67 – Review Version 3

GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated significant cardiovascular benefits beyond glycemic control. These agents stimulate insulin secretion in a glucose-dependent manner, suppress glucagon, slow gastric emptying, and promote weight loss. Cardiovascular outcome trials have shown significant reductions in major adverse cardiovascular events in high-risk patients.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 68 – Review Version 3

Antibiotic stewardship promotes appropriate antimicrobial use to improve patient outcomes while limiting resistance selection pressure. Key principles include prescribing antibiotics only when bacterial infection is confirmed or highly likely, selecting the narrowest spectrum effective agent, using the shortest evidence-based duration, and reviewing at 48 to 72 hours once microbiological results are available. Antimicrobial resistance is a global crisis threatening the foundations of modern medicine.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 69 – Review Version 3

Anaemia is defined as haemoglobin below the lower limit of normal for age and sex: below 130 g/L in adult men and below 120 g/L in adult women. The main causes are classified by mechanism: iron deficiency (most common worldwide, from blood loss, poor intake, or malabsorption), vitamin B12 or folate deficiency (macrocytic anaemia from impaired DNA synthesis), haemolysis (destruction of red blood cells from autoimmune, mechanical, or hereditary causes), and bone marrow failure or infiltration. History, full blood count, reticulocyte count, and iron studies guide initial investigation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 70 – Review Version 3

Primary hypertension has no identifiable single cause, with genetic predisposition, dietary sodium, obesity, physical inactivity, and chronic stress all contributing. Secondary causes account for 5 to 10 percent of cases.. Hypertension, defined as blood pressure at or above 130/80 mmHg by ACC/AHA guidelines, affects over 1.3 billion adults worldwide and is the leading modifiable risk factor for cardiovascular disease, stroke, and renal failure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 71 – Review Version 3

The PREDIMED trial demonstrated significant reductions in major cardiovascular events in high-risk patients following a Mediterranean diet compared with a low-fat diet. It also reduces risk of type 2 diabetes, colorectal and breast cancer, cognitive decline, and all-cause mortality.. The Mediterranean diet is characterised by high consumption of plant foods, olive oil as the primary fat, moderate fish and seafood, and low red and processed meat consumption.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 72 – Review Version 3

Occupational health medicine focuses on the relationship between work and health in both directions: how work can cause or worsen health conditions (occupational disease), and how health conditions affect the ability to work. Common occupational diseases include occupational asthma (from isocyanates, flour dust, or latex), vibration white finger from hand-arm vibration syndrome, noise-induced hearing loss, mesothelioma from asbestos exposure, contact dermatitis, and musculoskeletal disorders from manual handling or repetitive strain. Fitness for work assessments consider functional capacity, job demands, reasonable workplace adjustments, and prognosis. Statutory sick pay and fit notes (formerly sick notes) are important administrative tools in managing work absence.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 73 – Review Version 3

Hand hygiene is the single most effective measure to prevent healthcare-associated infections. The WHO five moments framework identifies critical junctures: before patient contact, before an aseptic procedure, after body fluid exposure, after patient contact, and after contact with patient surroundings. Alcohol-based hand rub is preferred for most clinical situations due to superior speed and convenience, while soap and water is required for *Clostridioides difficile* and norovirus.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 74 – Review Version 3

Diabetic nephropathy affects approximately 40 percent of diabetic patients and is the leading cause of end-stage renal disease globally. Early detection through annual screening for urinary albumin-to-creatinine ratio and eGFR is essential. ACE inhibitors and ARBs are first-line for hypertension in diabetic patients with albuminuria due to their independent nephroprotective properties beyond blood pressure lowering.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 75 – Review Version 3

Asthma is a chronic inflammatory airway disease characterised by variable reversible airflow obstruction, bronchial hyperresponsiveness, and airway remodelling. Inhaled corticosteroids are the cornerstone of controller therapy, reducing airway inflammation and risk of exacerbations. Short-acting beta-2 agonists provide rapid bronchodilation for symptom relief. Management follows a stepwise approach guided by symptom control and exacerbation frequency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 76 – Review Version 3

Time-in-range, defined as glucose between 70 and 180 mg/dL, is emerging as a key glycemic metric alongside HbA1c. Higher time-in-range is associated with reduced rates of diabetic microvascular and macrovascular complications.. Continuous glucose monitoring systems provide real-time glucose readings every few minutes, enabling patients and clinicians to make more informed treatment decisions.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 77 – Review Version 3

ACE inhibitors block conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. ARBs block angiotensin II at its receptor. Both reduce proteinuria and slow nephropathy progression. They should not be combined due to increased adverse effects without additional blood pressure benefit.. ACE inhibitors and ARBs are preferred antihypertensives in patients with diabetes, chronic kidney disease, or heart failure with reduced ejection fraction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 78 – Review Version 3

The Surviving Sepsis Campaign Hour-1 Bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, starting 30 mL/kg intravenous crystalloid for hypotension or elevated lactate, and vasopressors for refractory hypotension. Each hour of delay in antibiotic administration increases mortality by approximately 7 percent.. Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 79 – Review Version 3

Sertraline has the strongest evidence base in depression, while escitalopram and citalopram have good tolerability profiles. Full therapeutic benefit requires 4 to 6 weeks and treatment should continue for at least 6 to 12 months after remission to prevent relapse.. SSRIs are the first-line pharmacological treatment for depression and anxiety disorders due to their balance of efficacy, tolerability, and safety in overdose compared with older tricyclics.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 80 – Review Version 3

Wound healing progresses through four overlapping phases: haemostasis (immediate platelet plug and clot formation), inflammation (neutrophil and macrophage infiltration clearing debris and bacteria), proliferation (fibroblast migration producing collagen, angiogenesis, and re-epithelialisation), and remodelling (collagen reorganisation over months to years). Factors impairing wound healing include diabetes (neuropathy, ischaemia, and impaired immune response), malnutrition (particularly protein, zinc, and vitamin C deficiency), corticosteroids, smoking (causing vasoconstriction and reduced oxygen delivery), infection, and foreign bodies. Assessment should determine the wound bed, edges, surrounding skin, exudate, infection signs, and underlying cause.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 81 – Review Version 3

True resistance must be distinguished from pseudoresistance caused by poor adherence, white coat effect, or suboptimal measurement. Secondary causes should be investigated. Spironolactone (an aldosterone antagonist) as a fourth agent provides the most evidence for treating true resistant hypertension.. Resistant hypertension is blood pressure above target despite optimal doses of three antihypertensive agents including a diuretic.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 82 – Review Version 3

Antimicrobial resistance is driven by overuse and misuse of antibiotics in human medicine and agriculture, selecting for resistant bacterial strains. The WHO has identified antibiotic resistance as one of the greatest threats to global health. ESKAPE organisms (Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter species) account for the majority of serious healthcare-associated infections with resistance threatening treatment options.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 83 – Review Version 3

Pain assessment in clinical practice uses validated tools appropriate to the patient's communication ability. The numerical rating scale (0-10) is the most commonly used in cognitively intact adults. The visual analogue scale uses a 100mm line from no pain to worst imaginable pain. The Wong-Baker FACES scale is appropriate for children and patients with cognitive impairment. Multidimensional assessment tools such as the Brief Pain Inventory assess pain intensity and its impact on function and quality of life. Chronic pain assessment should include physical, psychological, and social dimensions using a biopsychosocial framework, with regular reassessment using the same validated tool for consistency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 84 – Review Version 3

Chronic liver disease progresses through stages from simple steatosis through steatohepatitis, fibrosis, cirrhosis, and finally hepatocellular carcinoma. Common causes include alcohol-related liver disease, non-alcoholic fatty liver disease (increasingly prevalent with obesity epidemic), chronic hepatitis B and C, autoimmune hepatitis, primary biliary cholangitis, and haemochromatosis. Cirrhosis complications include portal hypertension causing varices and ascites, hepatic encephalopathy, spontaneous bacterial peritonitis, hepatorenal syndrome, and coagulopathy. Six-monthly ultrasound and AFP for hepatocellular carcinoma surveillance is recommended in all patients with established cirrhosis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 85 – Review Version 3

The Mediterranean diet is characterised by high consumption of plant foods, olive oil as the primary fat, moderate fish and seafood, and low red and processed meat consumption. The PREDIMED trial demonstrated significant reductions in major cardiovascular events in high-risk patients following a Mediterranean diet compared with a low-fat diet. It also reduces risk of type 2 diabetes, colorectal and breast cancer, cognitive decline, and all-cause mortality.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 86 – Review Version 3

The biochemical triad includes hyperglycemia, metabolic acidosis, and ketonemia. Management requires fluid resuscitation, insulin infusion after potassium correction, electrolyte replacement particularly potassium, and identification of the precipitating cause such as infection or omitted insulin.. Diabetic ketoacidosis is a life-threatening acute complication caused by absolute or relative insulin deficiency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 87 – Review Version 3

Heart failure with reduced ejection fraction is managed with quadruple therapy: an ACE inhibitor, ARB, or ARNI (sacubitril/valsartan), a beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT-2 inhibitor. Each class has independently demonstrated mortality reduction in randomised trials. Achieving all four drug classes at target doses reduces all-cause mortality by more than 70 percent compared with no treatment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 88 – Review Version 3

Depression affects over 280 million people globally and is characterised by persistent low mood, anhedonia, fatigue, cognitive impairment, sleep disturbance, appetite changes, and suicidal ideation in severe cases. The biopsychosocial model recognises neurobiological contributions including dysregulation of serotonin, norepinephrine, and dopamine pathways, alongside psychological factors such as negative cognitive schemas and social factors including adverse life events and isolation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 89 – Review Version 3

Score 0-1 indicates low mortality risk suitable for home treatment. Score 2 warrants hospital admission. Score 3 or above indicates high mortality risk requiring hospital admission and possible ICU assessment.. Community-acquired pneumonia severity is assessed using the CURB-65 score, assigning one point each for confusion, urea above 7 mmol/L, respiratory rate 30 or more, blood pressure below 90/60 mmHg, and age 65 or over.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 90 – Review Version 3

Fibromyalgia is a centralised pain syndrome characterised by widespread musculoskeletal pain, fatigue, sleep disturbance, cognitive difficulties (fibro fog), and often comorbid depression and anxiety. It results from central sensitisation where the nervous system amplifies pain signals. Diagnosis is clinical using the 2010 ACR diagnostic criteria, requiring widespread pain index and symptom severity scores above thresholds. Management is multidisciplinary: aerobic exercise and physiotherapy are the most evidence-based interventions, CBT addresses maladaptive pain behaviours, and low-dose amitriptyline or duloxetine can improve pain and sleep. Opioids are not recommended and may worsen central sensitisation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 91 – Review Version 3

Thyroid hormones T3 (triiodothyronine) and T4 (thyroxine) regulate metabolic rate, growth, and development. Hypothyroidism causes fatigue, weight gain, cold intolerance, constipation, dry skin, and bradycardia, and is most commonly caused by Hashimoto's thyroiditis (autoimmune destruction). Hyperthyroidism causes weight loss despite increased appetite, heat intolerance, palpitations, tremor, and anxiety, most commonly from Graves' disease (TSH receptor stimulating antibodies). TSH is the most sensitive screening test for thyroid dysfunction, rising in hypothyroidism and falling in hyperthyroidism.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 92 – Review Version 3

Lithium remains the gold standard mood stabiliser for long-term prevention of both manic and depressive episodes, with robust evidence for suicide risk reduction. Valproate is effective for mania prevention; lamotrigine particularly targets depressive episodes. Close monitoring of lithium levels, renal function, and thyroid function is essential given its narrow therapeutic index.. Bipolar disorder features recurrent mania or hypomania alternating with depressive episodes.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 93 – Review Version 3

Delirium is an acute neuropsychiatric syndrome characterised by disturbance of attention and awareness with fluctuating course, developing acutely over hours to days. It affects up to 30 percent of hospitalised elderly patients and is associated with increased mortality, longer hospital stay, and accelerated cognitive decline. Precipitating factors include infection, metabolic disturbance, medications particularly anticholinergics and opioids, pain, urinary retention, constipation, dehydration, and unfamiliar environment. Management prioritises identifying and treating underlying causes, non-pharmacological reorientation strategies, and early mobilisation. Antipsychotics are used cautiously for severe agitation only.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 94 – Review Version 3

Obesity is characterised by excess adiposity impairing health. BMI provides simple screening: overweight 25 to 29.9, obesity class I 30 to 34.9, class II 35 to 39.9, class III 40 or above. Waist circumference better captures central adiposity and metabolic risk. Management involves comprehensive lifestyle intervention, pharmacotherapy in eligible patients including GLP-1 receptor agonists which can reduce weight by 10 to 20 percent, and bariatric surgery for severe obesity.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 95 – Review Version 3

Early detection through annual screening for urinary albumin-to-creatinine ratio and eGFR is essential. ACE inhibitors and ARBs are first-line for hypertension in diabetic patients with albuminuria due to their independent nephroprotective properties beyond blood pressure lowering. Diabetic nephropathy affects approximately 40 percent of diabetic patients and is the leading cause of end-stage renal disease globally.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 96 – Review Version 3

White coat hypertension affects 15 to 30 percent of referred patients and involves elevated readings in clinical settings but normal levels at home or on ambulatory monitoring. Ambulatory blood pressure monitoring over 24 hours provides a more accurate assessment of true blood pressure burden. Masked hypertension, the converse, carries cardiovascular risk similar to sustained hypertension and should not be overlooked.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 97 – Review Version 3

NSAIDs including ibuprofen, naproxen, and diclofenac inhibit COX-1 and COX-2 enzymes, reducing prostaglandin synthesis. COX-1 inhibition impairs gastric mucosal protection and platelet aggregation, increasing gastrointestinal bleeding risk particularly in elderly patients, those on corticosteroids or anticoagulants, or those with previous peptic ulcer disease. All NSAIDs carry cardiovascular and renal risks, particularly in volume-depleted patients or those with pre-existing kidney disease.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 98 – Review Version 3

In depression, CBT targets negative automatic thoughts and behavioural withdrawal through behavioural activation. In anxiety disorders, it incorporates exposure and response prevention, worry postponement, and attentional training. Its benefits persist beyond treatment completion, making it superior to medication alone for long-term relapse prevention.. Cognitive behavioural therapy is a structured evidence-based psychological treatment helping patients identify connections between thoughts, emotions, and behaviours.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 99 – Review Version 3

The pathophysiology involves impaired insulin signaling in peripheral tissues, particularly skeletal muscle, liver, and adipose tissue, leading to chronic hyperglycemia. Lifestyle modifications including dietary changes, physical activity, and weight management remain the cornerstone of initial management before pharmacological intervention is considered.. Type 2 diabetes mellitus is a chronic metabolic disorder characterized by insulin resistance and progressive beta-cell dysfunction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 100 – Review Version 3

Metformin is the first-line pharmacological agent for type 2 diabetes in most international guidelines. It works primarily by inhibiting hepatic gluconeogenesis through activation of AMP-activated protein kinase (AMPK), reducing fasting plasma glucose without causing hypoglycemia. Common side effects include gastrointestinal disturbances, which can be minimized by titrating the dose gradually and taking it with food.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 101 – Review Version 3

Beyond glycemic control, they demonstrate heart failure hospitalisation reduction and nephroprotective effects. They are now recommended early in patients with established cardiovascular disease or chronic kidney disease regardless of glycemic status.. SGLT-2 inhibitors including empagliflozin, dapagliflozin, and canagliflozin reduce blood glucose by inhibiting renal glucose reabsorption, causing glycosuria.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 102 – Review Version 3

Without intervention, 5 to 10 percent progress to type 2 diabetes each year. Intensive lifestyle intervention with 5 to 7 percent weight loss and 150 minutes weekly of moderate exercise reduces progression by up to 58 percent, making early identification and intervention critical.. Prediabetes is

characterized by fasting glucose 100-125 mg/dL, 2-hour OGTT glucose 140-199 mg/dL, or HbA1c 5.7-6.4 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 103 – Review Version 3

Gestational diabetes mellitus complicates 6 to 9 percent of pregnancies and is associated with adverse outcomes including macrosomia, birth trauma, and neonatal hypoglycemia. Screening is performed between 24 and 28 weeks using an oral glucose tolerance test. Management prioritises dietary modification and blood glucose monitoring, with insulin or metformin added if glycemic targets are not achieved.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 104 – Review Version 3

Time to reperfusion is the critical determinant of outcome: primary percutaneous coronary intervention is the preferred strategy when achievable within 120 minutes of first medical contact. Immediate treatment includes aspirin, a P2Y₁₂ inhibitor, anticoagulation, and infarct-related artery reperfusion.. Acute myocardial infarction results from rupture of an atherosclerotic plaque in a coronary artery leading to thrombus formation and vessel occlusion.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 105 – Review Version 3

Drug-drug interactions occur through pharmacokinetic mechanisms altering absorption, distribution, metabolism, or excretion, or pharmacodynamic mechanisms with additive or antagonistic effects. CYP3A4 enzyme interactions are particularly clinically significant: clarithromycin inhibits CYP3A4 and dramatically raises plasma levels of statins (risking rhabdomyolysis), ciclosporin, and benzodiazepines. Warfarin has hundreds of documented interactions and requires careful monitoring with any medication change.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 106 – Review Version 3

The Glasgow Coma Scale assesses conscious level through three components: eye opening (1-4), verbal response (1-5), and motor response (1-6), giving a total of 3 to 15. GCS 15 indicates full consciousness; GCS 8 or below indicates coma and the need to protect the airway. It is used in acute neurological and trauma assessment, with trends over time being more informative than a single measurement. The AVPU scale (Alert, Voice, Pain, Unresponsive) provides a simpler bedside assessment for initial triage. Causes of reduced conscious level are memorised using the AEIOU TIPS acronym: Alcohol/Acidosis, Epilepsy/Electrolytes, Infection, Opioids/Overdose, Uraemia, Trauma/Temperature, Insulin, Psychiatric, Stroke/Shock.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 107 – Review Version 3

Resistant hypertension is blood pressure above target despite optimal doses of three antihypertensive agents including a diuretic. True resistance must be distinguished from pseudoresistance caused by poor adherence, white coat effect, or suboptimal measurement. Secondary causes should be investigated. Spironolactone (an aldosterone antagonist) as a fourth agent provides the most evidence for treating true resistant hypertension.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 108 – Review Version 3

The coagulation cascade involves two pathways converging on a common final pathway. The extrinsic pathway, triggered by tissue factor exposure after vessel injury, activates factor X through factor VII. The intrinsic pathway, activated by contact with damaged surfaces, activates factor X through factors XII, XI, IX, and VIII. Factor Xa with factor Va forms prothrombinase, converting prothrombin to thrombin, which converts fibrinogen to fibrin and amplifies coagulation through positive feedback. Anticoagulants including warfarin, heparins, and direct oral anticoagulants target different steps in this cascade.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 109 – Review Version 3

The liver is the primary site of drug metabolism, predominantly through cytochrome P450 enzyme-mediated oxidation, reduction, and hydrolysis reactions (Phase I) followed by conjugation reactions making metabolites more water-soluble for excretion (Phase II). First-pass metabolism reduces bioavailability of oral medications before they reach systemic circulation. Liver disease impairs drug metabolism, requiring dose reductions for hepatically metabolised drugs. Enzyme inducers such as rifampicin and carbamazepine increase drug metabolism, reducing efficacy of co-medications. Enzyme inhibitors such as ciprofloxacin and fluoxetine reduce metabolism, risking toxicity of co-medications.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 110 – Review Version 3

Skin integrity assessment is fundamental to pressure ulcer prevention in at-risk patients. The Braden Scale assesses six risk factors: sensory perception, moisture, activity level, mobility, nutrition, and friction and shear. Scores range from 6 to 23, with lower scores indicating higher risk. Preventive interventions for high-risk patients include regular repositioning every 2 hours, pressure-redistributing mattresses and cushions, skin hygiene and moisturisation, optimisation of nutrition and hydration, and protection of heels with foam boots. Pressure ulcers are classified by depth from Stage 1 (non-blanchable erythema of intact skin) to Stage 4 (full thickness tissue loss exposing bone, tendon, or muscle).

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 1 – Review Version 4

Patient-centred care is an approach that respects and responds to individual patient preferences, needs, and values, ensuring that patient values guide all clinical decisions. Its core dimensions include respect for patient values and expressed needs, emotional support and relief of fear and anxiety, information and communication in accessible language, coordination and integration of care across settings, physical comfort and pain management, involvement of family and friends as appropriate, and access to care and continuity. Evidence consistently shows that patient-centred care improves patient satisfaction, treatment adherence, health outcomes, and reduces unnecessary investigations and readmissions.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 2 – Review Version 4

Prediabetes is characterized by fasting glucose 100-125 mg/dL, 2-hour OGTT glucose 140-199 mg/dL, or HbA1c 5.7-6.4 percent. Without intervention, 5 to 10 percent progress to type 2 diabetes each year. Intensive lifestyle intervention with 5 to 7 percent weight loss and 150 minutes weekly of moderate exercise reduces progression by up to 58 percent, making early identification and intervention critical.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 3 – Review Version 4

Hypertension, defined as blood pressure at or above 130/80 mmHg by ACC/AHA guidelines, affects over 1.3 billion adults worldwide and is the leading modifiable risk factor for cardiovascular disease, stroke, and renal failure. Primary hypertension has no identifiable single cause, with genetic predisposition, dietary sodium, obesity, physical inactivity, and chronic stress all contributing. Secondary causes account for 5 to 10 percent of cases.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 4 – Review Version 4

Thiazide diuretics such as chlorthalidone and hydrochlorothiazide are effective antihypertensives that reduce blood volume through natriuresis. They are particularly effective in Black patients and elderly patients with isolated systolic hypertension. Metabolic side effects include hypokalemia, hyperuricemia, and mild glucose and lipid disturbances. Chlorthalidone has longer duration of action and better evidence for cardiovascular event reduction compared with hydrochlorothiazide.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 5 – Review Version 4

SGLT-2 inhibitors including empagliflozin, dapagliflozin, and canagliflozin reduce blood glucose by inhibiting renal glucose reabsorption, causing glycosuria. Beyond glycemic control, they demonstrate heart failure hospitalisation reduction and nephroprotective effects. They are now recommended early in patients with established cardiovascular disease or chronic kidney disease regardless of glycemic status.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications,

contraindications, and multidisciplinary care approaches.

Topic 6 – Review Version 4

Oedema represents excess fluid accumulation in the interstitial compartment and is classified by distribution and cause. Bilateral leg oedema suggests systemic causes including heart failure (elevated venous pressure with sodium retention), nephrotic syndrome (protein loss reducing oncotic pressure), liver cirrhosis (reduced albumin synthesis and portal hypertension), or hypothyroidism causing myxoedema. Unilateral leg oedema suggests local causes including deep vein thrombosis, venous insufficiency, lymphoedema, or infection. Investigation follows the distribution: bilateral oedema warrants cardiac, renal, liver, and thyroid assessment; unilateral oedema warrants duplex ultrasound to exclude DVT.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 7 – Review Version 4

The DASH diet reduces systolic blood pressure by 8 to 14 mmHg in large randomized trials. It emphasises fruits, vegetables, whole grains, low-fat dairy, and reduced saturated fat and sodium. Limiting sodium below 2.3g daily provides additional reduction. Regular aerobic exercise reduces systolic pressure by a further 5 to 8 mmHg. These lifestyle measures can reduce or delay pharmacological treatment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 8 – Review Version 4

Urinary tract infections are classified as uncomplicated (affecting otherwise healthy, non-pregnant women with normal urinary tracts) or complicated (occurring in men, pregnant women, those with structural abnormalities, immunosuppression, or catheter-related). *Escherichia coli* causes approximately 80 percent of community-acquired UTIs. Uncomplicated lower UTI in women is treated with a 3-day course of trimethoprim or nitrofurantoin based on local resistance patterns. Recurrent UTIs (three or more per year) warrant investigation for predisposing factors and may be managed with low-dose prophylactic antibiotics, post-coital prophylaxis, or self-start therapy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 9 – Review Version 4

These agents stimulate insulin secretion in a glucose-dependent manner, suppress glucagon, slow gastric emptying, and promote weight loss. Cardiovascular outcome trials have shown significant reductions in major adverse cardiovascular events in high-risk patients. GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated significant cardiovascular benefits beyond glycemic control.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 10 – Review Version 4

Screening is performed between 24 and 28 weeks using an oral glucose tolerance test. Management prioritises dietary modification and blood glucose monitoring, with insulin or metformin

added if glycemic targets are not achieved.. Gestational diabetes mellitus complicates 6 to 9 percent of pregnancies and is associated with adverse outcomes including macrosomia, birth trauma, and neonatal hypoglycemia.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 11 – Review Version 4

COPD is characterised by progressive largely irreversible airflow limitation, predominantly caused by cigarette smoking or biomass fuel exposure. Spirometric confirmation with FEV1/FVC below 0.70 post-bronchodilator is required for diagnosis. Emphysema involves alveolar wall destruction reducing gas exchange, while chronic bronchitis is productive cough for at least three months in two consecutive years. Smoking cessation is the only intervention that alters disease progression.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 12 – Review Version 4

WHO recommends adults perform 150 to 300 minutes of moderate-intensity aerobic activity or 75 to 150 minutes of vigorous-intensity activity weekly, plus muscle-strengthening activities twice weekly. Regular physical activity reduces cardiovascular disease, type 2 diabetes, depression, several cancers, and all-cause mortality. Even modest increases from a sedentary baseline provide significant health benefits.. Physical inactivity contributes to 3.2 million deaths annually globally.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 13 – Review Version 4

Medication reconciliation is the process of creating the most accurate possible list of all medications a patient is taking, including dose, frequency, and route, and comparing it against the physician's admission, transfer, or discharge orders to identify and resolve discrepancies. It reduces medication errors at care transitions, which are responsible for approximately 60 percent of serious medication errors in hospital settings. Discrepancies found during reconciliation include omitted medications, wrong doses, duplicated medications, and medications continued that should have been stopped. All newly hospitalised patients should have medication reconciliation performed by a pharmacist within 24 hours of admission.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 14 – Review Version 4

Each class has independently demonstrated mortality reduction in randomised trials. Achieving all four drug classes at target doses reduces all-cause mortality by more than 70 percent compared with no treatment.. Heart failure with reduced ejection fraction is managed with quadruple therapy: an ACE inhibitor, ARB, or ARNI (sacubitril/valsartan), a beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT-2 inhibitor.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 15 – Review Version 4

Hypertensive emergencies involve markedly elevated blood pressure above 180/120 mmHg with acute target organ damage including hypertensive encephalopathy, aortic dissection, acute coronary syndrome, or acute kidney injury. They require immediate ICU admission and parenteral antihypertensive therapy with careful organ-specific blood pressure targets. Hypertensive urgency involves severely elevated pressure without organ damage and can be managed with oral agents and close follow-up.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 16 – Review Version 4

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection. The Surviving Sepsis Campaign Hour-1 Bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, starting 30 mL/kg intravenous crystalloid for hypotension or elevated lactate, and vasopressors for refractory hypotension. Each hour of delay in antibiotic administration increases mortality by approximately 7 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 17 – Review Version 4

High-intensity statins including atorvastatin 40 to 80mg and rosuvastatin 20 to 40mg are recommended for patients with established cardiovascular disease, targeting LDL-C below 1.4 mmol/L in very high-risk patients. Beyond cholesterol lowering, statins have plaque-stabilising and anti-inflammatory effects.. Statins inhibit HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis, reducing LDL cholesterol by 30 to 55 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 18 – Review Version 4

It progresses from non-proliferative stages with microaneurysms and retinal hemorrhages to proliferative retinopathy with neovascularization risking vitreous hemorrhage. Annual dilated fundal examination enables early detection. Intravitreal anti-VEGF injections and laser photocoagulation are effective treatments for sight-threatening retinopathy.. Diabetic retinopathy is the leading cause of new blindness in working-age adults.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 19 – Review Version 4

Pulmonary rehabilitation is a supervised multidisciplinary programme for patients with chronic respiratory disease combining exercise training, education, and psychosocial support. It consistently improves exercise capacity, quality of life, and dyspnoea scores without altering FEV1. It reduces acute exacerbations and hospital readmissions. It is recommended for COPD patients with MRC dyspnoea grade 3 or above and following hospitalisation for acute exacerbation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications,

contraindications, and multidisciplinary care approaches.

Topic 20 – Review Version 4

Palliative care focuses on improving quality of life for patients and families facing life-limiting illness through prevention and relief of suffering by early identification and impeccable assessment and treatment of pain and other physical, psychosocial, and spiritual problems. It is applicable from diagnosis of a life-limiting condition alongside curative treatment, not only at end of life. The World Health Organization emphasises that palliative care affirms life and regards dying as a normal process, neither hastening nor postponing death. Advance care planning, including discussion of resuscitation preferences, preferred place of death, and lasting power of attorney for health and welfare, is an important component.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 21 – Review Version 4

Cognitive behavioural therapy is a structured evidence-based psychological treatment helping patients identify connections between thoughts, emotions, and behaviours. In depression, CBT targets negative automatic thoughts and behavioural withdrawal through behavioural activation. In anxiety disorders, it incorporates exposure and response prevention, worry postponement, and attentional training. Its benefits persist beyond treatment completion, making it superior to medication alone for long-term relapse prevention.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 22 – Review Version 4

Statins inhibit HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis, reducing LDL cholesterol by 30 to 55 percent. High-intensity statins including atorvastatin 40 to 80mg and rosuvastatin 20 to 40mg are recommended for patients with established cardiovascular disease, targeting LDL-C below 1.4 mmol/L in very high-risk patients. Beyond cholesterol lowering, statins have plaque-stabilising and anti-inflammatory effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 23 – Review Version 4

Dehydration occurs when fluid losses exceed intake, reducing total body water. Mild dehydration (1-2 percent body weight) causes thirst and reduced concentration; moderate dehydration (3-5 percent) causes headache, dizziness, and reduced urine output; severe dehydration (above 6 percent) causes tachycardia, hypotension, and confusion. In adults, urine colour is a practical guide: pale yellow indicates adequate hydration; dark amber indicates dehydration. Oral rehydration is appropriate for mild to moderate dehydration using water or oral rehydration salts. Intravenous fluid resuscitation is required for severe dehydration or when oral intake is impossible. Elderly patients are at particular risk due to reduced thirst sensation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 24 – Review Version 4

Key principles include prescribing antibiotics only when bacterial infection is confirmed or highly likely, selecting the narrowest spectrum effective agent, using the shortest evidence-based duration, and reviewing at 48 to 72 hours once microbiological results are available. Antimicrobial resistance is a global crisis threatening the foundations of modern medicine.. Antibiotic stewardship promotes appropriate antimicrobial use to improve patient outcomes while limiting resistance selection pressure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 25 – Review Version 4

ACE inhibitors and ARBs are preferred antihypertensives in patients with diabetes, chronic kidney disease, or heart failure with reduced ejection fraction. ACE inhibitors block conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. ARBs block angiotensin II at its receptor. Both reduce proteinuria and slow nephropathy progression. They should not be combined due to increased adverse effects without additional blood pressure benefit.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 26 – Review Version 4

The principles of medical ethics provide a framework for clinical decision-making when values conflict. Autonomy refers to the patient's right to make informed decisions about their care, requiring valid informed consent. Beneficence is the duty to act in the patient's best interest. Non-maleficence requires avoiding unnecessary harm and balancing risks against benefits. Justice involves fair allocation of healthcare resources and equitable treatment of all patients. When these principles conflict, as in a patient refusing beneficial treatment (autonomy versus beneficence), the framework guides structured ethical analysis. Mental capacity to make specific decisions must be assessed before overriding patient choice, using the Mental Capacity Act 2005 framework in England and Wales.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 27 – Review Version 4

Peripheral diabetic neuropathy affects up to 50 percent of patients with long-standing diabetes, presenting as distal symmetric polyneuropathy with pain, tingling, numbness, and loss of protective sensation. Annual foot examinations and patient education on foot care are essential preventive measures. Pharmacological management of neuropathic pain includes duloxetine, pregabalin, and amitriptyline.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 28 – Review Version 4

Polypharmacy, the concurrent use of five or more medications, affects approximately 40 percent of adults over 65 and significantly increases risk of adverse drug reactions, drug-drug interactions, medication non-adherence, falls, cognitive impairment, and preventable hospitalisation. Regular structured medication reviews using the Beers Criteria or STOPP/START criteria identify potentially inappropriate medications and opportunities for safe deprescribing.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 29 – Review Version 4

Deep vein thrombosis and pulmonary embolism constitute venous thromboembolism, a major cause of preventable hospital mortality. DVT typically involves the leg deep veins, with clot propagation risking embolisation to the pulmonary vasculature. Risk factors include immobility, surgery, malignancy, pregnancy, oral contraceptive use, thrombophilia, and obesity. Treatment with anticoagulation prevents clot extension and embolisation while the body's fibrinolytic system dissolves the thrombus.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 30 – Review Version 4

Calcium channel blockers such as amlodipine are effective antihypertensives with particular benefit in elderly patients, Black patients, and isolated systolic hypertension. They reduce vascular smooth muscle tone by blocking voltage-gated calcium channels. Non-dihydropyridine CCBs such as diltiazem and verapamil also reduce heart rate, useful in atrial fibrillation but contraindicated in heart failure with reduced ejection fraction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 31 – Review Version 4

Dyspepsia (upper abdominal discomfort, bloating, early satiety, or nausea) affects up to 40 percent of the population and has many causes. Functional dyspepsia (no organic cause on investigation) is most common, managed with lifestyle advice, PPI or H2 blocker trial, and consideration of *Helicobacter pylori* testing and eradication if positive. Alarm features requiring urgent endoscopy include unintentional weight loss, dysphagia, persistent vomiting, haematemesis or melaena, iron-deficiency anaemia, and epigastric mass. Any new onset dyspepsia over age 55 warrants early endoscopy. *H. pylori* eradication using triple therapy (PPI, amoxicillin, and clarithromycin for 7 days) achieves eradication in approximately 80 percent of cases.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 32 – Review Version 4

Diabetic retinopathy is the leading cause of new blindness in working-age adults. It progresses from non-proliferative stages with microaneurysms and retinal hemorrhages to proliferative retinopathy with neovascularization risking vitreous hemorrhage. Annual dilated fundal examination enables early detection. Intravitreal anti-VEGF injections and laser photocoagulation are effective treatments for sight-threatening retinopathy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 33 – Review Version 4

BMI provides simple screening: overweight 25 to 29.9, obesity class I 30 to 34.9, class II 35 to 39.9, class III 40 or above. Waist circumference better captures central adiposity and metabolic risk. Management involves comprehensive lifestyle intervention, pharmacotherapy in eligible patients including GLP-1 receptor agonists which can reduce weight by 10 to 20 percent, and bariatric surgery for severe obesity.. Obesity is characterised by excess adiposity impairing health.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 34 – Review Version 4

COX-1 inhibition impairs gastric mucosal protection and platelet aggregation, increasing gastrointestinal bleeding risk particularly in elderly patients, those on corticosteroids or anticoagulants, or those with previous peptic ulcer disease. All NSAIDs carry cardiovascular and renal risks, particularly in volume-depleted patients or those with pre-existing kidney disease.. NSAIDs including ibuprofen, naproxen, and diclofenac inhibit COX-1 and COX-2 enzymes, reducing prostaglandin synthesis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 35 – Review Version 4

It works primarily by inhibiting hepatic gluconeogenesis through activation of AMP-activated protein kinase (AMPK), reducing fasting plasma glucose without causing hypoglycemia. Common side effects include gastrointestinal disturbances, which can be minimized by titrating the dose gradually and taking it with food.. Metformin is the first-line pharmacological agent for type 2 diabetes in most international guidelines.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 36 – Review Version 4

The WHO five moments framework identifies critical junctures: before patient contact, before an aseptic procedure, after body fluid exposure, after patient contact, and after contact with patient surroundings. Alcohol-based hand rub is preferred for most clinical situations due to superior speed and convenience, while soap and water is required for *Clostridioides difficile* and norovirus.. Hand hygiene is the single most effective measure to prevent healthcare-associated infections.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 37 – Review Version 4

They reduce vascular smooth muscle tone by blocking voltage-gated calcium channels. Non-dihydropyridine CCBs such as diltiazem and verapamil also reduce heart rate, useful in atrial fibrillation but contraindicated in heart failure with reduced ejection fraction.. Calcium channel blockers such as amlodipine are effective antihypertensives with particular benefit in elderly patients, Black patients, and isolated systolic hypertension.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 38 – Review Version 4

Physical inactivity contributes to 3.2 million deaths annually globally. WHO recommends adults perform 150 to 300 minutes of moderate-intensity aerobic activity or 75 to 150 minutes of vigorous-intensity activity weekly, plus muscle-strengthening activities twice weekly. Regular physical activity reduces cardiovascular disease, type 2 diabetes, depression, several cancers, and all-cause mortality. Even modest increases from a sedentary baseline provide significant health benefits.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 39 – Review Version 4

DVT typically involves the leg deep veins, with clot propagation risking embolisation to the pulmonary vasculature. Risk factors include immobility, surgery, malignancy, pregnancy, oral contraceptive use, thrombophilia, and obesity. Treatment with anticoagulation prevents clot extension and embolisation while the body's fibrinolytic system dissolves the thrombus.. Deep vein thrombosis and pulmonary embolism constitute venous thromboembolism, a major cause of preventable hospital mortality.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 40 – Review Version 4

Atrial fibrillation is the most common sustained cardiac arrhythmia, causing irregular ventricular response from chaotic atrial electrical activity. The primary risks are stroke from atrial thrombus and heart failure from tachycardia-mediated cardiomyopathy. Management involves rate control, rhythm control in appropriate patients, and anticoagulation with direct oral anticoagulants guided by CHA2DS2-VASc scoring.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 41 – Review Version 4

The primary risks are stroke from atrial thrombus and heart failure from tachycardia-mediated cardiomyopathy. Management involves rate control, rhythm control in appropriate patients, and anticoagulation with direct oral anticoagulants guided by CHA2DS2-VASc scoring.. Atrial fibrillation is the most common sustained cardiac arrhythmia, causing irregular ventricular response from chaotic atrial electrical activity.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 42 – Review Version 4

SSRIs are the first-line pharmacological treatment for depression and anxiety disorders due to their balance of efficacy, tolerability, and safety in overdose compared with older tricyclics. Sertraline has the strongest evidence base in depression, while escitalopram and citalopram have good tolerability profiles. Full therapeutic benefit requires 4 to 6 weeks and treatment should continue for at least 6 to 12 months after remission to prevent relapse.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 43 – Review Version 4

Bipolar disorder features recurrent mania or hypomania alternating with depressive episodes. Lithium remains the gold standard mood stabiliser for long-term prevention of both manic and depressive episodes, with robust evidence for suicide risk reduction. Valproate is effective for mania prevention; lamotrigine particularly targets depressive episodes. Close monitoring of lithium levels, renal function, and thyroid function is essential given its narrow therapeutic index.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 44 – Review Version 4

Type 2 diabetes mellitus is a chronic metabolic disorder characterized by insulin resistance and progressive beta-cell dysfunction. The pathophysiology involves impaired insulin signaling in peripheral tissues, particularly skeletal muscle, liver, and adipose tissue, leading to chronic hyperglycemia. Lifestyle modifications including dietary changes, physical activity, and weight management remain the cornerstone of initial management before pharmacological intervention is considered.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 45 – Review Version 4

Annual foot examinations and patient education on foot care are essential preventive measures. Pharmacological management of neuropathic pain includes duloxetine, pregabalin, and amitriptyline. Peripheral diabetic neuropathy affects up to 50 percent of patients with long-standing diabetes, presenting as distal symmetric polyneuropathy with pain, tingling, numbness, and loss of protective sensation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 46 – Review Version 4

The renin-angiotensin-aldosterone system (RAAS) is a key hormonal cascade regulating blood pressure and fluid balance. Renin, secreted by juxtaglomerular cells in response to reduced renal perfusion, converts angiotensinogen to angiotensin I. ACE converts angiotensin I to the active angiotensin II, a potent vasoconstrictor that also stimulates aldosterone release causing sodium and water retention. RAAS blockade with ACE inhibitors or ARBs is central to managing hypertension, heart failure, and diabetic nephropathy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 47 – Review Version 4

Septic arthritis is a medical emergency requiring same-day joint aspiration and intravenous antibiotics. It typically presents with acute monoarthritis with fever, a hot swollen joint, and severely restricted movement. The knee is most commonly affected. *Staphylococcus aureus* is the most

common causative organism in adults. Diagnosis is confirmed by synovial fluid analysis showing turbid fluid with white cell count above 50,000 cells per microlitre with predominantly neutrophils, and Gram stain and culture. Delay in treatment risks rapid cartilage destruction and permanent joint damage.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 48 – Review Version 4

The WHO has identified antibiotic resistance as one of the greatest threats to global health. ESKAPE organisms (Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter species) account for the majority of serious healthcare-associated infections with resistance threatening treatment options.. Antimicrobial resistance is driven by overuse and misuse of antibiotics in human medicine and agriculture, selecting for resistant bacterial strains.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 49 – Review Version 4

They are particularly effective in Black patients and elderly patients with isolated systolic hypertension. Metabolic side effects include hypokalemia, hyperuricemia, and mild glucose and lipid disturbances. Chlorthalidone has longer duration of action and better evidence for cardiovascular event reduction compared with hydrochlorothiazide.. Thiazide diuretics such as chlorthalidone and hydrochlorothiazide are effective antihypertensives that reduce blood volume through natriuresis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 50 – Review Version 4

Inhaled corticosteroids are the cornerstone of controller therapy, reducing airway inflammation and risk of exacerbations. Short-acting beta-2 agonists provide rapid bronchodilation for symptom relief. Management follows a stepwise approach guided by symptom control and exacerbation frequency.. Asthma is a chronic inflammatory airway disease characterised by variable reversible airflow obstruction, bronchial hyperresponsiveness, and airway remodelling.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 51 – Review Version 4

Chronic kidney disease is classified by cause, GFR category, and albuminuria category using the CGA staging system. GFR categories range from G1 (normal, above 90 mL/min) to G5 (kidney failure, below 15 mL/min). Albuminuria categories range from A1 (normal to mildly increased, ACR below 3 mg/mmol) to A3 (severely increased, ACR above 30 mg/mmol). High risk of progression is conferred by lower GFR combined with higher albuminuria. Management aims to slow progression through blood pressure control (target below 130/80 mmHg), RAAS blockade in proteinuric CKD, and SGLT-2 inhibitors now recommended across CKD stages for their proven renoprotective effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 52 – Review Version 4

Community-acquired pneumonia severity is assessed using the CURB-65 score, assigning one point each for confusion, urea above 7 mmol/L, respiratory rate 30 or more, blood pressure below 90/60 mmHg, and age 65 or over. Score 0-1 indicates low mortality risk suitable for home treatment. Score 2 warrants hospital admission. Score 3 or above indicates high mortality risk requiring hospital admission and possible ICU assessment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 53 – Review Version 4

It consistently improves exercise capacity, quality of life, and dyspnoea scores without altering FEV1. It reduces acute exacerbations and hospital readmissions. It is recommended for COPD patients with MRC dyspnoea grade 3 or above and following hospitalisation for acute exacerbation.. Pulmonary rehabilitation is a supervised multidisciplinary programme for patients with chronic respiratory disease combining exercise training, education, and psychosocial support.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 54 – Review Version 4

Diabetic ketoacidosis is a life-threatening acute complication caused by absolute or relative insulin deficiency. The biochemical triad includes hyperglycemia, metabolic acidosis, and ketonemia. Management requires fluid resuscitation, insulin infusion after potassium correction, electrolyte replacement particularly potassium, and identification of the precipitating cause such as infection or omitted insulin.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 55 – Review Version 4

Acute myocardial infarction results from rupture of an atherosclerotic plaque in a coronary artery leading to thrombus formation and vessel occlusion. Time to reperfusion is the critical determinant of outcome: primary percutaneous coronary intervention is the preferred strategy when achievable within 120 minutes of first medical contact. Immediate treatment includes aspirin, a P2Y12 inhibitor, anticoagulation, and infarct-related artery reperfusion.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 56 – Review Version 4

The HbA1c target for most adults with type 2 diabetes is below 7.0 percent, though individualized targets should account for age, comorbidities, hypoglycemia risk, and patient preference. In elderly or frail patients, a less stringent target of 7.5 to 8.0 percent may be appropriate to avoid hypoglycemia and its associated falls and cardiac risks.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 57 – Review Version 4

They require immediate ICU admission and parenteral antihypertensive therapy with careful organ-specific blood pressure targets. Hypertensive urgency involves severely elevated pressure without organ damage and can be managed with oral agents and close follow-up.. Hypertensive emergencies involve markedly elevated blood pressure above 180/120 mmHg with acute target organ damage including hypertensive encephalopathy, aortic dissection, acute coronary syndrome, or acute kidney injury.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 58 – Review Version 4

CYP3A4 enzyme interactions are particularly clinically significant: clarithromycin inhibits CYP3A4 and dramatically raises plasma levels of statins (risking rhabdomyolysis), ciclosporin, and benzodiazepines. Warfarin has hundreds of documented interactions and requires careful monitoring with any medication change.. Drug-drug interactions occur through pharmacokinetic mechanisms altering absorption, distribution, metabolism, or excretion, or pharmacodynamic mechanisms with additive or antagonistic effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 59 – Review Version 4

Vitamin D deficiency is highly prevalent, affecting approximately 1 billion people globally, and results from inadequate sun exposure, dietary insufficiency, or malabsorption. It causes rickets in children and osteomalacia in adults, presenting with bone pain, proximal muscle weakness, and pathological fractures. Severe vitamin D deficiency also impairs immune function and has been associated with increased risk of autoimmune disease, cardiovascular disease, and certain cancers. Supplementation guidelines recommend 400 IU daily for all adults in the UK and higher doses (1000-4000 IU) for those at high risk including elderly care home residents, those with dark skin or little sun exposure, and pregnant women.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 60 – Review Version 4

Spirometric confirmation with FEV1/FVC below 0.70 post-bronchodilator is required for diagnosis. Emphysema involves alveolar wall destruction reducing gas exchange, while chronic bronchitis is productive cough for at least three months in two consecutive years. Smoking cessation is the only intervention that alters disease progression.. COPD is characterised by progressive largely irreversible airflow limitation, predominantly caused by cigarette smoking or biomass fuel exposure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 61 – Review Version 4

Ambulatory blood pressure monitoring over 24 hours provides a more accurate assessment of true blood pressure burden. Masked hypertension, the converse, carries cardiovascular risk similar to sustained hypertension and should not be overlooked.. White coat hypertension affects 15 to 30 percent of referred patients and involves elevated readings in clinical settings but normal levels at home or on ambulatory monitoring.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 62 – Review Version 4

Chronic pain is defined as pain persisting beyond the normal expected healing time, typically more than 3 months. It affects approximately 28 million adults in the UK and has a significant biopsychosocial impact. It is now understood as a complex condition involving peripheral and central sensitisation, psychological factors including fear-avoidance beliefs and catastrophising, and social factors including work and relationship impacts. The modern pain management paradigm has shifted from a largely pharmacological approach to a multidisciplinary biopsychosocial model incorporating physiotherapy, psychological therapies particularly CBT and acceptance and commitment therapy, lifestyle optimisation, and judicious use of analgesics at the lowest effective dose.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 63 – Review Version 4

It emphasises fruits, vegetables, whole grains, low-fat dairy, and reduced saturated fat and sodium. Limiting sodium below 2.3g daily provides additional reduction. Regular aerobic exercise reduces systolic pressure by a further 5 to 8 mmHg. These lifestyle measures can reduce or delay pharmacological treatment.. The DASH diet reduces systolic blood pressure by 8 to 14 mmHg in large randomized trials.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 64 – Review Version 4

Informed consent is a legal and ethical requirement for medical procedures and treatment. For consent to be valid, the patient must have capacity to make the decision, the consent must be given voluntarily without coercion, and the patient must be provided with sufficient information about the proposed treatment including its benefits, risks, alternatives, and consequences of no treatment. Montgomery v Lanarkshire Health Board (2015) established in UK law that patients must be informed of all material risks that a reasonable patient would consider significant. Documentation of the consent process is essential. Consent for research requires additional safeguards including Research Ethics Committee approval.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 65 – Review Version 4

Inflammatory markers including C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR) are non-specific indicators of inflammation used to support diagnosis and monitor treatment response. CRP rises and falls rapidly within hours to days, making it more useful for monitoring acute conditions. ESR rises more slowly and reflects plasma viscosity. Both are elevated in

infection, autoimmune disease, malignancy, and tissue necrosis. Very high CRP above 100 mg/L suggests bacterial infection or severe inflammation. Procalcitonin is more specific for bacterial infection and is used in some settings to guide antibiotic prescribing. Normal inflammatory markers do not exclude serious pathology, particularly in immunocompromised patients.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 66 – Review Version 4

Continuous glucose monitoring systems provide real-time glucose readings every few minutes, enabling patients and clinicians to make more informed treatment decisions. Time-in-range, defined as glucose between 70 and 180 mg/dL, is emerging as a key glycemic metric alongside HbA1c. Higher time-in-range is associated with reduced rates of diabetic microvascular and macrovascular complications.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 67 – Review Version 4

GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated significant cardiovascular benefits beyond glycemic control. These agents stimulate insulin secretion in a glucose-dependent manner, suppress glucagon, slow gastric emptying, and promote weight loss. Cardiovascular outcome trials have shown significant reductions in major adverse cardiovascular events in high-risk patients.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 68 – Review Version 4

Antibiotic stewardship promotes appropriate antimicrobial use to improve patient outcomes while limiting resistance selection pressure. Key principles include prescribing antibiotics only when bacterial infection is confirmed or highly likely, selecting the narrowest spectrum effective agent, using the shortest evidence-based duration, and reviewing at 48 to 72 hours once microbiological results are available. Antimicrobial resistance is a global crisis threatening the foundations of modern medicine.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 69 – Review Version 4

Anaemia is defined as haemoglobin below the lower limit of normal for age and sex: below 130 g/L in adult men and below 120 g/L in adult women. The main causes are classified by mechanism: iron deficiency (most common worldwide, from blood loss, poor intake, or malabsorption), vitamin B12 or folate deficiency (macrocytic anaemia from impaired DNA synthesis), haemolysis (destruction of red blood cells from autoimmune, mechanical, or hereditary causes), and bone marrow failure or infiltration. History, full blood count, reticulocyte count, and iron studies guide initial investigation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 70 – Review Version 4

Primary hypertension has no identifiable single cause, with genetic predisposition, dietary sodium, obesity, physical inactivity, and chronic stress all contributing. Secondary causes account for 5 to 10 percent of cases.. Hypertension, defined as blood pressure at or above 130/80 mmHg by ACC/AHA guidelines, affects over 1.3 billion adults worldwide and is the leading modifiable risk factor for cardiovascular disease, stroke, and renal failure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 71 – Review Version 4

The PREDIMED trial demonstrated significant reductions in major cardiovascular events in high-risk patients following a Mediterranean diet compared with a low-fat diet. It also reduces risk of type 2 diabetes, colorectal and breast cancer, cognitive decline, and all-cause mortality.. The Mediterranean diet is characterised by high consumption of plant foods, olive oil as the primary fat, moderate fish and seafood, and low red and processed meat consumption.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 72 – Review Version 4

Occupational health medicine focuses on the relationship between work and health in both directions: how work can cause or worsen health conditions (occupational disease), and how health conditions affect the ability to work. Common occupational diseases include occupational asthma (from isocyanates, flour dust, or latex), vibration white finger from hand-arm vibration syndrome, noise-induced hearing loss, mesothelioma from asbestos exposure, contact dermatitis, and musculoskeletal disorders from manual handling or repetitive strain. Fitness for work assessments consider functional capacity, job demands, reasonable workplace adjustments, and prognosis. Statutory sick pay and fit notes (formerly sick notes) are important administrative tools in managing work absence.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 73 – Review Version 4

Hand hygiene is the single most effective measure to prevent healthcare-associated infections. The WHO five moments framework identifies critical junctures: before patient contact, before an aseptic procedure, after body fluid exposure, after patient contact, and after contact with patient surroundings. Alcohol-based hand rub is preferred for most clinical situations due to superior speed and convenience, while soap and water is required for *Clostridioides difficile* and norovirus.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 74 – Review Version 4

Diabetic nephropathy affects approximately 40 percent of diabetic patients and is the leading cause of end-stage renal disease globally. Early detection through annual screening for urinary albumin-to-creatinine ratio and eGFR is essential. ACE inhibitors and ARBs are first-line for hypertension in diabetic patients with albuminuria due to their independent nephroprotective properties beyond blood pressure lowering.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 75 – Review Version 4

Asthma is a chronic inflammatory airway disease characterised by variable reversible airflow obstruction, bronchial hyperresponsiveness, and airway remodelling. Inhaled corticosteroids are the cornerstone of controller therapy, reducing airway inflammation and risk of exacerbations. Short-acting beta-2 agonists provide rapid bronchodilation for symptom relief. Management follows a stepwise approach guided by symptom control and exacerbation frequency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 76 – Review Version 4

Time-in-range, defined as glucose between 70 and 180 mg/dL, is emerging as a key glycemic metric alongside HbA1c. Higher time-in-range is associated with reduced rates of diabetic microvascular and macrovascular complications.. Continuous glucose monitoring systems provide real-time glucose readings every few minutes, enabling patients and clinicians to make more informed treatment decisions.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 77 – Review Version 4

ACE inhibitors block conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. ARBs block angiotensin II at its receptor. Both reduce proteinuria and slow nephropathy progression. They should not be combined due to increased adverse effects without additional blood pressure benefit.. ACE inhibitors and ARBs are preferred antihypertensives in patients with diabetes, chronic kidney disease, or heart failure with reduced ejection fraction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 78 – Review Version 4

The Surviving Sepsis Campaign Hour-1 Bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, starting 30 mL/kg intravenous crystalloid for hypotension or elevated lactate, and vasopressors for refractory hypotension. Each hour of delay in antibiotic administration increases mortality by approximately 7 percent.. Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 79 – Review Version 4

Sertraline has the strongest evidence base in depression, while escitalopram and citalopram have good tolerability profiles. Full therapeutic benefit requires 4 to 6 weeks and treatment should continue for at least 6 to 12 months after remission to prevent relapse.. SSRIs are the first-line pharmacological treatment for depression and anxiety disorders due to their balance of efficacy, tolerability, and safety in overdose compared with older tricyclics.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 80 – Review Version 4

Wound healing progresses through four overlapping phases: haemostasis (immediate platelet plug and clot formation), inflammation (neutrophil and macrophage infiltration clearing debris and bacteria), proliferation (fibroblast migration producing collagen, angiogenesis, and re-epithelialisation), and remodelling (collagen reorganisation over months to years). Factors impairing wound healing include diabetes (neuropathy, ischaemia, and impaired immune response), malnutrition (particularly protein, zinc, and vitamin C deficiency), corticosteroids, smoking (causing vasoconstriction and reduced oxygen delivery), infection, and foreign bodies. Assessment should determine the wound bed, edges, surrounding skin, exudate, infection signs, and underlying cause.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 81 – Review Version 4

True resistance must be distinguished from pseudoresistance caused by poor adherence, white coat effect, or suboptimal measurement. Secondary causes should be investigated. Spironolactone (an aldosterone antagonist) as a fourth agent provides the most evidence for treating true resistant hypertension.. Resistant hypertension is blood pressure above target despite optimal doses of three antihypertensive agents including a diuretic.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 82 – Review Version 4

Antimicrobial resistance is driven by overuse and misuse of antibiotics in human medicine and agriculture, selecting for resistant bacterial strains. The WHO has identified antibiotic resistance as one of the greatest threats to global health. ESKAPE organisms (Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter species) account for the majority of serious healthcare-associated infections with resistance threatening treatment options.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 83 – Review Version 4

Pain assessment in clinical practice uses validated tools appropriate to the patient's communication ability. The numerical rating scale (0-10) is the most commonly used in cognitively intact adults. The visual analogue scale uses a 100mm line from no pain to worst imaginable pain. The Wong-Baker FACES scale is appropriate for children and patients with cognitive impairment. Multidimensional assessment tools such as the Brief Pain Inventory assess pain intensity and its impact on function and quality of life. Chronic pain assessment should include physical, psychological, and social dimensions using a biopsychosocial framework, with regular reassessment using the same validated tool for consistency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 84 – Review Version 4

Chronic liver disease progresses through stages from simple steatosis through steatohepatitis, fibrosis, cirrhosis, and finally hepatocellular carcinoma. Common causes include alcohol-related liver disease, non-alcoholic fatty liver disease (increasingly prevalent with obesity epidemic), chronic hepatitis B and C, autoimmune hepatitis, primary biliary cholangitis, and haemochromatosis. Cirrhosis complications include portal hypertension causing varices and ascites, hepatic encephalopathy, spontaneous bacterial peritonitis, hepatorenal syndrome, and coagulopathy. Six-monthly ultrasound and AFP for hepatocellular carcinoma surveillance is recommended in all patients with established cirrhosis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 85 – Review Version 4

The Mediterranean diet is characterised by high consumption of plant foods, olive oil as the primary fat, moderate fish and seafood, and low red and processed meat consumption. The PREDIMED trial demonstrated significant reductions in major cardiovascular events in high-risk patients following a Mediterranean diet compared with a low-fat diet. It also reduces risk of type 2 diabetes, colorectal and breast cancer, cognitive decline, and all-cause mortality.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 86 – Review Version 4

The biochemical triad includes hyperglycemia, metabolic acidosis, and ketonemia. Management requires fluid resuscitation, insulin infusion after potassium correction, electrolyte replacement particularly potassium, and identification of the precipitating cause such as infection or omitted insulin.. Diabetic ketoacidosis is a life-threatening acute complication caused by absolute or relative insulin deficiency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 87 – Review Version 4

Heart failure with reduced ejection fraction is managed with quadruple therapy: an ACE inhibitor, ARB, or ARNI (sacubitril/valsartan), a beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT-2 inhibitor. Each class has independently demonstrated mortality reduction in randomised trials. Achieving all four drug classes at target doses reduces all-cause mortality by more than 70 percent compared with no treatment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 88 – Review Version 4

Depression affects over 280 million people globally and is characterised by persistent low mood, anhedonia, fatigue, cognitive impairment, sleep disturbance, appetite changes, and suicidal ideation in severe cases. The biopsychosocial model recognises neurobiological contributions including dysregulation of serotonin, norepinephrine, and dopamine pathways, alongside psychological factors such as negative cognitive schemas and social factors including adverse life events and isolation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 89 – Review Version 4

Score 0-1 indicates low mortality risk suitable for home treatment. Score 2 warrants hospital admission. Score 3 or above indicates high mortality risk requiring hospital admission and possible ICU assessment.. Community-acquired pneumonia severity is assessed using the CURB-65 score, assigning one point each for confusion, urea above 7 mmol/L, respiratory rate 30 or more, blood pressure below 90/60 mmHg, and age 65 or over.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 90 – Review Version 4

Fibromyalgia is a centralised pain syndrome characterised by widespread musculoskeletal pain, fatigue, sleep disturbance, cognitive difficulties (fibro fog), and often comorbid depression and anxiety. It results from central sensitisation where the nervous system amplifies pain signals. Diagnosis is clinical using the 2010 ACR diagnostic criteria, requiring widespread pain index and symptom severity scores above thresholds. Management is multidisciplinary: aerobic exercise and physiotherapy are the most evidence-based interventions, CBT addresses maladaptive pain behaviours, and low-dose amitriptyline or duloxetine can improve pain and sleep. Opioids are not recommended and may worsen central sensitisation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 91 – Review Version 4

Thyroid hormones T3 (triiodothyronine) and T4 (thyroxine) regulate metabolic rate, growth, and development. Hypothyroidism causes fatigue, weight gain, cold intolerance, constipation, dry skin, and bradycardia, and is most commonly caused by Hashimoto's thyroiditis (autoimmune destruction). Hyperthyroidism causes weight loss despite increased appetite, heat intolerance, palpitations, tremor, and anxiety, most commonly from Graves' disease (TSH receptor stimulating antibodies). TSH is the most sensitive screening test for thyroid dysfunction, rising in hypothyroidism and falling in hyperthyroidism.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 92 – Review Version 4

Lithium remains the gold standard mood stabiliser for long-term prevention of both manic and depressive episodes, with robust evidence for suicide risk reduction. Valproate is effective for mania prevention; lamotrigine particularly targets depressive episodes. Close monitoring of lithium levels, renal function, and thyroid function is essential given its narrow therapeutic index.. Bipolar disorder features recurrent mania or hypomania alternating with depressive episodes.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 93 – Review Version 4

Delirium is an acute neuropsychiatric syndrome characterised by disturbance of attention and awareness with fluctuating course, developing acutely over hours to days. It affects up to 30 percent of hospitalised elderly patients and is associated with increased mortality, longer hospital stay, and accelerated cognitive decline. Precipitating factors include infection, metabolic disturbance, medications particularly anticholinergics and opioids, pain, urinary retention, constipation, dehydration, and unfamiliar environment. Management prioritises identifying and treating underlying causes, non-pharmacological reorientation strategies, and early mobilisation. Antipsychotics are used cautiously for severe agitation only.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 94 – Review Version 4

Obesity is characterised by excess adiposity impairing health. BMI provides simple screening: overweight 25 to 29.9, obesity class I 30 to 34.9, class II 35 to 39.9, class III 40 or above. Waist circumference better captures central adiposity and metabolic risk. Management involves comprehensive lifestyle intervention, pharmacotherapy in eligible patients including GLP-1 receptor agonists which can reduce weight by 10 to 20 percent, and bariatric surgery for severe obesity.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 95 – Review Version 4

Early detection through annual screening for urinary albumin-to-creatinine ratio and eGFR is essential. ACE inhibitors and ARBs are first-line for hypertension in diabetic patients with albuminuria due to their independent nephroprotective properties beyond blood pressure lowering. Diabetic nephropathy affects approximately 40 percent of diabetic patients and is the leading cause of end-stage renal disease globally.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 96 – Review Version 4

White coat hypertension affects 15 to 30 percent of referred patients and involves elevated readings in clinical settings but normal levels at home or on ambulatory monitoring. Ambulatory blood pressure monitoring over 24 hours provides a more accurate assessment of true blood pressure burden. Masked hypertension, the converse, carries cardiovascular risk similar to sustained hypertension and should not be overlooked.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 97 – Review Version 4

NSAIDs including ibuprofen, naproxen, and diclofenac inhibit COX-1 and COX-2 enzymes, reducing prostaglandin synthesis. COX-1 inhibition impairs gastric mucosal protection and platelet aggregation, increasing gastrointestinal bleeding risk particularly in elderly patients, those on corticosteroids or anticoagulants, or those with previous peptic ulcer disease. All NSAIDs carry cardiovascular and renal risks, particularly in volume-depleted patients or those with pre-existing kidney disease.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 98 – Review Version 4

In depression, CBT targets negative automatic thoughts and behavioural withdrawal through behavioural activation. In anxiety disorders, it incorporates exposure and response prevention, worry postponement, and attentional training. Its benefits persist beyond treatment completion, making it superior to medication alone for long-term relapse prevention.. Cognitive behavioural therapy is a structured evidence-based psychological treatment helping patients identify connections between thoughts, emotions, and behaviours.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 99 – Review Version 4

The pathophysiology involves impaired insulin signaling in peripheral tissues, particularly skeletal muscle, liver, and adipose tissue, leading to chronic hyperglycemia. Lifestyle modifications including dietary changes, physical activity, and weight management remain the cornerstone of initial management before pharmacological intervention is considered.. Type 2 diabetes mellitus is a chronic metabolic disorder characterized by insulin resistance and progressive beta-cell dysfunction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 100 – Review Version 4

Metformin is the first-line pharmacological agent for type 2 diabetes in most international guidelines. It works primarily by inhibiting hepatic gluconeogenesis through activation of AMP-activated protein kinase (AMPK), reducing fasting plasma glucose without causing hypoglycemia. Common side effects include gastrointestinal disturbances, which can be minimized by titrating the dose gradually and taking it with food.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 101 – Review Version 4

Beyond glycemic control, they demonstrate heart failure hospitalisation reduction and nephroprotective effects. They are now recommended early in patients with established cardiovascular disease or chronic kidney disease regardless of glycemic status.. SGLT-2 inhibitors including empagliflozin, dapagliflozin, and canagliflozin reduce blood glucose by inhibiting renal glucose reabsorption, causing glycosuria.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 102 – Review Version 4

Without intervention, 5 to 10 percent progress to type 2 diabetes each year. Intensive lifestyle intervention with 5 to 7 percent weight loss and 150 minutes weekly of moderate exercise reduces progression by up to 58 percent, making early identification and intervention critical.. Prediabetes is

characterized by fasting glucose 100-125 mg/dL, 2-hour OGTT glucose 140-199 mg/dL, or HbA1c 5.7-6.4 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 103 – Review Version 4

Gestational diabetes mellitus complicates 6 to 9 percent of pregnancies and is associated with adverse outcomes including macrosomia, birth trauma, and neonatal hypoglycemia. Screening is performed between 24 and 28 weeks using an oral glucose tolerance test. Management prioritises dietary modification and blood glucose monitoring, with insulin or metformin added if glycemic targets are not achieved.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 104 – Review Version 4

Time to reperfusion is the critical determinant of outcome: primary percutaneous coronary intervention is the preferred strategy when achievable within 120 minutes of first medical contact. Immediate treatment includes aspirin, a P2Y₁₂ inhibitor, anticoagulation, and infarct-related artery reperfusion.. Acute myocardial infarction results from rupture of an atherosclerotic plaque in a coronary artery leading to thrombus formation and vessel occlusion.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 105 – Review Version 4

Drug-drug interactions occur through pharmacokinetic mechanisms altering absorption, distribution, metabolism, or excretion, or pharmacodynamic mechanisms with additive or antagonistic effects. CYP3A4 enzyme interactions are particularly clinically significant: clarithromycin inhibits CYP3A4 and dramatically raises plasma levels of statins (risking rhabdomyolysis), ciclosporin, and benzodiazepines. Warfarin has hundreds of documented interactions and requires careful monitoring with any medication change.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 106 – Review Version 4

The Glasgow Coma Scale assesses conscious level through three components: eye opening (1-4), verbal response (1-5), and motor response (1-6), giving a total of 3 to 15. GCS 15 indicates full consciousness; GCS 8 or below indicates coma and the need to protect the airway. It is used in acute neurological and trauma assessment, with trends over time being more informative than a single measurement. The AVPU scale (Alert, Voice, Pain, Unresponsive) provides a simpler bedside assessment for initial triage. Causes of reduced conscious level are memorised using the AEIOU TIPS acronym: Alcohol/Acidosis, Epilepsy/Electrolytes, Infection, Opioids/Overdose, Uraemia, Trauma/Temperature, Insulin, Psychiatric, Stroke/Shock.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 107 – Review Version 4

Resistant hypertension is blood pressure above target despite optimal doses of three antihypertensive agents including a diuretic. True resistance must be distinguished from pseudoresistance caused by poor adherence, white coat effect, or suboptimal measurement. Secondary causes should be investigated. Spironolactone (an aldosterone antagonist) as a fourth agent provides the most evidence for treating true resistant hypertension.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 108 – Review Version 4

The coagulation cascade involves two pathways converging on a common final pathway. The extrinsic pathway, triggered by tissue factor exposure after vessel injury, activates factor X through factor VII. The intrinsic pathway, activated by contact with damaged surfaces, activates factor X through factors XII, XI, IX, and VIII. Factor Xa with factor Va forms prothrombinase, converting prothrombin to thrombin, which converts fibrinogen to fibrin and amplifies coagulation through positive feedback. Anticoagulants including warfarin, heparins, and direct oral anticoagulants target different steps in this cascade.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 109 – Review Version 4

The liver is the primary site of drug metabolism, predominantly through cytochrome P450 enzyme-mediated oxidation, reduction, and hydrolysis reactions (Phase I) followed by conjugation reactions making metabolites more water-soluble for excretion (Phase II). First-pass metabolism reduces bioavailability of oral medications before they reach systemic circulation. Liver disease impairs drug metabolism, requiring dose reductions for hepatically metabolised drugs. Enzyme inducers such as rifampicin and carbamazepine increase drug metabolism, reducing efficacy of co-medications. Enzyme inhibitors such as ciprofloxacin and fluoxetine reduce metabolism, risking toxicity of co-medications.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 110 – Review Version 4

Skin integrity assessment is fundamental to pressure ulcer prevention in at-risk patients. The Braden Scale assesses six risk factors: sensory perception, moisture, activity level, mobility, nutrition, and friction and shear. Scores range from 6 to 23, with lower scores indicating higher risk. Preventive interventions for high-risk patients include regular repositioning every 2 hours, pressure-redistributing mattresses and cushions, skin hygiene and moisturisation, optimisation of nutrition and hydration, and protection of heels with foam boots. Pressure ulcers are classified by depth from Stage 1 (non-blanchable erythema of intact skin) to Stage 4 (full thickness tissue loss exposing bone, tendon, or muscle).

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 1 – Review Version 5

Patient-centred care is an approach that respects and responds to individual patient preferences, needs, and values, ensuring that patient values guide all clinical decisions. Its core dimensions include respect for patient values and expressed needs, emotional support and relief of fear and anxiety, information and communication in accessible language, coordination and integration of care across settings, physical comfort and pain management, involvement of family and friends as appropriate, and access to care and continuity. Evidence consistently shows that patient-centred care improves patient satisfaction, treatment adherence, health outcomes, and reduces unnecessary investigations and readmissions.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 2 – Review Version 5

Prediabetes is characterized by fasting glucose 100-125 mg/dL, 2-hour OGTT glucose 140-199 mg/dL, or HbA1c 5.7-6.4 percent. Without intervention, 5 to 10 percent progress to type 2 diabetes each year. Intensive lifestyle intervention with 5 to 7 percent weight loss and 150 minutes weekly of moderate exercise reduces progression by up to 58 percent, making early identification and intervention critical.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 3 – Review Version 5

Hypertension, defined as blood pressure at or above 130/80 mmHg by ACC/AHA guidelines, affects over 1.3 billion adults worldwide and is the leading modifiable risk factor for cardiovascular disease, stroke, and renal failure. Primary hypertension has no identifiable single cause, with genetic predisposition, dietary sodium, obesity, physical inactivity, and chronic stress all contributing. Secondary causes account for 5 to 10 percent of cases.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 4 – Review Version 5

Thiazide diuretics such as chlorthalidone and hydrochlorothiazide are effective antihypertensives that reduce blood volume through natriuresis. They are particularly effective in Black patients and elderly patients with isolated systolic hypertension. Metabolic side effects include hypokalemia, hyperuricemia, and mild glucose and lipid disturbances. Chlorthalidone has longer duration of action and better evidence for cardiovascular event reduction compared with hydrochlorothiazide.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 5 – Review Version 5

SGLT-2 inhibitors including empagliflozin, dapagliflozin, and canagliflozin reduce blood glucose by inhibiting renal glucose reabsorption, causing glycosuria. Beyond glycemic control, they demonstrate heart failure hospitalisation reduction and nephroprotective effects. They are now recommended early in patients with established cardiovascular disease or chronic kidney disease regardless of glycemic status.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications,

contraindications, and multidisciplinary care approaches.

Topic 6 – Review Version 5

Oedema represents excess fluid accumulation in the interstitial compartment and is classified by distribution and cause. Bilateral leg oedema suggests systemic causes including heart failure (elevated venous pressure with sodium retention), nephrotic syndrome (protein loss reducing oncotic pressure), liver cirrhosis (reduced albumin synthesis and portal hypertension), or hypothyroidism causing myxoedema. Unilateral leg oedema suggests local causes including deep vein thrombosis, venous insufficiency, lymphoedema, or infection. Investigation follows the distribution: bilateral oedema warrants cardiac, renal, liver, and thyroid assessment; unilateral oedema warrants duplex ultrasound to exclude DVT.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 7 – Review Version 5

The DASH diet reduces systolic blood pressure by 8 to 14 mmHg in large randomized trials. It emphasises fruits, vegetables, whole grains, low-fat dairy, and reduced saturated fat and sodium. Limiting sodium below 2.3g daily provides additional reduction. Regular aerobic exercise reduces systolic pressure by a further 5 to 8 mmHg. These lifestyle measures can reduce or delay pharmacological treatment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 8 – Review Version 5

Urinary tract infections are classified as uncomplicated (affecting otherwise healthy, non-pregnant women with normal urinary tracts) or complicated (occurring in men, pregnant women, those with structural abnormalities, immunosuppression, or catheter-related). *Escherichia coli* causes approximately 80 percent of community-acquired UTIs. Uncomplicated lower UTI in women is treated with a 3-day course of trimethoprim or nitrofurantoin based on local resistance patterns. Recurrent UTIs (three or more per year) warrant investigation for predisposing factors and may be managed with low-dose prophylactic antibiotics, post-coital prophylaxis, or self-start therapy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 9 – Review Version 5

These agents stimulate insulin secretion in a glucose-dependent manner, suppress glucagon, slow gastric emptying, and promote weight loss. Cardiovascular outcome trials have shown significant reductions in major adverse cardiovascular events in high-risk patients. GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated significant cardiovascular benefits beyond glycemic control.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 10 – Review Version 5

Screening is performed between 24 and 28 weeks using an oral glucose tolerance test. Management prioritises dietary modification and blood glucose monitoring, with insulin or metformin

added if glycemic targets are not achieved.. Gestational diabetes mellitus complicates 6 to 9 percent of pregnancies and is associated with adverse outcomes including macrosomia, birth trauma, and neonatal hypoglycemia.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 11 – Review Version 5

COPD is characterised by progressive largely irreversible airflow limitation, predominantly caused by cigarette smoking or biomass fuel exposure. Spirometric confirmation with FEV1/FVC below 0.70 post-bronchodilator is required for diagnosis. Emphysema involves alveolar wall destruction reducing gas exchange, while chronic bronchitis is productive cough for at least three months in two consecutive years. Smoking cessation is the only intervention that alters disease progression.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 12 – Review Version 5

WHO recommends adults perform 150 to 300 minutes of moderate-intensity aerobic activity or 75 to 150 minutes of vigorous-intensity activity weekly, plus muscle-strengthening activities twice weekly. Regular physical activity reduces cardiovascular disease, type 2 diabetes, depression, several cancers, and all-cause mortality. Even modest increases from a sedentary baseline provide significant health benefits.. Physical inactivity contributes to 3.2 million deaths annually globally.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 13 – Review Version 5

Medication reconciliation is the process of creating the most accurate possible list of all medications a patient is taking, including dose, frequency, and route, and comparing it against the physician's admission, transfer, or discharge orders to identify and resolve discrepancies. It reduces medication errors at care transitions, which are responsible for approximately 60 percent of serious medication errors in hospital settings. Discrepancies found during reconciliation include omitted medications, wrong doses, duplicated medications, and medications continued that should have been stopped. All newly hospitalised patients should have medication reconciliation performed by a pharmacist within 24 hours of admission.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 14 – Review Version 5

Each class has independently demonstrated mortality reduction in randomised trials. Achieving all four drug classes at target doses reduces all-cause mortality by more than 70 percent compared with no treatment.. Heart failure with reduced ejection fraction is managed with quadruple therapy: an ACE inhibitor, ARB, or ARNI (sacubitril/valsartan), a beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT-2 inhibitor.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 15 – Review Version 5

Hypertensive emergencies involve markedly elevated blood pressure above 180/120 mmHg with acute target organ damage including hypertensive encephalopathy, aortic dissection, acute coronary syndrome, or acute kidney injury. They require immediate ICU admission and parenteral antihypertensive therapy with careful organ-specific blood pressure targets. Hypertensive urgency involves severely elevated pressure without organ damage and can be managed with oral agents and close follow-up.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 16 – Review Version 5

Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection. The Surviving Sepsis Campaign Hour-1 Bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, starting 30 mL/kg intravenous crystalloid for hypotension or elevated lactate, and vasopressors for refractory hypotension. Each hour of delay in antibiotic administration increases mortality by approximately 7 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 17 – Review Version 5

High-intensity statins including atorvastatin 40 to 80mg and rosuvastatin 20 to 40mg are recommended for patients with established cardiovascular disease, targeting LDL-C below 1.4 mmol/L in very high-risk patients. Beyond cholesterol lowering, statins have plaque-stabilising and anti-inflammatory effects.. Statins inhibit HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis, reducing LDL cholesterol by 30 to 55 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 18 – Review Version 5

It progresses from non-proliferative stages with microaneurysms and retinal hemorrhages to proliferative retinopathy with neovascularization risking vitreous hemorrhage. Annual dilated fundal examination enables early detection. Intravitreal anti-VEGF injections and laser photocoagulation are effective treatments for sight-threatening retinopathy.. Diabetic retinopathy is the leading cause of new blindness in working-age adults.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 19 – Review Version 5

Pulmonary rehabilitation is a supervised multidisciplinary programme for patients with chronic respiratory disease combining exercise training, education, and psychosocial support. It consistently improves exercise capacity, quality of life, and dyspnoea scores without altering FEV1. It reduces acute exacerbations and hospital readmissions. It is recommended for COPD patients with MRC dyspnoea grade 3 or above and following hospitalisation for acute exacerbation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications,

contraindications, and multidisciplinary care approaches.

Topic 20 – Review Version 5

Palliative care focuses on improving quality of life for patients and families facing life-limiting illness through prevention and relief of suffering by early identification and impeccable assessment and treatment of pain and other physical, psychosocial, and spiritual problems. It is applicable from diagnosis of a life-limiting condition alongside curative treatment, not only at end of life. The World Health Organization emphasises that palliative care affirms life and regards dying as a normal process, neither hastening nor postponing death. Advance care planning, including discussion of resuscitation preferences, preferred place of death, and lasting power of attorney for health and welfare, is an important component.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 21 – Review Version 5

Cognitive behavioural therapy is a structured evidence-based psychological treatment helping patients identify connections between thoughts, emotions, and behaviours. In depression, CBT targets negative automatic thoughts and behavioural withdrawal through behavioural activation. In anxiety disorders, it incorporates exposure and response prevention, worry postponement, and attentional training. Its benefits persist beyond treatment completion, making it superior to medication alone for long-term relapse prevention.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 22 – Review Version 5

Statins inhibit HMG-CoA reductase, the rate-limiting enzyme in cholesterol synthesis, reducing LDL cholesterol by 30 to 55 percent. High-intensity statins including atorvastatin 40 to 80mg and rosuvastatin 20 to 40mg are recommended for patients with established cardiovascular disease, targeting LDL-C below 1.4 mmol/L in very high-risk patients. Beyond cholesterol lowering, statins have plaque-stabilising and anti-inflammatory effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 23 – Review Version 5

Dehydration occurs when fluid losses exceed intake, reducing total body water. Mild dehydration (1-2 percent body weight) causes thirst and reduced concentration; moderate dehydration (3-5 percent) causes headache, dizziness, and reduced urine output; severe dehydration (above 6 percent) causes tachycardia, hypotension, and confusion. In adults, urine colour is a practical guide: pale yellow indicates adequate hydration; dark amber indicates dehydration. Oral rehydration is appropriate for mild to moderate dehydration using water or oral rehydration salts. Intravenous fluid resuscitation is required for severe dehydration or when oral intake is impossible. Elderly patients are at particular risk due to reduced thirst sensation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 24 – Review Version 5

Key principles include prescribing antibiotics only when bacterial infection is confirmed or highly likely, selecting the narrowest spectrum effective agent, using the shortest evidence-based duration, and reviewing at 48 to 72 hours once microbiological results are available. Antimicrobial resistance is a global crisis threatening the foundations of modern medicine.. Antibiotic stewardship promotes appropriate antimicrobial use to improve patient outcomes while limiting resistance selection pressure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 25 – Review Version 5

ACE inhibitors and ARBs are preferred antihypertensives in patients with diabetes, chronic kidney disease, or heart failure with reduced ejection fraction. ACE inhibitors block conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. ARBs block angiotensin II at its receptor. Both reduce proteinuria and slow nephropathy progression. They should not be combined due to increased adverse effects without additional blood pressure benefit.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 26 – Review Version 5

The principles of medical ethics provide a framework for clinical decision-making when values conflict. Autonomy refers to the patient's right to make informed decisions about their care, requiring valid informed consent. Beneficence is the duty to act in the patient's best interest. Non-maleficence requires avoiding unnecessary harm and balancing risks against benefits. Justice involves fair allocation of healthcare resources and equitable treatment of all patients. When these principles conflict, as in a patient refusing beneficial treatment (autonomy versus beneficence), the framework guides structured ethical analysis. Mental capacity to make specific decisions must be assessed before overriding patient choice, using the Mental Capacity Act 2005 framework in England and Wales.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 27 – Review Version 5

Peripheral diabetic neuropathy affects up to 50 percent of patients with long-standing diabetes, presenting as distal symmetric polyneuropathy with pain, tingling, numbness, and loss of protective sensation. Annual foot examinations and patient education on foot care are essential preventive measures. Pharmacological management of neuropathic pain includes duloxetine, pregabalin, and amitriptyline.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 28 – Review Version 5

Polypharmacy, the concurrent use of five or more medications, affects approximately 40 percent of adults over 65 and significantly increases risk of adverse drug reactions, drug-drug interactions, medication non-adherence, falls, cognitive impairment, and preventable hospitalisation. Regular structured medication reviews using the Beers Criteria or STOPP/START criteria identify potentially inappropriate medications and opportunities for safe deprescribing.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 29 – Review Version 5

Deep vein thrombosis and pulmonary embolism constitute venous thromboembolism, a major cause of preventable hospital mortality. DVT typically involves the leg deep veins, with clot propagation risking embolisation to the pulmonary vasculature. Risk factors include immobility, surgery, malignancy, pregnancy, oral contraceptive use, thrombophilia, and obesity. Treatment with anticoagulation prevents clot extension and embolisation while the body's fibrinolytic system dissolves the thrombus.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 30 – Review Version 5

Calcium channel blockers such as amlodipine are effective antihypertensives with particular benefit in elderly patients, Black patients, and isolated systolic hypertension. They reduce vascular smooth muscle tone by blocking voltage-gated calcium channels. Non-dihydropyridine CCBs such as diltiazem and verapamil also reduce heart rate, useful in atrial fibrillation but contraindicated in heart failure with reduced ejection fraction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 31 – Review Version 5

Dyspepsia (upper abdominal discomfort, bloating, early satiety, or nausea) affects up to 40 percent of the population and has many causes. Functional dyspepsia (no organic cause on investigation) is most common, managed with lifestyle advice, PPI or H2 blocker trial, and consideration of *Helicobacter pylori* testing and eradication if positive. Alarm features requiring urgent endoscopy include unintentional weight loss, dysphagia, persistent vomiting, haematemesis or melaena, iron-deficiency anaemia, and epigastric mass. Any new onset dyspepsia over age 55 warrants early endoscopy. *H. pylori* eradication using triple therapy (PPI, amoxicillin, and clarithromycin for 7 days) achieves eradication in approximately 80 percent of cases.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 32 – Review Version 5

Diabetic retinopathy is the leading cause of new blindness in working-age adults. It progresses from non-proliferative stages with microaneurysms and retinal hemorrhages to proliferative retinopathy with neovascularization risking vitreous hemorrhage. Annual dilated fundal examination enables early detection. Intravitreal anti-VEGF injections and laser photocoagulation are effective treatments for sight-threatening retinopathy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 33 – Review Version 5

BMI provides simple screening: overweight 25 to 29.9, obesity class I 30 to 34.9, class II 35 to 39.9, class III 40 or above. Waist circumference better captures central adiposity and metabolic risk. Management involves comprehensive lifestyle intervention, pharmacotherapy in eligible patients including GLP-1 receptor agonists which can reduce weight by 10 to 20 percent, and bariatric surgery for severe obesity.. Obesity is characterised by excess adiposity impairing health.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 34 – Review Version 5

COX-1 inhibition impairs gastric mucosal protection and platelet aggregation, increasing gastrointestinal bleeding risk particularly in elderly patients, those on corticosteroids or anticoagulants, or those with previous peptic ulcer disease. All NSAIDs carry cardiovascular and renal risks, particularly in volume-depleted patients or those with pre-existing kidney disease.. NSAIDs including ibuprofen, naproxen, and diclofenac inhibit COX-1 and COX-2 enzymes, reducing prostaglandin synthesis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 35 – Review Version 5

It works primarily by inhibiting hepatic gluconeogenesis through activation of AMP-activated protein kinase (AMPK), reducing fasting plasma glucose without causing hypoglycemia. Common side effects include gastrointestinal disturbances, which can be minimized by titrating the dose gradually and taking it with food.. Metformin is the first-line pharmacological agent for type 2 diabetes in most international guidelines.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 36 – Review Version 5

The WHO five moments framework identifies critical junctures: before patient contact, before an aseptic procedure, after body fluid exposure, after patient contact, and after contact with patient surroundings. Alcohol-based hand rub is preferred for most clinical situations due to superior speed and convenience, while soap and water is required for *Clostridioides difficile* and norovirus.. Hand hygiene is the single most effective measure to prevent healthcare-associated infections.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 37 – Review Version 5

They reduce vascular smooth muscle tone by blocking voltage-gated calcium channels. Non-dihydropyridine CCBs such as diltiazem and verapamil also reduce heart rate, useful in atrial fibrillation but contraindicated in heart failure with reduced ejection fraction.. Calcium channel blockers such as amlodipine are effective antihypertensives with particular benefit in elderly patients, Black patients, and isolated systolic hypertension.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 38 – Review Version 5

Physical inactivity contributes to 3.2 million deaths annually globally. WHO recommends adults perform 150 to 300 minutes of moderate-intensity aerobic activity or 75 to 150 minutes of vigorous-intensity activity weekly, plus muscle-strengthening activities twice weekly. Regular physical activity reduces cardiovascular disease, type 2 diabetes, depression, several cancers, and all-cause mortality. Even modest increases from a sedentary baseline provide significant health benefits.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 39 – Review Version 5

DVT typically involves the leg deep veins, with clot propagation risking embolisation to the pulmonary vasculature. Risk factors include immobility, surgery, malignancy, pregnancy, oral contraceptive use, thrombophilia, and obesity. Treatment with anticoagulation prevents clot extension and embolisation while the body's fibrinolytic system dissolves the thrombus.. Deep vein thrombosis and pulmonary embolism constitute venous thromboembolism, a major cause of preventable hospital mortality.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 40 – Review Version 5

Atrial fibrillation is the most common sustained cardiac arrhythmia, causing irregular ventricular response from chaotic atrial electrical activity. The primary risks are stroke from atrial thrombus and heart failure from tachycardia-mediated cardiomyopathy. Management involves rate control, rhythm control in appropriate patients, and anticoagulation with direct oral anticoagulants guided by CHA2DS2-VASc scoring.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 41 – Review Version 5

The primary risks are stroke from atrial thrombus and heart failure from tachycardia-mediated cardiomyopathy. Management involves rate control, rhythm control in appropriate patients, and anticoagulation with direct oral anticoagulants guided by CHA2DS2-VASc scoring.. Atrial fibrillation is the most common sustained cardiac arrhythmia, causing irregular ventricular response from chaotic atrial electrical activity.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 42 – Review Version 5

SSRIs are the first-line pharmacological treatment for depression and anxiety disorders due to their balance of efficacy, tolerability, and safety in overdose compared with older tricyclics. Sertraline has the strongest evidence base in depression, while escitalopram and citalopram have good tolerability profiles. Full therapeutic benefit requires 4 to 6 weeks and treatment should continue for at least 6 to 12 months after remission to prevent relapse.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 43 – Review Version 5

Bipolar disorder features recurrent mania or hypomania alternating with depressive episodes. Lithium remains the gold standard mood stabiliser for long-term prevention of both manic and depressive episodes, with robust evidence for suicide risk reduction. Valproate is effective for mania prevention; lamotrigine particularly targets depressive episodes. Close monitoring of lithium levels, renal function, and thyroid function is essential given its narrow therapeutic index.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 44 – Review Version 5

Type 2 diabetes mellitus is a chronic metabolic disorder characterized by insulin resistance and progressive beta-cell dysfunction. The pathophysiology involves impaired insulin signaling in peripheral tissues, particularly skeletal muscle, liver, and adipose tissue, leading to chronic hyperglycemia. Lifestyle modifications including dietary changes, physical activity, and weight management remain the cornerstone of initial management before pharmacological intervention is considered.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 45 – Review Version 5

Annual foot examinations and patient education on foot care are essential preventive measures. Pharmacological management of neuropathic pain includes duloxetine, pregabalin, and amitriptyline. Peripheral diabetic neuropathy affects up to 50 percent of patients with long-standing diabetes, presenting as distal symmetric polyneuropathy with pain, tingling, numbness, and loss of protective sensation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 46 – Review Version 5

The renin-angiotensin-aldosterone system (RAAS) is a key hormonal cascade regulating blood pressure and fluid balance. Renin, secreted by juxtaglomerular cells in response to reduced renal perfusion, converts angiotensinogen to angiotensin I. ACE converts angiotensin I to the active angiotensin II, a potent vasoconstrictor that also stimulates aldosterone release causing sodium and water retention. RAAS blockade with ACE inhibitors or ARBs is central to managing hypertension, heart failure, and diabetic nephropathy.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 47 – Review Version 5

Septic arthritis is a medical emergency requiring same-day joint aspiration and intravenous antibiotics. It typically presents with acute monoarthritis with fever, a hot swollen joint, and severely restricted movement. The knee is most commonly affected. *Staphylococcus aureus* is the most

common causative organism in adults. Diagnosis is confirmed by synovial fluid analysis showing turbid fluid with white cell count above 50,000 cells per microlitre with predominantly neutrophils, and Gram stain and culture. Delay in treatment risks rapid cartilage destruction and permanent joint damage.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 48 – Review Version 5

The WHO has identified antibiotic resistance as one of the greatest threats to global health. ESKAPE organisms (Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter species) account for the majority of serious healthcare-associated infections with resistance threatening treatment options.. Antimicrobial resistance is driven by overuse and misuse of antibiotics in human medicine and agriculture, selecting for resistant bacterial strains.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 49 – Review Version 5

They are particularly effective in Black patients and elderly patients with isolated systolic hypertension. Metabolic side effects include hypokalemia, hyperuricemia, and mild glucose and lipid disturbances. Chlorthalidone has longer duration of action and better evidence for cardiovascular event reduction compared with hydrochlorothiazide.. Thiazide diuretics such as chlorthalidone and hydrochlorothiazide are effective antihypertensives that reduce blood volume through natriuresis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 50 – Review Version 5

Inhaled corticosteroids are the cornerstone of controller therapy, reducing airway inflammation and risk of exacerbations. Short-acting beta-2 agonists provide rapid bronchodilation for symptom relief. Management follows a stepwise approach guided by symptom control and exacerbation frequency.. Asthma is a chronic inflammatory airway disease characterised by variable reversible airflow obstruction, bronchial hyperresponsiveness, and airway remodelling.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 51 – Review Version 5

Chronic kidney disease is classified by cause, GFR category, and albuminuria category using the CGA staging system. GFR categories range from G1 (normal, above 90 mL/min) to G5 (kidney failure, below 15 mL/min). Albuminuria categories range from A1 (normal to mildly increased, ACR below 3 mg/mmol) to A3 (severely increased, ACR above 30 mg/mmol). High risk of progression is conferred by lower GFR combined with higher albuminuria. Management aims to slow progression through blood pressure control (target below 130/80 mmHg), RAAS blockade in proteinuric CKD, and SGLT-2 inhibitors now recommended across CKD stages for their proven renoprotective effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 52 – Review Version 5

Community-acquired pneumonia severity is assessed using the CURB-65 score, assigning one point each for confusion, urea above 7 mmol/L, respiratory rate 30 or more, blood pressure below 90/60 mmHg, and age 65 or over. Score 0-1 indicates low mortality risk suitable for home treatment. Score 2 warrants hospital admission. Score 3 or above indicates high mortality risk requiring hospital admission and possible ICU assessment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 53 – Review Version 5

It consistently improves exercise capacity, quality of life, and dyspnoea scores without altering FEV1. It reduces acute exacerbations and hospital readmissions. It is recommended for COPD patients with MRC dyspnoea grade 3 or above and following hospitalisation for acute exacerbation.. Pulmonary rehabilitation is a supervised multidisciplinary programme for patients with chronic respiratory disease combining exercise training, education, and psychosocial support.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 54 – Review Version 5

Diabetic ketoacidosis is a life-threatening acute complication caused by absolute or relative insulin deficiency. The biochemical triad includes hyperglycemia, metabolic acidosis, and ketonemia. Management requires fluid resuscitation, insulin infusion after potassium correction, electrolyte replacement particularly potassium, and identification of the precipitating cause such as infection or omitted insulin.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 55 – Review Version 5

Acute myocardial infarction results from rupture of an atherosclerotic plaque in a coronary artery leading to thrombus formation and vessel occlusion. Time to reperfusion is the critical determinant of outcome: primary percutaneous coronary intervention is the preferred strategy when achievable within 120 minutes of first medical contact. Immediate treatment includes aspirin, a P2Y12 inhibitor, anticoagulation, and infarct-related artery reperfusion.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 56 – Review Version 5

The HbA1c target for most adults with type 2 diabetes is below 7.0 percent, though individualized targets should account for age, comorbidities, hypoglycemia risk, and patient preference. In elderly or frail patients, a less stringent target of 7.5 to 8.0 percent may be appropriate to avoid hypoglycemia and its associated falls and cardiac risks.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 57 – Review Version 5

They require immediate ICU admission and parenteral antihypertensive therapy with careful organ-specific blood pressure targets. Hypertensive urgency involves severely elevated pressure without organ damage and can be managed with oral agents and close follow-up.. Hypertensive emergencies involve markedly elevated blood pressure above 180/120 mmHg with acute target organ damage including hypertensive encephalopathy, aortic dissection, acute coronary syndrome, or acute kidney injury.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 58 – Review Version 5

CYP3A4 enzyme interactions are particularly clinically significant: clarithromycin inhibits CYP3A4 and dramatically raises plasma levels of statins (risking rhabdomyolysis), ciclosporin, and benzodiazepines. Warfarin has hundreds of documented interactions and requires careful monitoring with any medication change.. Drug-drug interactions occur through pharmacokinetic mechanisms altering absorption, distribution, metabolism, or excretion, or pharmacodynamic mechanisms with additive or antagonistic effects.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 59 – Review Version 5

Vitamin D deficiency is highly prevalent, affecting approximately 1 billion people globally, and results from inadequate sun exposure, dietary insufficiency, or malabsorption. It causes rickets in children and osteomalacia in adults, presenting with bone pain, proximal muscle weakness, and pathological fractures. Severe vitamin D deficiency also impairs immune function and has been associated with increased risk of autoimmune disease, cardiovascular disease, and certain cancers. Supplementation guidelines recommend 400 IU daily for all adults in the UK and higher doses (1000-4000 IU) for those at high risk including elderly care home residents, those with dark skin or little sun exposure, and pregnant women.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 60 – Review Version 5

Spirometric confirmation with FEV1/FVC below 0.70 post-bronchodilator is required for diagnosis. Emphysema involves alveolar wall destruction reducing gas exchange, while chronic bronchitis is productive cough for at least three months in two consecutive years. Smoking cessation is the only intervention that alters disease progression.. COPD is characterised by progressive largely irreversible airflow limitation, predominantly caused by cigarette smoking or biomass fuel exposure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 61 – Review Version 5

Ambulatory blood pressure monitoring over 24 hours provides a more accurate assessment of true blood pressure burden. Masked hypertension, the converse, carries cardiovascular risk similar to sustained hypertension and should not be overlooked.. White coat hypertension affects 15 to 30 percent of referred patients and involves elevated readings in clinical settings but normal levels at home or on ambulatory monitoring.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 62 – Review Version 5

Chronic pain is defined as pain persisting beyond the normal expected healing time, typically more than 3 months. It affects approximately 28 million adults in the UK and has a significant biopsychosocial impact. It is now understood as a complex condition involving peripheral and central sensitisation, psychological factors including fear-avoidance beliefs and catastrophising, and social factors including work and relationship impacts. The modern pain management paradigm has shifted from a largely pharmacological approach to a multidisciplinary biopsychosocial model incorporating physiotherapy, psychological therapies particularly CBT and acceptance and commitment therapy, lifestyle optimisation, and judicious use of analgesics at the lowest effective dose.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 63 – Review Version 5

It emphasises fruits, vegetables, whole grains, low-fat dairy, and reduced saturated fat and sodium. Limiting sodium below 2.3g daily provides additional reduction. Regular aerobic exercise reduces systolic pressure by a further 5 to 8 mmHg. These lifestyle measures can reduce or delay pharmacological treatment.. The DASH diet reduces systolic blood pressure by 8 to 14 mmHg in large randomized trials.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 64 – Review Version 5

Informed consent is a legal and ethical requirement for medical procedures and treatment. For consent to be valid, the patient must have capacity to make the decision, the consent must be given voluntarily without coercion, and the patient must be provided with sufficient information about the proposed treatment including its benefits, risks, alternatives, and consequences of no treatment. Montgomery v Lanarkshire Health Board (2015) established in UK law that patients must be informed of all material risks that a reasonable patient would consider significant. Documentation of the consent process is essential. Consent for research requires additional safeguards including Research Ethics Committee approval.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 65 – Review Version 5

Inflammatory markers including C-reactive protein (CRP) and erythrocyte sedimentation rate (ESR) are non-specific indicators of inflammation used to support diagnosis and monitor treatment response. CRP rises and falls rapidly within hours to days, making it more useful for monitoring acute conditions. ESR rises more slowly and reflects plasma viscosity. Both are elevated in

infection, autoimmune disease, malignancy, and tissue necrosis. Very high CRP above 100 mg/L suggests bacterial infection or severe inflammation. Procalcitonin is more specific for bacterial infection and is used in some settings to guide antibiotic prescribing. Normal inflammatory markers do not exclude serious pathology, particularly in immunocompromised patients.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 66 – Review Version 5

Continuous glucose monitoring systems provide real-time glucose readings every few minutes, enabling patients and clinicians to make more informed treatment decisions. Time-in-range, defined as glucose between 70 and 180 mg/dL, is emerging as a key glycemic metric alongside HbA1c. Higher time-in-range is associated with reduced rates of diabetic microvascular and macrovascular complications.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 67 – Review Version 5

GLP-1 receptor agonists such as semaglutide and liraglutide have demonstrated significant cardiovascular benefits beyond glycemic control. These agents stimulate insulin secretion in a glucose-dependent manner, suppress glucagon, slow gastric emptying, and promote weight loss. Cardiovascular outcome trials have shown significant reductions in major adverse cardiovascular events in high-risk patients.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 68 – Review Version 5

Antibiotic stewardship promotes appropriate antimicrobial use to improve patient outcomes while limiting resistance selection pressure. Key principles include prescribing antibiotics only when bacterial infection is confirmed or highly likely, selecting the narrowest spectrum effective agent, using the shortest evidence-based duration, and reviewing at 48 to 72 hours once microbiological results are available. Antimicrobial resistance is a global crisis threatening the foundations of modern medicine.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 69 – Review Version 5

Anaemia is defined as haemoglobin below the lower limit of normal for age and sex: below 130 g/L in adult men and below 120 g/L in adult women. The main causes are classified by mechanism: iron deficiency (most common worldwide, from blood loss, poor intake, or malabsorption), vitamin B12 or folate deficiency (macrocytic anaemia from impaired DNA synthesis), haemolysis (destruction of red blood cells from autoimmune, mechanical, or hereditary causes), and bone marrow failure or infiltration. History, full blood count, reticulocyte count, and iron studies guide initial investigation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 70 – Review Version 5

Primary hypertension has no identifiable single cause, with genetic predisposition, dietary sodium, obesity, physical inactivity, and chronic stress all contributing. Secondary causes account for 5 to 10 percent of cases.. Hypertension, defined as blood pressure at or above 130/80 mmHg by ACC/AHA guidelines, affects over 1.3 billion adults worldwide and is the leading modifiable risk factor for cardiovascular disease, stroke, and renal failure.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 71 – Review Version 5

The PREDIMED trial demonstrated significant reductions in major cardiovascular events in high-risk patients following a Mediterranean diet compared with a low-fat diet. It also reduces risk of type 2 diabetes, colorectal and breast cancer, cognitive decline, and all-cause mortality.. The Mediterranean diet is characterised by high consumption of plant foods, olive oil as the primary fat, moderate fish and seafood, and low red and processed meat consumption.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 72 – Review Version 5

Occupational health medicine focuses on the relationship between work and health in both directions: how work can cause or worsen health conditions (occupational disease), and how health conditions affect the ability to work. Common occupational diseases include occupational asthma (from isocyanates, flour dust, or latex), vibration white finger from hand-arm vibration syndrome, noise-induced hearing loss, mesothelioma from asbestos exposure, contact dermatitis, and musculoskeletal disorders from manual handling or repetitive strain. Fitness for work assessments consider functional capacity, job demands, reasonable workplace adjustments, and prognosis. Statutory sick pay and fit notes (formerly sick notes) are important administrative tools in managing work absence.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 73 – Review Version 5

Hand hygiene is the single most effective measure to prevent healthcare-associated infections. The WHO five moments framework identifies critical junctures: before patient contact, before an aseptic procedure, after body fluid exposure, after patient contact, and after contact with patient surroundings. Alcohol-based hand rub is preferred for most clinical situations due to superior speed and convenience, while soap and water is required for *Clostridioides difficile* and norovirus.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 74 – Review Version 5

Diabetic nephropathy affects approximately 40 percent of diabetic patients and is the leading cause of end-stage renal disease globally. Early detection through annual screening for urinary albumin-to-creatinine ratio and eGFR is essential. ACE inhibitors and ARBs are first-line for hypertension in diabetic patients with albuminuria due to their independent nephroprotective properties beyond blood pressure lowering.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 75 – Review Version 5

Asthma is a chronic inflammatory airway disease characterised by variable reversible airflow obstruction, bronchial hyperresponsiveness, and airway remodelling. Inhaled corticosteroids are the cornerstone of controller therapy, reducing airway inflammation and risk of exacerbations. Short-acting beta-2 agonists provide rapid bronchodilation for symptom relief. Management follows a stepwise approach guided by symptom control and exacerbation frequency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 76 – Review Version 5

Time-in-range, defined as glucose between 70 and 180 mg/dL, is emerging as a key glycemic metric alongside HbA1c. Higher time-in-range is associated with reduced rates of diabetic microvascular and macrovascular complications.. Continuous glucose monitoring systems provide real-time glucose readings every few minutes, enabling patients and clinicians to make more informed treatment decisions.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 77 – Review Version 5

ACE inhibitors block conversion of angiotensin I to angiotensin II, reducing vasoconstriction and aldosterone secretion. ARBs block angiotensin II at its receptor. Both reduce proteinuria and slow nephropathy progression. They should not be combined due to increased adverse effects without additional blood pressure benefit.. ACE inhibitors and ARBs are preferred antihypertensives in patients with diabetes, chronic kidney disease, or heart failure with reduced ejection fraction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 78 – Review Version 5

The Surviving Sepsis Campaign Hour-1 Bundle includes measuring lactate, obtaining blood cultures before antibiotics, administering broad-spectrum antibiotics, starting 30 mL/kg intravenous crystalloid for hypotension or elevated lactate, and vasopressors for refractory hypotension. Each hour of delay in antibiotic administration increases mortality by approximately 7 percent.. Sepsis is life-threatening organ dysfunction caused by a dysregulated host response to infection.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 79 – Review Version 5

Sertraline has the strongest evidence base in depression, while escitalopram and citalopram have good tolerability profiles. Full therapeutic benefit requires 4 to 6 weeks and treatment should continue for at least 6 to 12 months after remission to prevent relapse.. SSRIs are the first-line pharmacological treatment for depression and anxiety disorders due to their balance of efficacy, tolerability, and safety in overdose compared with older tricyclics.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 80 – Review Version 5

Wound healing progresses through four overlapping phases: haemostasis (immediate platelet plug and clot formation), inflammation (neutrophil and macrophage infiltration clearing debris and bacteria), proliferation (fibroblast migration producing collagen, angiogenesis, and re-epithelialisation), and remodelling (collagen reorganisation over months to years). Factors impairing wound healing include diabetes (neuropathy, ischaemia, and impaired immune response), malnutrition (particularly protein, zinc, and vitamin C deficiency), corticosteroids, smoking (causing vasoconstriction and reduced oxygen delivery), infection, and foreign bodies. Assessment should determine the wound bed, edges, surrounding skin, exudate, infection signs, and underlying cause.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 81 – Review Version 5

True resistance must be distinguished from pseudoresistance caused by poor adherence, white coat effect, or suboptimal measurement. Secondary causes should be investigated. Spironolactone (an aldosterone antagonist) as a fourth agent provides the most evidence for treating true resistant hypertension.. Resistant hypertension is blood pressure above target despite optimal doses of three antihypertensive agents including a diuretic.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 82 – Review Version 5

Antimicrobial resistance is driven by overuse and misuse of antibiotics in human medicine and agriculture, selecting for resistant bacterial strains. The WHO has identified antibiotic resistance as one of the greatest threats to global health. ESKAPE organisms (Enterococcus faecium, Staphylococcus aureus, Klebsiella pneumoniae, Acinetobacter baumannii, Pseudomonas aeruginosa, and Enterobacter species) account for the majority of serious healthcare-associated infections with resistance threatening treatment options.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 83 – Review Version 5

Pain assessment in clinical practice uses validated tools appropriate to the patient's communication ability. The numerical rating scale (0-10) is the most commonly used in cognitively intact adults. The visual analogue scale uses a 100mm line from no pain to worst imaginable pain. The Wong-Baker FACES scale is appropriate for children and patients with cognitive impairment. Multidimensional assessment tools such as the Brief Pain Inventory assess pain intensity and its impact on function and quality of life. Chronic pain assessment should include physical, psychological, and social dimensions using a biopsychosocial framework, with regular reassessment using the same validated tool for consistency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 84 – Review Version 5

Chronic liver disease progresses through stages from simple steatosis through steatohepatitis, fibrosis, cirrhosis, and finally hepatocellular carcinoma. Common causes include alcohol-related liver disease, non-alcoholic fatty liver disease (increasingly prevalent with obesity epidemic), chronic hepatitis B and C, autoimmune hepatitis, primary biliary cholangitis, and haemochromatosis. Cirrhosis complications include portal hypertension causing varices and ascites, hepatic encephalopathy, spontaneous bacterial peritonitis, hepatorenal syndrome, and coagulopathy. Six-monthly ultrasound and AFP for hepatocellular carcinoma surveillance is recommended in all patients with established cirrhosis.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 85 – Review Version 5

The Mediterranean diet is characterised by high consumption of plant foods, olive oil as the primary fat, moderate fish and seafood, and low red and processed meat consumption. The PREDIMED trial demonstrated significant reductions in major cardiovascular events in high-risk patients following a Mediterranean diet compared with a low-fat diet. It also reduces risk of type 2 diabetes, colorectal and breast cancer, cognitive decline, and all-cause mortality.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 86 – Review Version 5

The biochemical triad includes hyperglycemia, metabolic acidosis, and ketonemia. Management requires fluid resuscitation, insulin infusion after potassium correction, electrolyte replacement particularly potassium, and identification of the precipitating cause such as infection or omitted insulin.. Diabetic ketoacidosis is a life-threatening acute complication caused by absolute or relative insulin deficiency.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 87 – Review Version 5

Heart failure with reduced ejection fraction is managed with quadruple therapy: an ACE inhibitor, ARB, or ARNI (sacubitril/valsartan), a beta-blocker, a mineralocorticoid receptor antagonist, and an SGLT-2 inhibitor. Each class has independently demonstrated mortality reduction in randomised trials. Achieving all four drug classes at target doses reduces all-cause mortality by more than 70 percent compared with no treatment.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 88 – Review Version 5

Depression affects over 280 million people globally and is characterised by persistent low mood, anhedonia, fatigue, cognitive impairment, sleep disturbance, appetite changes, and suicidal ideation in severe cases. The biopsychosocial model recognises neurobiological contributions including dysregulation of serotonin, norepinephrine, and dopamine pathways, alongside psychological factors such as negative cognitive schemas and social factors including adverse life events and isolation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 89 – Review Version 5

Score 0-1 indicates low mortality risk suitable for home treatment. Score 2 warrants hospital admission. Score 3 or above indicates high mortality risk requiring hospital admission and possible ICU assessment.. Community-acquired pneumonia severity is assessed using the CURB-65 score, assigning one point each for confusion, urea above 7 mmol/L, respiratory rate 30 or more, blood pressure below 90/60 mmHg, and age 65 or over.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 90 – Review Version 5

Fibromyalgia is a centralised pain syndrome characterised by widespread musculoskeletal pain, fatigue, sleep disturbance, cognitive difficulties (fibro fog), and often comorbid depression and anxiety. It results from central sensitisation where the nervous system amplifies pain signals. Diagnosis is clinical using the 2010 ACR diagnostic criteria, requiring widespread pain index and symptom severity scores above thresholds. Management is multidisciplinary: aerobic exercise and physiotherapy are the most evidence-based interventions, CBT addresses maladaptive pain behaviours, and low-dose amitriptyline or duloxetine can improve pain and sleep. Opioids are not recommended and may worsen central sensitisation.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 91 – Review Version 5

Thyroid hormones T3 (triiodothyronine) and T4 (thyroxine) regulate metabolic rate, growth, and development. Hypothyroidism causes fatigue, weight gain, cold intolerance, constipation, dry skin, and bradycardia, and is most commonly caused by Hashimoto's thyroiditis (autoimmune destruction). Hyperthyroidism causes weight loss despite increased appetite, heat intolerance, palpitations, tremor, and anxiety, most commonly from Graves' disease (TSH receptor stimulating antibodies). TSH is the most sensitive screening test for thyroid dysfunction, rising in hypothyroidism and falling in hyperthyroidism.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 92 – Review Version 5

Lithium remains the gold standard mood stabiliser for long-term prevention of both manic and depressive episodes, with robust evidence for suicide risk reduction. Valproate is effective for mania prevention; lamotrigine particularly targets depressive episodes. Close monitoring of lithium levels, renal function, and thyroid function is essential given its narrow therapeutic index.. Bipolar disorder features recurrent mania or hypomania alternating with depressive episodes.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 93 – Review Version 5

Delirium is an acute neuropsychiatric syndrome characterised by disturbance of attention and awareness with fluctuating course, developing acutely over hours to days. It affects up to 30 percent of hospitalised elderly patients and is associated with increased mortality, longer hospital stay, and accelerated cognitive decline. Precipitating factors include infection, metabolic disturbance, medications particularly anticholinergics and opioids, pain, urinary retention, constipation, dehydration, and unfamiliar environment. Management prioritises identifying and treating underlying causes, non-pharmacological reorientation strategies, and early mobilisation. Antipsychotics are used cautiously for severe agitation only.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 94 – Review Version 5

Obesity is characterised by excess adiposity impairing health. BMI provides simple screening: overweight 25 to 29.9, obesity class I 30 to 34.9, class II 35 to 39.9, class III 40 or above. Waist circumference better captures central adiposity and metabolic risk. Management involves comprehensive lifestyle intervention, pharmacotherapy in eligible patients including GLP-1 receptor agonists which can reduce weight by 10 to 20 percent, and bariatric surgery for severe obesity.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 95 – Review Version 5

Early detection through annual screening for urinary albumin-to-creatinine ratio and eGFR is essential. ACE inhibitors and ARBs are first-line for hypertension in diabetic patients with albuminuria due to their independent nephroprotective properties beyond blood pressure lowering. Diabetic nephropathy affects approximately 40 percent of diabetic patients and is the leading cause of end-stage renal disease globally.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 96 – Review Version 5

White coat hypertension affects 15 to 30 percent of referred patients and involves elevated readings in clinical settings but normal levels at home or on ambulatory monitoring. Ambulatory blood pressure monitoring over 24 hours provides a more accurate assessment of true blood pressure burden. Masked hypertension, the converse, carries cardiovascular risk similar to sustained hypertension and should not be overlooked.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 97 – Review Version 5

NSAIDs including ibuprofen, naproxen, and diclofenac inhibit COX-1 and COX-2 enzymes, reducing prostaglandin synthesis. COX-1 inhibition impairs gastric mucosal protection and platelet aggregation, increasing gastrointestinal bleeding risk particularly in elderly patients, those on corticosteroids or anticoagulants, or those with previous peptic ulcer disease. All NSAIDs carry cardiovascular and renal risks, particularly in volume-depleted patients or those with pre-existing kidney disease.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 98 – Review Version 5

In depression, CBT targets negative automatic thoughts and behavioural withdrawal through behavioural activation. In anxiety disorders, it incorporates exposure and response prevention, worry postponement, and attentional training. Its benefits persist beyond treatment completion, making it superior to medication alone for long-term relapse prevention.. Cognitive behavioural therapy is a structured evidence-based psychological treatment helping patients identify connections between thoughts, emotions, and behaviours.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 99 – Review Version 5

The pathophysiology involves impaired insulin signaling in peripheral tissues, particularly skeletal muscle, liver, and adipose tissue, leading to chronic hyperglycemia. Lifestyle modifications including dietary changes, physical activity, and weight management remain the cornerstone of initial management before pharmacological intervention is considered.. Type 2 diabetes mellitus is a chronic metabolic disorder characterized by insulin resistance and progressive beta-cell dysfunction.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 100 – Review Version 5

Metformin is the first-line pharmacological agent for type 2 diabetes in most international guidelines. It works primarily by inhibiting hepatic gluconeogenesis through activation of AMP-activated protein kinase (AMPK), reducing fasting plasma glucose without causing hypoglycemia. Common side effects include gastrointestinal disturbances, which can be minimized by titrating the dose gradually and taking it with food.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 101 – Review Version 5

Beyond glycemic control, they demonstrate heart failure hospitalisation reduction and nephroprotective effects. They are now recommended early in patients with established cardiovascular disease or chronic kidney disease regardless of glycemic status.. SGLT-2 inhibitors including empagliflozin, dapagliflozin, and canagliflozin reduce blood glucose by inhibiting renal glucose reabsorption, causing glycosuria.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 102 – Review Version 5

Without intervention, 5 to 10 percent progress to type 2 diabetes each year. Intensive lifestyle intervention with 5 to 7 percent weight loss and 150 minutes weekly of moderate exercise reduces progression by up to 58 percent, making early identification and intervention critical.. Prediabetes is

characterized by fasting glucose 100-125 mg/dL, 2-hour OGTT glucose 140-199 mg/dL, or HbA1c 5.7-6.4 percent.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 103 – Review Version 5

Gestational diabetes mellitus complicates 6 to 9 percent of pregnancies and is associated with adverse outcomes including macrosomia, birth trauma, and neonatal hypoglycemia. Screening is performed between 24 and 28 weeks using an oral glucose tolerance test. Management prioritises dietary modification and blood glucose monitoring, with insulin or metformin added if glycemic targets are not achieved.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 104 – Review Version 5

Time to reperfusion is the critical determinant of outcome: primary percutaneous coronary intervention is the preferred strategy when achievable within 120 minutes of first medical contact. Immediate treatment includes aspirin, a P2Y₁₂ inhibitor, anticoagulation, and infarct-related artery reperfusion.. Acute myocardial infarction results from rupture of an atherosclerotic plaque in a coronary artery leading to thrombus formation and vessel occlusion.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 105 – Review Version 5

Drug-drug interactions occur through pharmacokinetic mechanisms altering absorption, distribution, metabolism, or excretion, or pharmacodynamic mechanisms with additive or antagonistic effects. CYP3A4 enzyme interactions are particularly clinically significant: clarithromycin inhibits CYP3A4 and dramatically raises plasma levels of statins (risking rhabdomyolysis), ciclosporin, and benzodiazepines. Warfarin has hundreds of documented interactions and requires careful monitoring with any medication change.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 106 – Review Version 5

The Glasgow Coma Scale assesses conscious level through three components: eye opening (1-4), verbal response (1-5), and motor response (1-6), giving a total of 3 to 15. GCS 15 indicates full consciousness; GCS 8 or below indicates coma and the need to protect the airway. It is used in acute neurological and trauma assessment, with trends over time being more informative than a single measurement. The AVPU scale (Alert, Voice, Pain, Unresponsive) provides a simpler bedside assessment for initial triage. Causes of reduced conscious level are memorised using the AEIOU TIPS acronym: Alcohol/Acidosis, Epilepsy/Electrolytes, Infection, Opioids/Overdose, Uraemia, Trauma/Temperature, Insulin, Psychiatric, Stroke/Shock.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 107 – Review Version 5

Resistant hypertension is blood pressure above target despite optimal doses of three antihypertensive agents including a diuretic. True resistance must be distinguished from pseudoresistance caused by poor adherence, white coat effect, or suboptimal measurement. Secondary causes should be investigated. Spironolactone (an aldosterone antagonist) as a fourth agent provides the most evidence for treating true resistant hypertension.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 108 – Review Version 5

The coagulation cascade involves two pathways converging on a common final pathway. The extrinsic pathway, triggered by tissue factor exposure after vessel injury, activates factor X through factor VII. The intrinsic pathway, activated by contact with damaged surfaces, activates factor X through factors XII, XI, IX, and VIII. Factor Xa with factor Va forms prothrombinase, converting prothrombin to thrombin, which converts fibrinogen to fibrin and amplifies coagulation through positive feedback. Anticoagulants including warfarin, heparins, and direct oral anticoagulants target different steps in this cascade.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 109 – Review Version 5

The liver is the primary site of drug metabolism, predominantly through cytochrome P450 enzyme-mediated oxidation, reduction, and hydrolysis reactions (Phase I) followed by conjugation reactions making metabolites more water-soluble for excretion (Phase II). First-pass metabolism reduces bioavailability of oral medications before they reach systemic circulation. Liver disease impairs drug metabolism, requiring dose reductions for hepatically metabolised drugs. Enzyme inducers such as rifampicin and carbamazepine increase drug metabolism, reducing efficacy of co-medications. Enzyme inhibitors such as ciprofloxacin and fluoxetine reduce metabolism, risking toxicity of co-medications.

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.

Topic 110 – Review Version 5

Skin integrity assessment is fundamental to pressure ulcer prevention in at-risk patients. The Braden Scale assesses six risk factors: sensory perception, moisture, activity level, mobility, nutrition, and friction and shear. Scores range from 6 to 23, with lower scores indicating higher risk. Preventive interventions for high-risk patients include regular repositioning every 2 hours, pressure-redistributing mattresses and cushions, skin hygiene and moisturisation, optimisation of nutrition and hydration, and protection of heels with foam boots. Pressure ulcers are classified by depth from Stage 1 (non-blanchable erythema of intact skin) to Stage 4 (full thickness tissue loss exposing bone, tendon, or muscle).

Key points: Review the epidemiology, pathophysiology, diagnosis, management, prevention, and patient education related to this topic. Consider current guidelines, common complications, contraindications, and multidisciplinary care approaches.